INCH-POUND
MIL-DTL-32772
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DETAIL SPECIFICATION

SWITCHGEAR, POWER, SHOCK-MOUNTED, LOW VOLTAGE, NAVAL SHIPBOARD

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1. SCOPE

1.1 Scope. This specification covers electrical power dead-front, shock-mounted, low voltage (see 6.3.9) switchgear for naval shipboard use.

1.2 Classification. Switchgear covered by this specification are of the following types:

a. Type I: Ship's service power switchgear group (see 6.3.13) for ships with alternating current (AC) power.

b. Type II: Emergency switchgear group for ships with AC power.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-208	-	Ink, Marking, Stencil, Opaque (Porous and Non-Porous Surfaces)
A-A-59125	-	Terminal Boards, Molded, Barrier Screw and Stud Types and Associated Accessories
A-A-59125/2	-	Terminal Boards, Molded, Barrier Screw Type, Class 38TB
A-A-59125/9	-	Terminal Boards, Molded, Barrier Stud Type, Class 3TB
A-A-59781	-	Light Emitting Diodes for Use as Indicator Lights

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-I-631	-	Insulation, Electrical, Synthetic-Resin Composition, Nonrigid
MIL-DTL-901	-	Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for
MIL-DTL-917	-	Electric Power Equipment, Basic Requirements for
MIL-DTL-1222	-	Studs, Bolts, Screws and Nuts for Applications Where a High Degree of Reliability is Required; General Specification for
MIL-I-1361	-	Instrument Auxiliaries, Electrical Measuring: Shunts, Resistors, and Transformers
MIL-R-2729	-	Regulator-Exciter Systems, Voltage, A.C. Generator, Naval Shipboard Use
MIL-I-3158	-	Insulation Tape, Electrical Glass-Fiber (Resin-Filled): and Cord, Fibrous-Glass
MIL-DTL-3661	-	Lampholders, Indicator Lights, Indicator Light Housings, and Indicator Light Lenses, General Specification for

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MIL-L-3661/38	-	Housings, Indicator-Light, Style LH80
MIL-L-3661/62	-	Lampholder, Lights, Indicator (Housing), Style LH95 (for D.C. Applications)
MIL-L-3661/63	-	Lampholder, Lights, Indicator (Housing), Style LH96
MIL-L-3661/64	-	Lampholder, Lights, Indicator (Housing), Style LH97
MIL-DTL-3661/65	-	Lampholder, Lights, Indicator (Housing), Style LH98
MIL-DTL-5541	-	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-DTL-15024	-	Plates and Tags for Identification of Equipment, General Specification for
MIL-P-15024/5	-	Plates, Identification
MIL-T-15108	-	Transformers, Power, Step-Down, Single-Phase, 60-Hertz, 1-Kilovoltampere Approximate Minimum Rating, Dry Type, Naval Shipboard
MIL-DTL-15109	-	Resistors and Rheostats, Naval Shipboard
MIL-PRF-15160	-	Fuses, Instrument, Power, and Telephone, General Specification for
MIL-DTL-16034	-	Meters, Electrical-Indicating (Switchboard and Portable Types), General Specification for
MIL-S-16104	-	Synchroscope
MIL-M-16125	-	Meters, Electrical, Frequency
MIL-T-16315	-	Transformers, Power, Step-Down (Miscellaneous, Naval Shipboard Use)
MIL-T-16366	-	Terminals, Electrical Lug and Conductor Splices, Crimp-Style
MIL-PRF-17773	-	Switch, Bus Transfer, Electric Power, Automatic
MIL-DTL-18240	-	Fastener Element, Self-Locking, Threaded Fastener, 250 °F Maximum
MIL-W-19088	-	Wattmeters, Switchboard Type, 4-1/2 Inch
MIL-PRF-19207	-	Fuseholders, Extractor Post Type, Blown Fuse, Indicating and Nonindicating, General Specification for
MIL-PRF-19207/1	-	Fuseholders, Extractor Post Type, Blown Fuse Indicating, Types FHL10U and FHL10G
MIL-PRF-19207/2	-	Fuseholders, Extractor Post Type, Blown Fuse Indicating, Types FHL11U and FHL11G
MIL-PRF-19207/3	-	Fuseholders, Extractor Post Type, Blown Fuse Indicating, Types FHL12U and FHL12G
MIL-PRF-19207/5	-	Fuseholders, Extractor Post Type, Blown Fuse Indicating, Type FHL14G
MIL-PRF-19207/21	-	Fuseholders, Extractor Post Type, Blown Fuse Indicating, Non-EMI/RFI Shielded and EMI/RFI Shielded, Type FHL32W and Type FHL32WS
MIL-PRF-19207/22	-	Fuseholders, Extractor Post Type, Blown Fuse Indicating, Type FHL33W
MIL-PRF-19207/23	-	Fuseholders, Extractor Post Type, Blown Fuse Indicating, Type FHL34W

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MIL-PRF-19207/24	-	Fuseholders, Extractor Post Type, Blown Fuse Indicating, Non-EMI/RFI Shielded and EMI/RFI Shielded, Type FHL35W and Type FHL35WS
MIL-DTL-21604	-	Switches, Rotary, Multipole and Selector, General Specification for
MIL-DTL-21604/5	-	Switch, Rotary, Multipole and Selector, 10 Ampere, Style JR
MIL-V-23151	-	Voltmeter, Expanded Scale Switchboard Type (Naval Shipboard Use)
MIL-M-23167	-	Meter, Frequency, Expanded Scale Switchboard Type (Naval Shipboard Use)
MIL-S-24188	-	Synchronizing Control Equipment, 60 Cycles, 450 Volts, Naval Shipboard
MIL-M-24350	-	Monitors, Reverse Power and Power-Sensing, Electrical Power (Naval Shipboard Use) (Metric)
MIL-I-24391	-	Insulation Tape, Electrical, Plastic, Pressure-Sensitive
MIL-DTL-24552	-	Terminals and Seals, Plugs and Switch, Casualty Power Systems, Receptacle Type, General Specification for
MIL-DTL-24552/1	-	Terminals and Seals, Plugs and Switch, Casualty Power, Receptacle Type, Back Connected 200-Ampere, 450-Volt, Alternating Current, Three-Phase, Symbol No. 1046
MIL-DTL-24552/2	-	Terminals and Seals, Plugs and Switch, Casualty Power, Receptacle Type, Upper and Lower Riser, 200-Ampere, 450-Volt, Alternating Current, Three-Phase, Symbol No. 1047
MIL-R-24563	-	Relay, Alternating Current, Power-Sensing
MIL-PRF-24712	-	Coatings, Powder, Thermosetting
MIL-DTL-28803	-	Display, Optoelectronic, Segmented Readouts, Backlighted, General Specification for
MIL-DTL-32656	-	Arc Fault Protection System, General Specification for
MIL-DTL-83522	-	Connectors, Fiber Optic, Single Ferrule, General Specification for
MIL-PRF-85045	-	Cables, Fiber Optic, General Specification for

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-108	-	Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment
MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)
MIL-STD-202	-	Electronic and Electrical Component Parts
MIL-STD-461	-	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
MIL-STD-740-2	-	Structureborne Vibratory Acceleration Measurements and Acceptance Criteria of Shipboard Equipment
MIL-STD-810	-	Environmental Engineering Considerations and Laboratory Tests

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MIL-STD-1310	-	Shipboard Bonding, Grounding, and Other Techniques for Electromagnetic Compatibility, Electromagnetic Pulse (EMP) Mitigation, and Safety
MIL-STD-1399-300-1	-	Low Voltage Electric Power, Alternating Current
MIL-STD-1472	-	Human Engineering
MIL-STD-1474	-	Noise Limits
MIL-STD-2003-2	-	Electric Plant Installation Standard Methods for Surface Ships and Submarines (Equipment)

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-454	-	General Guidelines for Electronic Equipment
MIL-HDBK-470	-	Designing and Developing Maintainable Products and Systems, Volume I

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

29 CFR 1910	-	Occupational Safety and Health Standards

(Copies of this document are available online at www.ecfr.gov.)

COMMITTEE ON NATIONAL SECURITY SYSTEMS INSTRUCTIONS (CNSSI)

CNSSI No. 1253	-	Security Categorization and Control Selection for National Security Systems

(Copies of this document are available online at www.cnss.gov/CNSS/issuances/Instructions.cfm.)

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY (NIST)

NIST SP 800-53	-	Security and Privacy Controls for Information Systems and Organizations
NIST SP 800-82	-	Guide to Industrial Control Systems (ICS) Security

(Copies of these documents are available online at www.nist.gov.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9074-AQ-GIB-010/248	-	Requirements for Welding and Brazing Procedure and Performance Qualification
S9074-AR-GIB-010/278	-	Requirements for Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels
T9070-AL-DPC-020/077-2	-	NAVSEA Hazardous Material Avoidance Process

(Copies of these documents are available online via Model Based Product Support (MBPS) at https://mbps.navseaplm.navy.mil/Windchill/app/. To gain access to MBPS, obtain an account with National Help Desk Service Management (NHDSM) at https://nhdsm.navair.navy.mil (a valid CAC is required to access this website) and submit a SAAR-N Request. Refer questions, inquiries, or problems to (888) 292-5919. These documents are available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASME

ASME Y14.38 - Abbreviations and Acronyms for Use in Product Definition and Related Documents: Engineering Product Definition and Related Documentation Practices

(Copies of this document are available online at www.asme.org.)

ASTM INTERNATIONAL

ASTM A131/A131M - Standard Specification for Structural Steel for Ships

ASTM A240/A240M - Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications

ASTM A666 - Standard Specification for Annealed or Cold-Worked Austenitic Stainless Steel Sheet, Strip, Plate, and Flat Bar

ASTM A1011/A1011M - Standard Specification for Steel, Sheet and Strip, Hot-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, and Ultra-High Strength

ASTM B36/B36M - Standard Specification for Brass Plate, Sheet, Strip, and Rolled Bar

ASTM B121/B121M - Standard Specification for Leaded Brass Plate, Sheet, Strip, and Rolled Bar

ASTM B127 - Standard Specification for Nickel-Copper Alloy Plate, Sheet, and Strip

ASTM B164 - Standard Specification for Nickel-Copper Alloy Rod, Bar, and Wire

ASTM B187/B187M - Standard Specification for Copper, Bus Bar, Rod, and Shapes and General Purpose Rod, Bar, and Shapes

ASTM B209/B209M - Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate

ASTM B221 - Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes

ASTM B564 - Standard Specification for Nickel Alloy Forgings

ASTM B700 - Standard Specification for Electrodeposited Coatings of Silver for Engineering Use

ASTM D709 - Standard Specification for Laminated Thermosetting Materials

ASTM F2362 - Standard Specification for Temperature Monitoring Equipment

(Copies of these documents are available online at www.astm.org.)

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS, INC. (IEEE)

IEEE C37.20.1 - IEEE Standard for Metal-Enclosed Low-Voltage (1000 Vac and below, 3200 Vdc and below) Power Circuit Breaker Switchgear

(Copies of this document are available online at www.ieee.org.)

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IPC

IPC-CC-830 - Qualification and Performance of Electrical Insulating Compound for Printed Wiring Assemblies

(Copies of this document are available online at www.ipc.org.)

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NAS 411-1 - Hazardous Materials Target List

NASM16228 - Nut, Self-Locking, Hexagon, Thin, UNC-3B, Non Metallic Insert, Austenitic Corrosion Resistant Steel, Nonmagnetic, 250 °F

NASM17829 - Nut, Self-Locking, Hexagon, Regular Height, 250 °F, Non-Metallic Insert, Non-CRES Steel

NASM17830 - Nut, Self-Locking, Hexagon-Regular, 250 °F and 450 °F, Non-Metallic Insert, 300 Series CRES

(Copies of these documents are available online at www.aia-aerospace.org.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ANSI/NEMA HP 5 - Electrical and Electronic Crosslinked, Modified Polyethylene (XLPE) Insulated 125°C Hook-Up Wire, Types L (600 V), LL (1,000 V), and LX (3,000 V)

ANSI/NEMA HP 6 - Electrical and Electronic Silicone and Silicone-Braided Insulated Hook-Up Wire Types S (600 V), ZHS (600 V), SS (1,000 V), ZHSS (1,000 V) and SSB Braided (1,000 V)

(Copies of these documents are available online at www.nema.org.)

SAE INTERNATIONAL

SAE-AS7928 - Terminals, Lug: Splices, Conductor: Crimp Style, Copper, General Specification for

SAE-AS33671 - Strap, Tiedown, Electrical Components, Adjustable, Self-Clinching, Plastic, Type I, Class 1

(Copies of these documents are available online at www.sae.org.)

UNDERWRITERS LABORATORIES (UL)

UL 94 - Tests for Flammability of Plastic Materials for Parts in Devices and Appliances

UL 489D - Outline of Investigation for US Naval Commercial-Off-the-Shelf (COTS) Designated Molded-Case Circuit Breakers, Molded-Case Switches, and Accessories for Use on Non Nuclear Naval Combatant Ships

UL 746C - Polymeric Materials - Use in Electrical Equipment Evaluations

UL 1066A - Outline of Investigation for US Naval Commercial Off the Shelf (COTS) Designated Three-Pole Low-Voltage AC Power Circuit Breakers, Drawout-Mounted Type, and Accessories/Functional Components for Use on Non-Nuclear Naval Combatant Ships

(Copies of these documents are available online at www.ul.com.)

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2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample switchboard (see 6.3.12) shall be subjected to first article inspection in accordance with 4.2.

3.2 General requirements.

3.2.1 Safety. The switchgear shall provide fail-safe features for safety of personnel during the installation, operation, maintenance, and repair or interchanging of a complete equipment assembly or component parts thereof. Equipment design for personnel safety shall be equal to or better than the appropriate requirements of the Occupational Safety and Health Act as identified in Title 29, Part 1910 of the Code of Federal Regulations (CFR). Unless otherwise specified herein, safety shall be in accordance with MIL-DTL-917.

3.2.1.1 Accidental contact. When energized during normal operations, the switchgear shall provide protection to personnel against accidental contact with voltages in excess of 30 volts root mean square (rms) or direct current (DC) with all removable panels in place and hinged doors closed.

3.2.1.2 Door interlocks. When specified (see 6.2), door interlocks shall de-energize accessible power circuits in a compartment (see 6.3.2) greater than 30 volts.

3.2.1.3 Tag-out requirements. Control circuits operating at greater than 30 volts shall have tag-out provisions for isolation and safety of connected equipment during maintenance.

3.2.1.3.1 Molded case type circuit breakers. Since molded case type commercial circuit breakers do not have holes drilled through the handle for purposes of hanging tags, other provisions shall be made for tag-out.

3.2.1.4 Maintenance test points/ground studs. When bus bars are fully insulated, easily accessible test points/ground studs shall be provided to perform initial voltage verification and insulation resistance maintenance checks. Ground studs shall have field replaceable boots that allow maintenance access.

3.2.2 Design life. All switchboards and switchboard equipment that are detailed in this specification shall have a minimum design life of not less than 30 years. Specific components not meeting this requirement shall be identified.

3.3 Construction requirements.

3.3.1 Materials and coatings. Materials and coatings contained in or on the switchboard shall meet the requirements of MIL-DTL-917 except as modified or added to by the requirements of this specification.

3.3.1.1 Prohibited materials. The switchboard shall not contain any chemicals categorized as “prohibited” in accordance with T9070-AL-DPC-020/077-2 for NAVSEA programs or NAS 411-1 for all other programs.

3.3.1.1.1 Polyvinyl chloride (PVC). PVC material shall not be used unless it is a part of or integral to the leads of a component or as a constituent or integrated element of a component.

3.3.1.1.2 Cadmium. Elemental cadmium shall not be present in any component of the final product.

3.3.1.1.3 Mercury. Mercury shall not be used in any process of manufacture or test of switchgear and its components.

3.3.1.2 Non-preferred component materials. The following component materials are non-preferred and shall not be used without prior approval by the Technical Authority (see 6.3.14):

a. Insulating materials that contain halogens (e.g., chlorine, fluorine, and bromine) which evolve gases during combustion.

b. Organic materials not inherently fungus-resistant or validated to be fungus-resistant. Refer to MIL-HDBK-454, Guideline 4.

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c. Corrosion-susceptible materials not in accordance with MIL-DTL-917 requirements.

d. Lead free solder used for printed circuit boards.

e. Type I compositions containing hexavalent chromium in accordance with MIL-DTL-5541.

3.3.1.3 Coatings.

3.3.1.3.1 Painting.

3.3.1.3.1.1 Enamel. Switchboards shall be prepared and painted in accordance with MIL-DTL-917, with the exception that only one coat of gray enamel shall be applied. Touching up is permitted for marks or scratches due to assembly, testing, or other factory handling.

3.3.1.3.1.2 Powder coating. Powder coating may be used as an alternative to enamel painting. The powder coating shall be in accordance with MIL-PRF-24712.

3.3.1.3.1.3 Grounding fasteners with paint. Grounding connections shall be free of enamel or powder coat paint. Toothed lockwashers or similar shall not be used to grind enamel or powder coat paint away under screws for grounding.

3.3.1.3.2 Chemical conversion coatings. Chemical conversion coatings may be used as a protective coating for aluminum frame and aluminum components internal to the switchboard. Coatings may also be used as a preparation for the paint process. Electrically-conductive class 3 coating may be used in grounding. When used, coatings shall meet the requirements of MIL-DTL-5541 for visually discernible color.

3.3.1.3.3 Non-preferred component coatings. Corrosion-susceptible coatings not in accordance with MIL-DTL-917 requirements are non-preferred and shall be identified (see 6.2).

3.3.1.3.4 Conformal coating. All circuit boards shall be conformal coated in accordance with IPC-CC-830.

3.3.2 Assembly. When the switchboard consists of multiple vertical units (see 6.3.15), they are bolted together to form a switchboard section (see 6.3.11). All surfaces shall be finished smooth, welded or bolted together, and reinforced where necessary with structural members for vertical units of the switchboard section.

3.3.3 Welding. Unless otherwise specified herein, requirements for welding and allied processes shall be in accordance with applicable American Welding Society (AWS) standards or NAVSEA S9074-AR-GIB-010/278.

3.3.4 Accessibility (see also 3.5.10).

3.3.4.1 Space. Space shall be provided for the installation, securing, and connection of cables. The space for cable routing shall include that required for the bending radius of the cable and for the installation of cable lugs.

3.3.4.2 Servicing. Equipment, bus work, fasteners, and electrical connections that may require servicing, repair, or replacement during the life of the switchboard shall be readily accessible for servicing, inspection, tightening, repair, or replacement. Access shall be maintained after all bus work, switchboard wiring, and ship cabling are installed. The enclosure side sheets, or portions thereof, shall be removable to provide installation and maintenance access.

3.3.5 Switchboard structure.

3.3.5.1 Structural materials and general construction methods. Switchboard structure may be either of steel or aluminum construction as specified (see 6.2). Steel conforming to ASTM A131/A131M shall be utilized in the fabrication of switchboard structures. Steel conforming to ASTM A1011/A1011M shall be used for plates and sheets. Aluminum alloy conforming to ASTM B221, alloy 5086-H111 shall be used for bars, rods, profiles, and tubes. Aluminum alloy conforming to ASTM B209/B209M, alloy 5052-H32, 5052-H34, or 5086-H32 shall be used for plates and sheets. When aluminum components are subsequently bent, 50XX grades of aluminum shall be used. Each section (see 6.3.11) of a switchgear group (see 6.3.13) may be constructed using alternate methods of construction that differ from the traditional welded frame.

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3.3.5.2 Compartmentation. Each unit shall be internally subdivided by barriers of either sheet metal, nonconductive material, or a combination of both to form separate compartments for the circuit breakers to isolate them from the bus work and cable connections and eliminate gaps between compartments. There shall be no gap between the barrier and top enclosure of the switchboard. The gap between the barrier and the horizontal plane at the bottom of the switchboard shall not exceed 3/8 inch. Gaps on any switchboard barrier sections not covered above shall not exceed 1/16 inch. If necessary, barriers may overlap. The power circuit breaker stationary component (cradle) is acceptable as a rear barrier in a power circuit breaker compartment. The power circuit breaker cradle may have vents present in their design. Cradle vents do not invalidate the compartmentation requirement. When barriers are installed, insulation distances shall be as specified in 3.3.6.2. A separate compartment shall be provided for each power circuit breaker, but all molded case circuit breakers on one unit may be in a common compartment. The compartments shall be constructed to prevent an arc or fault from spreading from one compartment to another or into the area occupied by the bus work. Bus disconnecting devices shall be located outside the circuit breaker compartment. Each power circuit breaker compartment shall be enclosed on the front by a removable panel section. On molded case circuit breaker compartments over 60 inches high, the front enclosure shall be made by two panels.

3.3.5.2.1 Compartmentation for switchgear equipped with arc fault detectors. Compartmentation for switchgear which utilizes Arc Fault Detection (AFD) shall be similar to that of 3.3.5.2 with the exception that the gap between the unit's internal subdivision barrier and the horizontal plane at the bottom of the switchboard shall not exceed 3/4 inch. All other gaps shall not exceed 3/16 inch.

3.3.5.3 Ventilation. Ventilation for compartments within a unit and for the unit itself shall be constructed, and the openings sized, so that the maximum internal temperature shall not exceed 167 °F (75 °C) when the unit is operated in an external ambient temperature of 122 °F (50 °C). Power circuit breakers shall be vented through the front of the unit or through the rear of the unit. Components and wiring required to be installed in the units shall operate to rated capacity in the above internal, ambient temperature without performance degradation. Ventilating louvers and grill openings shall minimize the probability of entrance of solids and shall protect personnel against injury from electric arcs. An operating test of applicable controls and communication functions shall be performed at temperature stabilization to verify functionality at high temperatures.

3.3.5.3.1 Louvers and other openings. Louvers and other openings shall not pass a rod having a diameter greater than 3/8 inch.

3.3.5.4 Thermal imaging windows. When specified (see 6.2), the number and locations of thermal imaging windows shall be sufficient to allow viewing of selected current-carrying joints, components, or expected hot spots to the maximum extent practicable. Thermal imaging windows shall meet the following requirements:

a. Shall be transparent in the visible light spectrum and in the 8-14 micron wavelength band to allow both visual and thermal imaging camera viewing of selected current-carrying joints, components, and expected hot spots within the switchboard. The windows shall maintain a uniform transmission factor when operating in the temperature range of 15 to 70 °C.

b. Shall have a minimum size of 3 inches in diameter to allow off-angle viewing. To the extent practicable, mesh, grates or other structures in the window that obscure viewing or limit the viewing angle should be avoided.

c. Shall meet applicable material requirements of UL 94 and UL 746C and shall be non-hygroscopic or shall be treated to prevent water absorption.

d. Shall be reliably secured in such a manner that it cannot become adrift during first article testing.

e. Shall have a permanently attached, hinged solid metal impact cover that meets all applicable requirements for the enclosure. The metal cover shall be secured or opened without tools for viewing access by non-removable fasteners and shall be able to remain secured in the closed position when the thermal imaging window is not being used for viewing.

3.3.5.4.1 Thermal imaging window assembly label. A label shall be provided near each thermal imaging window assembly stating the transmissivity of the viewing window perpendicular to the plane of the window and shall include the following statement:

"Metal cover shall remain closed and secured when window is not in use."

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3.3.5.5 Hinging. Front and rear doors shall have rolled or formed edges as required to provide a rigid frame unit and shall be of the hinged or bolted type. Hinge pins shall provide a bearing surface axially for positive alignment. Doors shall be provided with captive hand operable knobs for fastening in place. Panels shall be hinged to provide access to electrical equipment mounted thereon. Means shall be provided so that these hinged sections may be opened without removing the guard rail. Hinged panels shall be provided with door stops or positioning devices to stop the door in its opening swing and to hold it in the open position. The preferred side for hinging panels is the left-hand side when facing the front of the panel. A typical panel hinge assembly is shown on figure 1. Hinged panels shall be electrically grounded to the unit framework by a separate conductor. Hinges are not considered to be an acceptable grounding path.

3.3.5.6 Lifting. Provisions for lifting such as lifting channels, angles, and lifting eyes shall be provided on the top of each shipping section of sufficient strength and number for lifting, handling, and installation on the ship. Additionally, replacement bolts, with all necessary washers to maintain drip-proof integrity, shall be provided in order for the lifting components to be replaced when the switchboard is installed on the ship.

3.3.5.7 Removable sheets and panels. Removable sheets and panels on the front and rear shall be provided with lifting handles. Mounting and alignment devices shall be provided to facilitate installation and removal of removable enclosing panels. When needed to prevent incorrect assembly, alignment pins or devices shall be staggered or have a unique arrangement in order to prevent interchangeability of panel fronts or rears between units of the same switchboard section and units of other switchboards sections within the same switchboard group or ship's compartment.

3.3.5.8 Panel fastening. Hinged and removable panels on the front and rear of the unit shall be secured by thumb screws (see figure 2 and figure 3). Anchor devices for thumbscrews (see figure 4) that are used to attach front and rear panels to the switchboard framing shall use hex head or counter-sunk head bolts and self-locking nuts, or button socket-head or flat socket-head, self-locking screws in tapped holes. Welding, brazing, or riveting shall not be used for anchor devices.

3.3.5.9 Side sheets. The exterior side sheets of the enclosure, or portions thereof, shall be removable to provide installation and maintenance access. Removable side sheets are not considered structural members.

3.3.5.10 Guard rails. Each switchboard unit, except for distribution units (see 6.3.5), shall be provided with a guard rail. The guard rail shall have a sufficient number of brackets to provide support for personnel under all operating conditions of the ship. The guard rail shall be attached in such a manner that all panels can be removed and hinged panels opened without removal of the guard rail from the member to which it is attached. These guard rails shall be located so that they will not interfere with operation of the equipment and shall not extend more than 4 inches from the front of the panel. Guard rails shall be made from electrically insulating materials. See figure 5 for typical vertical guard rail mounting.

3.3.5.10.1 Removable guard rails. When specified (see 6.2), removable guard rail sections shall be provided on the rear framework of the switchboard to prevent personnel from being thrown against live buses caused by the motion of the ship when the rear enclosure is removed. Rails shall be required at proper levels when needed to prevent personnel from being thrown against live buses caused by the motion of the ship. Rails shall be mounted on the inside of the rear vertical members so as not to protrude beyond the rear or ends of the structure. Guard rail sections shall be easily removable for maintenance access, but shall be provided with securing bolts or other locking means to prevent dislodgement caused by shock. See figure 6 for typical removable guard rail mounting.

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3.3.5.11 Top enclosure. A sheet cover shall enclose the top of each switchboard section. The top shall prevent dripping water or falling objects from entering the switchboard enclosure or damaging instruments on the front panels. It shall extend approximately 4 inches beyond the face of the front panels and 4 inches beyond the rear of the switchboard structure. Clearance between the upper edge of the front panels and the underside of the top sheet, where it extends over the front, shall be at least 1/4 inch. Front and rear edges of the top sheet shall be turned up to form a 1-inch flange along the entire length of the section or switchboard to prevent water from spilling over the front or rear. To ensure watertight integrity at the corners, the turned-up front and rear flanges shall be extended around the corners and along the end of the top sheet approximately 8 inches. The edge of the top sheet shall be turned down to form a flange and shall overlap the side sheets of the switchboard to allow drainage over the end in a manner that prevents water from seeping down the inside of the side sheet. Cable/wiring that penetrates the top panel shall be passed through cable transits, stuffing tubes, or by means of other acceptable methods to maintain drip-proof integrity.

3.3.6 Insulation.

3.3.6.1 Insulating material. Insulating material for bus supports and spacers, fuse bases, and other similar parts shall be a minimum of class B (266 °F [130 °C]) in accordance with MIL-DTL-917. Insulating material shall be sealed to prevent the absorption of water. Primary conductor insulating material not in accordance with MIL-DTL-917 requirements may be used, but such material shall be subjected to flame testing of 4.4.3.15.

3.3.6.2 Insulation distances (creepage and clearance). Insulation distances in air and surface creepage distances on insulating materials for equipments shall be in accordance with applicable specifications. For all other live parts, such as bus work, and bus to component line-side and load-side stud connections on circuit breaker bases which have a frame size greater than 100 amperes, values shall meet the applicable creepage and clearance distances in table I. See 3.3.6.2.1 for 100-ampere circuit breaker connection requirements.

TABLE I. Insulation distances (creepage and clearance).

Voltage	In air (inches)	Insulation distances to ground or to opposite polarity surface creepage	
		Bottom and side surfaces (inches)	Top and flat surfaces 1/ (inches)
125 AC or DC	0.50	0.94	1.13
230 AC	0.63	1.20	1.50
500 AC or DC	0.78	1.56	2.00
1000 AC	1.31	2.50	3.34

NOTE:

1/ Values for top creepage apply to flat surfaces or to curved surfaces presenting sufficient irregularities to permit accumulation of dust and moisture. Values are not intended to apply to simply cylindrical surfaces having a radius of 3 inches or less where side creepage values apply.

3.3.6.2.1 Special insulation distance requirements for 100-ampere frame circuit breaker base connections. Bus to component stud line-side and load-side connections on 100-ampere frame circuit breaker bases shall meet the following requirements:

a. Bus connections within 4 inches of the bus/component stud connection shall meet the clearance distances of MIL-DTL-917.

b. Bus connections greater than 4 inches from the bus/component stud connection shall meet the clearance values of table I.

c. Creepage distance shall be in accordance with the values provided in table I.

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3.3.7 Fastener locking devices. Unless otherwise specified in the individual equipment specifications, locking devices shall be provided in accordance with 3.3.7.1 for electrical connections and mechanical assemblies in all switchboards, switchgear units, and in all equipment except where solderless types of connectors are used for conductors below 14 American Wire Gauge (AWG) (4000 circular mils [cmils]). A visual method to check bus bar and cable fastener tightness (such as torque check paint) shall be provided.

3.3.7.1 Acceptable locking devices. The following locking devices rated for 250 °F (121.1 °C) or higher shall be acceptable for use (clarifications noted as applicable):

a. Corrosion-resistant steel (CRES) self-locking nuts in accordance with NASM17830 or zinc-plated steel self-locking nuts in accordance with NASM17829. CRES self-locking nuts in accordance with NASM16228 may be used as an alternative subject to approval by the Technical Authority. Consideration shall be given as applicable when taking into account that NASM16228 has lower strength requirements than NASM17830.

b. Nut and jam nut for molded case circuit breaker or molded case switch mounting base cylindrical stud connections only.

c. In electrical connections where a self-locking nut or jam nut cannot be used, a fastener with a self-locking pellet, type N, in accordance with MIL-DTL-18240 shall be used as an alternative.

d. Anaerobic threadlocking compound shall only be used in structural or mechanical applications.

e. Toothed or split-ring lockwashers shall only be acceptable as a constituent or integrated element of a component such as a terminal board.

Tightening requirements for self-locking nuts, jam nuts, and self-locking pellet, type N fasteners used in bus bar joints or other applications shall be in accordance with 3.3.10.6.3.2.1.

3.3.7.2 Prohibited locking devices. The following locking devices shall not be used (clarifications noted as applicable):

a. Anaeorbic threadlocking compound shall not be used in electrical connections.

b. Belleville washers (plain).

c. Toothed and split ring lockwashers shall not be used except as described in 3.3.7.1.

d. A fastener with a self-locking pellet, type N, in accordance with MIL-DTL-18240 shall not be used in an electrical connection unless it is not feasible to use a self-locking nut or jam nut.

e. Clip-on types of nuts.

f. Single-thread engaging nuts formed by stamping a thread-engaging impression in a flat piece of metal.

g. Setscrews.

h. Nut and jam nut shall not be used except as described in 3.3.7.1.b.

3.3.8 Wiring of control and instrument circuits.

3.3.8.1 Control and instrument cable types. Cables used for control and instrument circuits, ground detector, indicator lights, communications devices, and other parts shall be in accordance with the following types as specified (see 6.2):

a. ANSI/NEMA HP 6, type SSB, 1000 volt, 392 °F (200 °C) or equivalent.

b. ANSI/NEMA HP 5, type L, 600 volt, 257 °F (125 °C) or equivalent.

c. ANSI/NEMA HP 5, type LL, 1000 volt, 257 °F (125 °C) or equivalent.

d. ANSI/NEMA HP 5, type LX, 3000 volt, 257 °F (125 °C) or equivalent.

Control and instrument wiring shall not be smaller than size AWG 16 but may be larger, if necessary, for current carrying capacity requirements. Communication device wiring shall not be smaller than size AWG 20. Control cable for low-level signals for electronic devices and load sharing shall be shielded type.

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3.3.8.2 Wire and terminal marking. Internal wiring of equipment shall be marked on both ends by fiber tags or by synthetic resin tubing type F, grade A of MIL-I-631. See figure 7 for typical marking. Markings shall agree with the designation shown on the switchboard drawings to facilitate checking of connections. It shall not be necessary to mark the terminal studs on instruments and control switches, but wiring diagrams shall clearly indicate the studs or contacts to which marked wires are connected. Terminal blocks shall be equipped with marking strips. These marking strips shall be permanently marked at each terminal with a number (for example, 1 to 12) and also to agree with the corresponding number on the wiring diagram. Terminal blocks in each unit shall be numbered TB1, TB2, and so forth.

3.3.8.3 Securing of control wiring. Control wiring shall adhere to the following requirements:

a. Wires shall be neatly formed into groups which shall be bundled or banded and supported or clamped in a manner that will prevent chafing of the insulation caused by vibration.

b. Bundling material shall be tie-wraps in accordance with SAE-AS33671, nonflammable cord, glass fiber tape in accordance with type SR-1 of MIL-I-3158, or pressure-sensitive adhesive plastic tape in accordance with MIL-I-24391. The last two materials are specifically adapted for serving switchgear control wiring in locations subject to abrasion. Glass fiber tape may be coated with adhesive on one side, but when used with this adhesive coating, it shall be covered with pressure sensitive adhesive plastic tape for permanent adhesion.

c. Wire groups running from hinged panels shall be formed and clamped so that sharp bends do not occur with the panel in either the open or closed position and shall be connected to terminal blocks on the fixed portion of the switchgear structure.

d. There shall be no splices in the wire and connections shall be made at the terminal studs of the devices or terminal boards.

e. Where wires run from one section of a switchboard to an adjacent one, and these sections are separated for shipping or installation purposes, a terminal board shall be provided on one of the units and located in order to facilitate completion of the interconnections during installation of the switchboard on the ship.

3.3.8.4 External control cable connections. External control cable connections to the switchboards shall terminate at terminal boards accessibly located near the top or bottom of the switchboard, corresponding to ship's cable entrance. Where calibrated shunt leads are required, they shall terminate on the instrument studs. An excess length of shunt leads shall be neatly coiled and secured to prevent interference with access to other equipment. Small wiring to circuit breaker trip coils shall run directly to the coils and associated contact circuits and shall not be routed through terminals of other apparatus. Taps from bus bars shall be made by through-bolting terminals to the bus bars.

3.3.8.5 Wire end connectors. Ends of each wire shall be connected to the apparatus studs by means of an SAE-AS7928, type II solderless terminal or by forming a wire loop around the apparatus stud or terminal screw and retaining the loop in a cup or crimped washer. The wire loop shall be soldered to secure the strands together. Locking devices in accordance with 3.3.7.1 shall be used to secure the nuts or screws connecting wire loops to the apparatus connection point. See figure 7 for illustration of wire connections.

3.3.8.6 Fiber optic connections.

3.3.8.6.1 Fiber optic cables. Fiber optic cables shall comply with the requirements of MIL-PRF-85045 or alternative specification approved by the Technical Authority.

3.3.8.6.2 Fiber optic connectors. Fiber optic connectors shall comply with the requirements of MIL-DTL-83522 or alternative specification approved by the Technical Authority.

3.3.8.7 Cable connector sizes. On switchboards having an AC bus structure designed for short circuit currents of 15,000 amperes or less, single-conductor insulated copper cable, size AWG 4, in accordance with ANSI/NEMA HP 6, type SSB, 392 °F (200 °C) or equivalent, shall be used for connections to 100-ampere frame size molded case type circuit breakers. On switchboard buses having a total connected DC generator capacity of not more than 150 kilowatts at 120 volts, similar cable connections shall be provided for 100-ampere frame size molded case type circuit breakers.

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3.3.8.8 Cable connector sizes for casualty power. Cable connecting a casualty power circuit breaker to the line-side of an AC generator circuit breaker shall be size AWG 1/0, in accordance with ANSI/NEMA HP 6, type SSB, 392 °F (200 °C) or equivalent. Casualty power terminal shall be enclosure type similar to that depicted by method 2D-3-2 in accordance with MIL-STD-2003-2. Cable connecting the casualty power circuit breaker and the casualty power terminal shall be size AWG 2 in accordance with ANSI/NEMA HP 6, type SSB, 392 °F (200 °C) or equivalent.

3.3.9 Power cables. The required shipboard cable, terminations, routing, recommended bracing, and the location of the cable connections shall be as specified (see 6.2).

3.3.9.1 Securing of power cables. Power cable groups shall adhere to the following requirements:

a. Cables shall be neatly formed into groups which shall be bundled or banded and supported or clamped in a manner that will prevent chafing of the insulation caused by vibration. Cables shall not be supported by bus bars. There shall be a minimum clearance of 1/4 inch between cables and bus bars to prevent abrasion under conditions of vibration, except at the bus bar terminating end of the cable.

b. Cable straps or metal clamps may be used to support or secure cables, but where metal is used it shall be covered with a flame-retardant, heat-resistant material which will prevent chafing of the cable insulation. Commercial nylon cable straps shall not be used to support or secure power cables to the framework or structure of the switchboard unit.

c. There shall be no power cable splices within the switchgear enclosure. Power cables shall be bolted directly to the bus bars or terminal studs of the device.

d. Cable lug terminals installed on all cables entering or within the switchboard shall be crimp type CLC or CLCG in accordance with MIL-T-16366. Prior to assembly and crimping, a light coating of NO-OX-ID "A-SPECIAL" (NSN 8030-00-598-5915) corrosion-preventive paste, or equal, shall be applied to the conductor and the interior of the terminal lug barrel. CLC or CLCG lug terminals shall be used with power circuit breakers or bus bar applications. Cable lug terminals shall not be installed on spare circuit breakers to which a cable is not attached. For all cables of 14 AWG (4000 cmils) and larger, cable lug terminals shall be secured by means of locking devices in accordance with 3.3.7.1.

3.3.10 Bus bars and bus bar joint connections.

3.3.10.1 Bus bar material. Rectangular copper bus bars with rounded corners or round edges, half hard temper in accordance with ASTM B187/B187M shall be used. Aluminum conductors or terminals shall not be used.

3.3.10.2 Bus bar sizes. Size of the bus bars shall be selected on a basis of the current carrying ratings shown in table II. In no case shall the bus bar sizes be smaller for a given current than the corresponding sizes listed in table II.

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TABLE II. Bus bar sizes and ampere rating.

Number of bars parallel	Size of bars (inches)	Cross-sectional area (square inches)	Copper bus, silver surface, AC ampere rating, 60 Hz	Copper bus, silver surface, DC ampere rating
1	1/2 x 1/8	0.063	140	140
	5/8 x 1/8	0.078	175	175
	3/4 x 1/8	0.094	210	210
	1 x 1/8	0.125	285	285
	1-1/2 x 1/8	0.188	425	425
	2 x 1/8	0.250	555	555
	3/4 x 3/16	0.140	265	265
	1 x 3/16	0.188	355	355
	1-1/2 x 3/16	0.278	550	550
	2 x 3/16	0.375	700	710
	3/4 x 1/4	0.188	295	295
	1 x 1/4	0.250	410	410
	1-1/2 x 1/4	0.375	600	600
	2 x 1/4	0.500	780	800
	2-1/2 x 1/4	0.625	1,000	1,050
	3 x 1/4	0.750	1,140	1,185
	4 x 1/4	1.000	1,425	1,490
	5 x 1/4	1.250	1,760	1,850
	6 x 1/4	1.500	2,100	2,190
2 (1/4 inch apart)	1 x 1/4	0.500	650	650
	1-1/2 x 1/4	0.750	950	950
	2 x 1/4	1.000	1,350	1,370
	2-1/2 x 1/4	1.250	1,610	1,610
	3 x 1/4	1.500	1,825	2,000
	4 x 1/4	2.000	2,280	2,530
	5 x 1/4	2.500	2,740	3,100
	6 x 1/4	3.000	3,140	3,630
3 (1/4 inch apart)	3 x 1/4	2.250	2,200	2,620
	4 x 1/4	3.000	2,660	3,110
	5 x 1/4	3.750	3,200	3,830
	6 x 1/4	4.500	3,600	4,560
4 (1/4 inch apart)	3 x 1/4	3.000	2,650	3,130
	4 x 1/4	4.000	3,020	3,870
	5 x 1/4	5.000	3,450	4,750

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TABLE II. Bus bar sizes and ampere rating – Continued.

Number of bars parallel	Size of bars (inches)	Cross-sectional area (square inches)	Copper bus, silver surface, AC ampere rating, 60 Hz	Copper bus, silver surface, DC ampere rating
4 (2 pairs/phase, 3/4 inch between pairs)	6 x 1/4	6.000	4,000	----
4 (2 pairs/phase, 2-1/2 inches between pairs	5 x 1/4	5.000	4,200	----
	6 x 1/4	6.000	5,000	----
	8 x 1/4	8.000	6,400	----

3.3.10.2.1 Optional bus bar geometry. Bus bars shall be of sufficient size to carry the rated current without exceeding the temperature rating of the bus bar, but in all circumstances shall be no smaller than and in accordance with the requirements of table II. Bus bars not conforming to the requirements of table II shall require additional temperature sensors during temperature rise testing to validate the design (see 4.4.3.8.2).

3.3.10.3 Bus bar capacity. The main bus bar rating shall not be less than the largest feeder circuit element rating of the power source supplying the main bus as specified (see 6.2). Either main bus taps (see 6.3.10) or branch bus taps (see 6.3.1) feeding three or more molded case circuit breakers shall be sized based on 75 percent of the sum of the active and spare circuit breaker frame sizes (maximum current rating) supplied, but not exceeding the main bus rating. Main bus taps and branch bus taps feeding one or two circuit breakers shall be sized based on 100 percent of the active and spare circuit breaker frame sizes supplied, but not exceeding the main bus rating.

3.3.10.4 Bus bar forming. Flat bends for 1/8-inch bus bars shall have an inside radius 1.5 times the thickness of the bus bar, and the ends of the bus bars shall be neatly finished. Flat bends for bus bars greater than 1/8 inch shall have an inside radius of no less than the thickness of the bus bar, and the ends of the bus bars shall be neatly finished. Edgewise bends of copper bus bars up to 2 inches in width may be made on an inside radius of 1 inch. Edgewise bends of copper bus bars over 2 inches in width, but not exceeding 4 inches in width, may be made on an inside radius of 2 inches. Bars shall be free from cracks or flaws at bends.

3.3.10.5 Bus bar insulation. When specified (see 6.2), bus bar insulation shall be rated for 266 °F (130 °C). Insulating materials that contain halogens (e.g., chlorine, fluorine, and bromine) which evolve gases during combustion, shall not be used. When insulating varnish is utilized, the coating shall be flexible, tough, moisture and acid resisting, and shall have a dielectric strength of at least 500 volts per mil of thickness. One coat 0.004 to 0.006 inch thick shall be applied. Before painting, burrs and sharp edges shall be removed and bus bars cleaned to remove dust, grease, oil, or other foreign material. Paint may be applied by spraying or flowed on with a soft bristle brush when spraying is impractical. Coatings shall be free from pinholes. Bus bars shall be painted before assembly up to 1/4 to 1/2 inch from joint area. All current-carrying parts, including bus joints and connections, shall be insulated upon completion of final shipboard assembly.

3.3.10.6 Bus bar joints.

3.3.10.6.1 Welding or brazing. Bus bars may be joined by welding or brazing; however, it shall be restricted to assemblies where disassembly will not be required for maintenance or repair of underlying assemblies. All welders and weld procedures for copper bus bars shall be in accordance with NAVSEA S9074-AQ-GIB-010/248. Brazing procedures shall be in accordance with their applicable AWS specification.

3.3.10.6.2 Bolted bus bar joint insulation. When bus bars are insulated, bolted bus bar joints shall have field replaceable boots that allow maintenance access to bus bolts. Boots shall be rated for 266 °F (130 °C). Insulating materials that contain halogens (e.g., chlorine, fluorine, and bromine) which evolve gases during combustion, shall not be used.

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3.3.10.6.3 Connection hardware. Nuts and bolts used in bus bar joints shall be of the same material. CRES nuts shall be used with CRES bolts, silicon bronze nuts shall be used with silicon bronze bolts, and zinc-plated steel nuts shall be used with zinc-plated steel bolts in accordance with 3.3.10.6.3.1 and 3.3.10.6.3.2. Silicon bronze hardware shall be used in bus bar joints 4000 amperes and above. Washers of material similar to the nuts and bolts shall be used under all bolt heads and nuts adjacent to the conductors. See figure 8 for typical bus bar joint hardware connections. See figure 9 for typical bus bar mountings.

3.3.10.6.3.1 Bolts. Bolts used in bus bar joints shall be of the coarse thread series made of one of the following material combinations:

a. CRES, grades 304 or 316, in accordance with MIL-DTL-1222.

b. Silicon bronze, grades 651 or 655, in accordance with MIL-DTL-1222.

c. Zinc-plated steel, grade 5, in accordance with MIL-DTL-1222.

3.3.10.6.3.1.1 Number of bolts/tightening of bolts for bus bar joints. The number of bolts for bus bar joints shall be as shown in table III. Torque values for bolts in bus bar joints shall be as shown in table IV. Bolts for bus bar joints may be lubricated.

3.3.10.6.3.2 Nuts. Nuts used in bus bar joints shall be of the coarse thread series made of one of the following material combinations:

a. CRES, grades 304 or 316, in accordance with MIL-DTL-1222, NASM17830, or NASM16228 as applicable (see 3.3.7.1).

b. Silicon bronze, grades 651 or 655, in accordance with MIL-DTL-1222.

c. Zinc-plated steel, grade 5, in accordance with MIL-DTL-1222 or NASM17829 as applicable (see 3.3.7.1).

3.3.10.6.3.2.1 Tightening nuts for bus bar joints.

3.3.10.6.3.2.1.1 Tightening sequence and incremental torque. The tightening sequence for nuts shall be in accordance with figure 8. Ten percent of the specified torque (see 3.3.10.6.3.2.1.2 shall first be applied to all nuts in accordance with the tightening sequence to ensure the parts are solidly together. Torque shall be then applied to all nuts at 25, 50, 75, and 100 percent of the specified torque in accordance with the tightening sequence. Finally, the tightening sequence shall be reversed and 100 percent of required torque shall be applied to all nuts.

3.3.10.6.3.2.1.2 Self-locking nuts. Prevailing torque self-locking fasteners shall be torqued to 100 percent of applicable table IV values, plus the prevailing torque that is measured when the nut is turned on the bolt in its unloaded state. All full height nuts that are not prevailing torque self-locking fasteners shall be torqued to 100 percent of applicable table IV values.

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TABLE III. Bolting of connection – bars/diameter and number of bolts.

Size bus bars (in.)	Size bus bars (inches)											
	1/2 (inch)	5/8 (inch)	3/4 (inch)	1 (inch)	1-1/2 (inch)	2 (inch)	2-1/2 (inch)	3 (inch)	4 (inch)	5 (inch)	6 (inch)	8 (inch)
1/2	One 1/4	-	-	-	-	-	-	-	-	-	-	-
5/8	One 1/4	One 3/8	-	-	-	-	-	-	-	-	-	-
3/4	One 1/4	One 3/8	One 3/8	-	-	-	-	-	-	-	-	-
1	Two 1/4	One 3/8	One 3/8	One 1/2	-	-	-	-	-	-	-	-
1-1/2	Two 1/4	Two 3/8	Two 3/8	One 1/2	One 5/8	-	-	-	-	-	-	-
2	Two 1/4	Two 3/8	Two 3/8	Two 3/8	One 5/8	One 5/8	-	-	-	-	-	-
2-1/2	Two 1/4	Two 3/8	Two 3/8	Two 3/8	Two 1/2	Two 1/2	Two 1/2	-	-	-	-	-
3	Two 1/4	Two 3/8	Two 3/8	Two 1/2	Two 1/2	Two 1/2	Two 5/8	Two 5/8	-	-	-	-
4	Two 1/4	Two 3/8	Two 3/8	Two 1/2	Two 5/8	Two 5/8	Two 5/8	Two 5/8	Four 5/8	-	-	-
5	Two 1/4	Two 3/8	Two 3/8	Two 1/2	Two 5/8	Two 5/8	Two 5/8	Three 5/8	Five 5/8	Five 5/8	-	-
6	Two 1/4	Two 3/8	Two 3/8	Two 1/2	Two 5/8	Two 5/8	Two 5/8	Three 5/8	Five 5/8	Five 5/8	Five 5/8	-
8	Two 1/4	Two 3/8	Two 3/8	Two 1/2	Two 5/8	Two 5/8	Two 5/8	Three 5/8	Five 5/8	Five 5/8	Five 5/8	Six 5/8

TABLE IV. Torque on bus bar joints.

Torque				
Bolt diameter (inches)	CRES and zinc plated steel		Silicon bronze/copper	
	Min.-max. lubricated	Min.-max. non-lubricated	Min.-max. lubricated	Min.-max. non-lubricated
1/4	2-3 foot-pounds (24-36 inch-pounds)	3-4 foot-pounds (36-48 inch-pounds)	-	-
5/16	6-7 foot-pounds (72-84 inch-pounds)	9-10 foot-pounds (108-120 inch-pounds)	-	-
3/8	14-16 foot-pounds (168-192 inch-pounds)	21-24 foot-pounds (252-288 inch-pounds)	10-11 foot-pounds (120-132 inch-pounds)	15-16 foot-pounds (180-192 inch-pounds)
1/2	30-33 foot-pounds (360-396 inch-pounds)	45-50 foot-pounds (540-600 inch-pounds)	21-23 foot-pounds (252-276 inch-pounds)	31-34 foot-pounds (372-408 inch-pounds)
5/8	50-55 foot-pounds (600-660 inch-pounds)	75-80 foot-pounds (900-960 inch-pounds)	35-39 foot-pounds (420-468 inch-pounds)	52-58 foot-pounds (624-696 inch-pounds)

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3.3.10.6.4 Holes in bus bars. Holes in bus bars for bolting may be either punched or drilled, but the contact area shall be smoothed and burr free before silver surfacing. Where punching is used, the operation shall not appreciably indent the surface of the bus bar in the vicinity of the hole. Slotted holes shall not be used. Holes in bus bars for standard bolts shall have the dimensions specified in table V.

TABLE V. Bus bar hole size.

Bolt size (inch)	Hole size (inch)
3/8	7/16
1/2	9/16
5/8	11/16

When 5-inch or larger bus bars are joined together, the bolt hole size may be enlarged to provide 1/8-inch clearance. See table III for the diameter and number of holes required.

3.3.10.6.5 Preparation of bus bar joints. Bus bar joints shall be specially prepared before bolting as follows:

a. A thin film of NO-OX-ID "A-SPECIAL" (NSN 8030-00-598-5915) corrosion-preventive paste, or equal, shall be applied to the mating faces of the joint. Paste may be applied to the hardware. If paste is applied to the hardware, the hardware shall be treated as lubricated for torqueing purposes.

b. Without removing the paste, joints shall be bolted together using applicable torque values from table IV.

c. After joints are bolted together, paste shall be removed from all exposed surfaces with a clean, dry cloth.

3.3.10.6.6 Silver surfacing of bus bar joints. Silver surfacing of bus bars joints shall be electrodeposited type 1, grade A, B, or C silver coatings as specified in ASTM B700 with the exception that the silver plate shall not be less than 0.0002 inch thick. Contact surfaces of bus bars shall be silvered up to 1 inch past the joint area. The entire bus may be silver-surfaced.

3.3.10.7 Bus supports. The construction, number, spacing, and mounting of the bus bar supports shall be based on the magnitudes of forces and stresses encountered on the buses being supported when subjected to maximum rms asymmetrical short circuit currents, as specified (see 6.2). Values of maximum stress in the outside fibers shall be based on the yield strength of the bus bar support and not upon the tensile strength of the bus bar support. Support shall prevent contact of live parts with each other or ground during short circuit, high-impact shock, or vibration. Any permanent deformation or displacement of the bus bars following a short circuit shall not reduce the specified clearance distances by more than 30 percent of their original values.

3.3.11 Identification plates, information plates, and marking. Identification and information plates on the outside of the switchboard shall be in accordance with types A, B, C, F, or H of MIL-DTL-15024 and MIL-P-15024/5. Plastic identification and information plates shall be light gray. Beveled edges are not required. Inscriptions or markings for all type plates shall be black, except for danger, warning, and caution plates which shall have words in red for emphasis or be red with white markings. Identification and information plates inside the switchboard shall be in accordance with type F of MIL-DTL-15024 and MIL-P-15024/5.

3.3.11.1 Location. Identification and information plates shall be placed either directly above or directly below the equipment to which it refers.

3.3.11.2 Size. In general, identification and information plates shall be of the size as listed in MIL-DTL-15024 and MIL-P-15024/5.

3.3.11.2.1 Thickness. Thickness of identification and information plates shall be in accordance with MIL-DTL-15024 and MIL-P-15024/5.

3.3.11.2.2 Shape. Information plates for rheostat handwheels and other rotary devices where required shall be circular or some other shape to suit the equipment.

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3.3.11.3 Information. Identification plates with information consistent with the following shall be provided at or near the top of the front enclosure of each switchgear section or for each unit, if required individually:

a. Name or functional designation of item.

b. Manufacturer's name.

c. Manufacturer's drawing number (front view).

d. Year manufactured.

e. NAVSEA drawing number (front view), if assigned.

f. Stock number, if assigned.

g. NAVSEA number of manual, if assigned.

h. Serial number.

3.3.11.4 Abbreviations. Unless lack of space prevents spelling out the word, abbreviations of words on identification and information plates shall be avoided except in the case of words which have abbreviations in common use in accordance with ASME Y14.38.

3.3.11.5 Installation. Identification and information plates shall be fastened securely to such parts of the equipment as ordinarily will not be renewed during its service life. Identification and information plates shall be attached in accordance with MIL-DTL-15024 and MIL-P-15024/5. Metal and laminated identification and information plates and mimic bus may be attached with an adhesive.

3.3.11.6 Detail application requirements. Information plates shall be installed immediately above each circuit breaker and shall contain the following information:

a. Circuit name.

b. Circuit designation and number.

c. Circuit breaker element rating.

d. Certification data sheet number, as applicable.

3.3.11.7 Blank plates. Information plates without markings shall be provided for mounted spare circuit breakers.

3.3.11.8 Instrument identification. Instruments shall be marked with the name of circuits; marking may appear on the instrument or on a separately mounted information plate. Switchboard devices and component assemblies shall be provided with identification sufficient to identify each component from the information contained on the applicable switchboard drawings.

3.3.11.9 Fuse marking. Fuses for each switchboard control circuit, instrument, and instrument transformer shall have an information plate showing the name or designation of circuit, phase (if applicable), fuse ampere rating, and type designations.

3.3.11.10 Marking switchboard buses. Buses shall be clearly marked in a visible location with their polarity or phase identification. If marking is accomplished by painting (stencil) application, the ink shall be in accordance with A-A-208, type III. Information plates for buses may be similar to the warning plates specified in 3.3.11.12.

3.3.11.11 Phase identification. Phase identification of AC buses shall be indicated by the capital letters A, B, or C, as applicable. A neutral bus of a four-wire, three-phase circuit where used shall be indicated by the capital letter N. Letters A, B, and C in that order shall indicate the phase sequence.

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3.3.11.12 Bus bars. Bus bars facing and closest to the rear of the unit shall be stenciled "DANGER _____ VOLTS". Applicable voltage shall be entered. Letters shall be red and not less than 3/4 inch high. As an alternative to stenciling, bus bars may be provided with a warning plate bearing the same legend. Letters shall be not less than 3/4 inch high. The plate shall be pressure-sensitive, adhesive-backed elastomeric film with a carrier or separator sheet applied over the adhesive. Film shall consist of a white printed opaque polyvinyl fluoride film 0.002-inch thick laminated to a 0.001-inch clear polyethylene terephthalate film. Plate adhesive shall consist of an acrylic base vinyl polymer of a pressure sensitive type. A minimum of two warning stencils or plates per unit shall be provided, one in the upper and one in the lower section.

3.3.11.13 Circuit breaker information plate. An information plate, containing instructions for positioning the circuit breaker in the connected, test, disconnected, and withdrawn positions, shall be provided, including steps and precautions to discharge the opening and closing springs. Additional information plates shall identify operating devices and positions, give pertinent instructions for operation, call attention to special precautions, and call attention to environmental warnings. If possible, all of the information above should be included on one label plate.

3.3.11.14 Programmable items. Equipment that is software programmable shall indicate the identifying number and revision of the software program which has been loaded into memory. The preferred method is to provide either a local or a remote display that is under the control of the software program. However, when the use of a display is not practical, the equipment enclosure shall be marked with the identifying number and revision of the software program. The identifying number shall be preceded by the words "software program".

3.3.12 Ship's service power switchgear group for ships having AC power.

3.3.12.1 AC generator control unit for switchgear groups having an electric plant control panel (EPCP) (see 6.3.7) or having both an EPCP and an electric plant control console (EPCC). (See figure 10 for a typical one-line diagram.) The following shall be mounted on or within the unit:

a. One generator circuit breaker.

b. One control power transformer (450/120 volts), connected to generator side of circuit breaker, for generator circuit breaker and governor motor control circuits, if required.

c. Potential transformers for metering and relaying, electric governor, and synchronizing, in number as required.

d. Current transformers for metering and relaying and for electric governor, in number as required. Protective device when metering is remote.

e. One current transformer with protective device for remote metering, where remote metering is required.

f. Dead-front fuse panel for fusing of the potential and control transformer primary and secondary circuits and any control circuits extending outside of the section. Dead-front fuseholders shall be in accordance with MIL-PRF-19207 or alternative specification approved by the Technical Authority.

g. Space, mounting, and wiring for the voltage regulator equipment, including associated current and potential transformers as specified (see 6.2). Equipment shall be furnished with the generators.

h. One circuit breaker, molded case type, 250-ampere frame size. The casualty power circuit breaker shall be connected to the generator side of the generator circuit breaker. When the generator capacity is 500 kilowatts or greater, two casualty power circuit breakers shall be furnished.

i. One casualty power terminal, back-connected type, in accordance with MIL-DTL-24552, connected through the molded case circuit breaker, item (h). When the generator capacity is 500 kilowatts or greater, two casualty power terminals, back-connected type, shall be furnished.

j. One reverse power relay, in accordance with MIL-M-24350 or alternative specification approved by the Technical Authority, to prevent the generator from motoring on the bus after loss of driving power from the prime mover.

k. One indicator light (blue) to indicate that the generator circuit breaker is closed.

l. One generator running and voltage available auxiliary relay for control of associated indicator lights.

m. Space, mounting, and wiring for the electric governor equipment, if required (see 6.2). Equipment shall be furnished with the generator set.

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3.3.12.1.1 AC generator control unit for switchgear groups on ships provided with a centralized machinery control system including an EPCC, but not an EPCP. The following shall be mounted within the unit:

a. One generator circuit breaker, electrically-operated.

b. One control power transformer (450/120 volts), connected to generator side of generator circuit breaker, for generator breaker and governor motor control circuits, if required.

c. Potential transformer for metering and relaying, electric governor, and synchronizing, in number as required.

d. Current transformers, for metering and relaying, and for electric governor, in number as required. Protective device when metering is remote.

e. One current transformer with protective device for remote metering.

f. Dead-front fuse panel for fusing of the potential and control transformer primary and secondary circuits and any control circuits extending outside of the section. Dead-front fuseholders shall be in accordance with MIL-PRF-19207 or alternative specification approved by the Technical Authority.

g. Space, mounting, and wiring for the voltage regulator equipment, including associated current and potential transformers as specified (see 6.2). Equipment shall be furnished with the generators.

h. Relays for remote control of the generator circuit breaker. Relays shall be energized from EPCC power.

i. Load sensing relays in accordance with MIL-R-24563, MIL-M-24350, or alternative specification approved by the Technical Authority to sense generator overload and activate the load shedding system, if required (see 6.2).

j. One molded case type circuit breaker (see 3.6.1.1), instantaneous element only, for power supply to the pump supplying cooling (sea) water to the generator diesel or gas turbine prime mover. Circuit breaker shall be connected on the generator side of the generator circuit breaker. A red warning instruction plate shall be provided stating that the circuit breaker shall always be in the closed position when the gas turbine or diesel are set up for starting.

k. One circuit breaker (see 3.6.1.1), molded case type, 250-ampere frame size. Casualty power circuit breaker shall be connected to the generator side of the generator circuit breaker. When the generator capacity is 500 kilowatts or greater, two casualty power circuit breakers shall be furnished.

l. One casualty power terminal, back-connected type, in accordance with MIL-DTL-24552 connected through the molded case circuit breaker, item (k). When the generator capacity is 500 kilowatts or greater, two casualty power terminals, back-connected type, shall be furnished.

m. One reverse power relay, in accordance with MIL-M-24350 or alternative specification approved by the Technical Authority, to prevent the generator from motoring on the bus after loss of driving power from prime mover.

n. One generator running and voltage available auxiliary relay for control of associated indicator lights. This relay is not required when a supervisory control system (SCS) is incorporated in the EPCC, and the signal can be derived from generator voltage and speed signals.

o. Space, mounting, and wiring for the electric governor equipment, if required (see 6.2). Equipment shall be furnished with the generator set.

p. One type I synchronizing system for each switchgear group. The synchronizing control system may be mounted in the bus tie unit instead of the generator control unit.

q. One type II automatic paralleling device, if required (see 6.2). The automatic paralleling system may be mounted in the bus tie unit instead of the generator control unit.

3.3.12.1.1.1 Front panel. The following shall be mounted on the front panel of the generator unit:

a. One AC ammeter for reading one phase of generator current.

b. One AC voltmeter with transfer switch for reading one phase of generator potential, one phase of each bus tie voltage, and one phase of shore power voltage for the switchgear group containing shore power connection.

c. One polyphase wattmeter for reading generator output.

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d. One frequency meter with transfer switch for reading frequency of generator and each bus tie frequency and shore power for the switchgear group containing shore power connection.

e. One control switch for control of the generator circuit breaker.

f. Control switches as required (see 6.2) in conjunction with the governor speed regulation of the generator prime mover.

g. One local/remote control transfer switch for the purpose of transferring control from the local station to the EPCC. Monitoring and indicating circuits shall not be transferred except as noted below. Circuits transferred by this switch shall include:

(1) Generator starting and stopping. Remote generator starting shall be transferred by the switch. Remote generator stopping shall be locked out when the switch is in the local position. Local stopping and remote emergency stopping shall be effective regardless of the position of the switch.

(2) Generator output voltage control.

(3) Prime mover governor manual control.

(4) Voltage regulator mode selection.

(5) Governor mode selection.

(6) Circuit breaker control of all circuit breakers which are controlled at the EPCC. This shall include control of the synchronizing control system. Local automatic protection such as reverse power protection shall remain operative regardless of the position of the local/remote transfer switch.

(7) Elements of the generator green light circuit which are duplicated locally and at the EPCC.

(8) Local/remote transfer switch position indication.

h. Local/remote transfer switch position indicator light (white).

i. One control switch with indicator light (white) for control of the generator space heaters when provided with the generator as specified (see 6.2).

j. Control switches, rheostats, and similar equipment that are designed for panel mounting and are required in conjunction with the voltage regulator equipment mounted in the generator unit shall be mounted and operated from the front panel as specified (see 6.2).

k. For installation in which a bus tie unit is not provided, one set of three ground indicator lights (clear) with a control switch to indicate ground on the main bus or its connected feeders.

l. One synchroscope and one set of synchronizing lights (clear) with selector switch for synchronizing with the main bus, the main bus with the bus tie circuit, and the bus tie with shore power switchgear group containing shore power connection.

(1) For synchronizing monitor devices (type I) and automatic paralleling devices (type II) the following applies:

(a) System selector switch for selecting the system to be paralleled. This switch may be combined with the synchroscope selector switch.

(b) An indicator light (white) to indicate that the synchronizing control system is energized.

(2) For synchronizing monitor devices the following applies:

(a) Synchronizing control monitor operating mode selector switch having Operating, Test, and Off positions.

(3) For automatic paralleling devices the following applies:

(a) One operating mode selector switch having Automatic, Manual Permissive, Test, and Off positions.

(b) One pushbutton control switch for initiation of the automatic paralleling operation.

(c) An indicator light (red) to indicate that the synchronizing control equipment is unable to close the circuit breaker when in the automatic mode.

(d) A pushbutton to interrupt the automatic paralleling cycle to completion of paralleling.

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(e) Indicating devices as required by MIL-S-24188 or alternative specification approved by the Technical Authority.

m. Test receptacles with test plugs for connecting portable instruments to the metering current and potential circuits. Removal of the receptacle protective cover shall automatically short circuit the secondary winding of metering current transformers so that only the meters are electrically removed from the circuit.

n. One temperature indicator with selector switch and accessories, when temperature detectors are provided with the generator as specified (see 6.2).

o. One indicator light (blue) to indicate that the generator circuit breaker is closed.

p. One AC ammeter for each casualty power terminal furnished for reading one phase of casualty power current.

3.3.12.1.2 AC bus tie unit for switchgear groups on ships provided with a centralized machinery control system including an EPCC, but not an EPCP. The following shall be mounted within the unit:

a. One electrically-operated bus tie circuit breaker.

b. One feeder circuit breaker.

c. Necessary disconnecting devices for performing isolating functions as may be required (see 6.2).

d. Potential transformers, for metering and relaying, and for synchronizing, in number as required for the bus connections of the switchgear group.

e. One current transformer for metering the current interchange through the bus tie circuit breaker.

f. Dead-front fuse panel for fusing the potential and control power transformer primary and secondary circuits. Dead-front fuseholders shall be in accordance with MIL-PRF-19207 or alternative specification approved by the Technical Authority.

g. Transfer relays to automatically transfer the source of circuit breaker closing power between the switchboard bus and bus tie.

h. Relays for remote control of the bus tie circuit breaker. Relays shall be energized from EPCC power.

Bus tie units of the switchgear group containing the shore power connection shall be designated as a “bus tie and shore power unit”. In addition, this unit shall substitute item (i) for the feeder circuit breaker, and items (j), (k), and (1) as follows shall be added:

i. Circuit breakers, molded case type with a fuse or current limiting molded case type without fuse, motor operated, 400-ampere frame size, to be furnished on only one bus tie unit of any switchgear group. One shall be provided for each shore power receptacle. Shore power circuit shall be connected to the bus tie circuit so that power may be fed to either the local or remote switchboard through a bus tie circuit breaker. Where more than four circuit breakers are required, a separate switchgear unit may be provided.

j. Current transformers for metering the total shore power current.

k. Potential transformers for metering and synchronizing, in number as required.

l. Relays for remote control of the shore power circuit breaker motor operators. Motor operators shall be controlled from the local switchboard.

3.3.12.1.2.1 Front panel. The following shall be mounted on the front panel of the bus tie unit:

a. One control switch for control of the bus tie circuit breaker.

b. Indicator lights (blue) to show the closed position of the bus tie circuit breaker and the power type (see 3.6.1.1) feeder circuit breaker.

c. One set of three ground indicator lights (clear) with a control switch to indicate ground conditions on the main bus or any of its connected feeders.

d. One AC ammeter for indicating the bus tie current.

e. One zero neutral wattmeter for reading bus tie power and direction of power flow.

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3.3.12.1.2.2 Bus tie and shore power unit. When this unit is designated a bus tie and shore power unit, the following additional devices shall be mounted on the front panel of the unit:

a. One shore power circuit breaker control switch for operating all shore power circuit breakers simultaneously.

b. One indicator light (white) per shore power receptacle to indicate the shore power feeder is energized. One indicator light (white) to indicate the bus feeder is energized. These indicator lights shall be mounted near the shore power circuit breakers.

c. Indicator lights (blue) to indicate that each shore power circuit breaker is closed.

d. One switchboard type phase sequence indicator with a multiposition selector switch for use in checking phase rotation and orientation of shore power connections (see figure 11). Two indicator lights (green) to indicate phase orientation.

e. One transfer switch for use with the generator unit frequency meter for reading the frequency of the bus tie circuit and shore power.

f. One reference selector switch for use in checking phase orientation of shore power connections.

g. One AC ammeter for reading shore power current.

h. For shore power switchboard units with more than four shore power breakers installed, one multiposition selector switch to allow one of a selected number of different shore power receptacles to act as the master circuit. Necessary devices, including potential transformers, shall be provided on this receptacle circuit for metering and synchronizing of shore power and the switchboard bus across the shore power breakers.

3.3.12.2 AC generator control unit for switchgear groups not having an EPCP or an EPCC. (See figure 12 for a typical one-line diagram.) Devices listed in 3.3.12.1, except item (l), shall be mounted on or within the unit.

3.3.12.2.1 Front panel. The following devices shall be mounted on the front panel of the generator unit:

a. One AC ammeter for reading one phase of generator current.

b. One AC voltmeter with transfer switch for reading one phase of generator potential, one phase of each bus tie voltage, and one phase of shore power voltage for the switchgear group containing shore power connection.

c. One polyphase wattmeter for reading generator output.

d. One frequency meter for reading the frequency of the generator.

e. One control switch for control of the generator circuit breaker where an electrically-operated breaker is used.

f. Control switches, rheostats, and similar devices, as required (see 6.2) in conjunction with the governor speed regulation of the generator prime mover.

g. One transfer switch to transfer the governor control from the local control switch to the remote control switch on the AC bus tie unit (or control unit) of the control switchgear group. It shall also control the lamp (see 3.3.12.4.1(c)) on the bus tie unit or control unit to indicate when the remote switch is in control. This shall be provided only when the unit has control of a remote generator.

h. One control switch with indicator light (white) for control of the generator space heaters when provided with the generator as specified (see 6.2).

i. Control switches, rheostats, and similar equipment that are for panel mounting and are required in conjunction with the voltage regulator equipment mounted in the generator unit shall be mounted and operated from the front panel (see 6.2).

j. For installations in which a bus tie unit is not provided, one set of three indicator lights (clear) with a control switch, to indicate ground conditions on the main bus or any of its feeders, shall be mounted on the front panel of the generator unit.

k. One synchroscope and one set of synchronizing lights (clear) with transfer switch for synchronizing the generator with the main bus, and the main bus with the bus tie circuit, and the bus tie with shore power for the switchgear group containing shore power connection mounted and operated from the front panel.

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l. One type I synchronizing control equipment shall be provided for each single or two generator switchgroup. The synchronizing control monitor may be mounted in the bus tie unit instead of the generator control unit.

m. One type II automatic paralleling device, if required (see 6.2). The automatic paralleling device may be mounted in the bus tie unit instead of the generator control unit.

n. Test receptacles with test plugs for connecting portable instruments to the metering current and potential circuits. Removal of the receptacle protective cover shall automatically short circuit the secondary winding of metering current transformers, so that only the meters are electrically removed from the secondary circuit.

o. One temperature indicator with selector switch and accessories, when temperature detectors are provided with the generator as specified (see 6.2).

3.3.12.3 AC bus tie unit for switchgear groups having an EPCP, or having both an EPCP and EPCC. The following shall be mounted within the unit:

a. One bus tie circuit breaker.

b. One feeder circuit breaker.

c. Necessary disconnecting devices for performing isolation functions as may be required (see 6.2).

d. Potential transformers, for metering and relaying, and for synchronizing, in number as required for the bus connections of the switchgear group.

e. One current transformer for metering the current interchange through the bus tie circuit breaker.

f. Dead-front fuse panel for fusing the potential and control power transformer primary and secondary circuits. Dead-front fuseholders shall be in accordance with MIL-PRF-19207 or alternative specification approved by the Technical Authority.

g. Indicator light (blue) to show the closed position of the bus tie and feeder circuit breaker.

Bus tie units of the switchgear group containing the shore power connection shall be designated as a “bus tie and shore power unit”. In addition, this unit shall substitute item (h) for the feeder circuit breaker, and add items (i) and (j) below:

h. Circuit breakers, molded case type with fuse or current limiting molded case type without fuse, electrically-operated, 400-ampere frame size shall be furnished on only one bus tie unit of any switchgear group. One shall be provided for each shore power receptacle. The shore power circuit shall be connected to the bus tie circuit so that power may be fed to either the local or remote switchboard through a bus tie circuit breaker. Where more than four circuit breakers are required, a separate switchgear unit may be provided.

i. Current transformers for metering the total shore power current.

j. Potential transformers for metering and synchronizing, in number as required.

3.3.12.4 AC bus tie unit for switchgear groups not having an EPCP or EPCC. The following shall be mounted within the unit:

a. One bus tie circuit breaker.

b. One feeder circuit breaker.

c. Necessary disconnecting devices for performing switching functions as may be required.

d. Potential transformers for metering and relaying, and for synchronizing, in number as required for the bus connections of the switchgear group.

e. Dead-front fuse panel for fusing the control circuits fed from the control power bus. Dead-front fuseholders shall be in accordance with MIL-PRF-19207 or alternative specification approved by the Technical Authority.

f. Dead-front fuse panel for fusing the potential and control power transformer primary and secondary circuits. Dead-front fuseholders shall be in accordance with MIL-PRF-19207 or alternative specification approved by the Technical Authority.

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3.3.12.4.1 Front panel. The following shall be mounted on the front panel of the bus tie unit:

a. One polyphase wattmeter, calibrated to indicate the three-phase power output of remote ship's service generator.

b. One AC ammeter for indicating the current of remote ship's service generator and from the shore power connection.

c. Remote control switches or rheostats and indicator lights (white), as required, to control the speed of the prime movers of the remote generators. The control switch power shall be connected through the transfer switches (see 3.3.12.2.1(g)) on the generator units of the remote switchgear groups, and the indicating lamps shall be connected to indicate when the transfer of control is accomplished. This shall be provided when the unit has control of a remote generator.

d. One composite transfer switch for reading the power and current of each remote generator by means of (a) and (b) above and the shore power current by means of (b) above.

e. One control switch for control of the bus tie circuit breaker where an electrically-operated circuit breaker is used.

f. Indicator lights (blue) to show the closed position of the bus tie circuit breaker and the power type feeder circuit breaker.

g. One frequency meter for reading the frequency of the bus tie circuit.

h. One transfer switch, with two white indicator lights, for connecting the switchboard control bus to the local 450-volt ship's service bus, or the 450-volt bus of the corresponding (forward or aft) emergency switchboard. Transfer may be accomplished automatically; however, in such cases one source of power shall be the local 450-volt ship's service bus and the second source shall be a 450-volt bus tie supply.

i. One set of three indicator lights (clear) with a control switch to indicate ground conditions on the main bus or any of its connected feeders.

j. One zero neutral wattmeter for reading bus tie power and direction of power flow.

3.3.12.4.2 Bus tie and shore power unit. One bus tie unit of the switchgear group containing the shore power connection shall be designated as a "bus tie and shore power unit". In addition to the devices specified in 3.3.12.4 and 3.3.12.4.1, this unit shall substitute item (a) for the feeder circuit breaker, and the additional items (b) through (h) shall be provided.

a. Circuit breakers, molded case type with fuse or current limiting molded case type without fuse, electrically-operated, 400-ampere frame size shall be furnished on only one unit of the switchgear group. One shall be provided for each shore power receptacle. The shore power circuit shall be connected to the bus tie circuit so that power may be fed to either the local or remote switchboard through a bus tie circuit breaker.

b. One shore power circuit breaker control switch for operating all shore power circuit breakers simultaneously.

c. One indicator light (white) per shore power receptacle to indicate the shore power feeder is energized. One indicator light (white) to indicate the bus feeder is energized. These indicator lights shall be mounted near the shore power circuit breakers.

d. Indicator lights (blue) to indicate that each shore power circuit breaker is closed.

e. Current transformers for metering the total shore power current.

f. One switchboard type phase sequence indicator with a multiposition selector switch for use in checking phase rotation and orientation of shore power connections.

g. Potential transformers for metering and synchronizing, in number as required.

h. One transfer switch for use with item (g) of 3.3.12.4.1 for reading the frequency of the bus tie circuit and shore power (see figure 11).

i. One reference selector switch for use in checking phase orientation of shore power connections.

j. Two indicator lights (green) for use with (f) and (i) above.

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k. For shore power switchboard units with more than four shore power breakers installed, one multiposition selector switch to allow one of a selected number of different shore power receptacles to act as the master circuit. Necessary devices, including potential transformers, shall be provided on this receptacle circuit for metering and synchronizing of shore power and the switchboard bus across the shore power breakers.

3.3.12.5 Electric plant control panels.

3.3.12.5.1 Desk type EPCP. Desk type EPCP (see 6.3.4 and figure 13) shall be one of the following bench types:

a. A horizontal or inclined desk top for plant operation controls.

b. Inclined desk top for plant operation controls with apron section below.

Desk type EPCP shall have vertical panels for the mounting of instruments, displays, and associated controls. When specified (see 6.2), desk type EPCP for surface ships shall be provided with a chair that can be mounted so that it can be swung out of the way when not in use. Requirements for desk type EPCP shall be as specified (see 6.2).

3.3.12.5.1.1 Control switches. Control switches for the electrically-operated circuit breakers controlled from the EPCP shall be mounted on the desk top and arranged within convenient reach of the operator. Circuit breaker control switches and their corresponding switches in the mimic bus system shall be located so that their relative positions as to their locations on the EPCP are similar. Automatic voltage adjusting rheostats shall be mounted close to the meters being affected by their adjustment. Large equipment which requires infrequent personnel attention shall be located in the apron portion to conserve panel and desk space. The portion of the apron panel used for mounting the equipment shall not project beyond the front edge of the desktop. Space shall be provided for filing the reduced size drawing. Space shall be provided for the mounting of the interior communication (IC) equipment as specified (see 6.2).

3.3.12.5.2 Vertical EPCP. Vertical EPCP (see 6.3.16) shall be arranged to orient instruments, displays, and their associated controls near the top, other plant operation controls toward the middle, and larger, infrequently used equipment at the bottom. Requirements for vertical type EPCP shall be as specified (see 6.2).

3.3.12.5.3 Surface ship EPCP. The following shall be mounted on the EPCP for surface ships:

a. Control switches, rheostats, and similar equipment that are for panel mounting and are required in conjunction with the voltage regulator equipment mounted in the generator unit shall be mounted on and operated from the EPCP (see 6.2).

b. One AC ammeter for each local generator for reading one phase of generator current.

c. One AC voltmeter for each local generator for reading one phase of generator potential.

d. One AC voltmeter with transfer switch for reading single-phase voltage of the switchboard bus and of the bus tie feeders and shore power for switchgear groups containing shore power connection.

e. One polyphase wattmeter for reading generator output for each local generator.

f. One polyphase wattmeter with transfer switch to indicate the three-phase power output of the remote ship's service generator (shall be provided where a switchgear group is to have control of a remote generator or generators).

g. One temperature indicator with selector switch and accessories when temperature detectors are provided with the generator as specified (see 6.2).

h. One frequency meter for reading the frequency of the generator for each local generator. For units controlling two local generators, each frequency meter shall be provided with a transfer switch for reading the frequency of one generator and one bus tie circuit and shore power, in which case 3.3.12.5.3(i) shall be omitted.

i. One frequency meter with transfer switch for reading the frequency of the bus and of each bus tie circuit and shore power for the switchgear group containing shore power connection. For units controlling two local generators, each frequency meter (see 3.3.12.5.3(h)) shall be provided with a transfer switch for reading the frequency of one generator and one bus tie circuit and shore power, in which case 3.3.12.5.3(i) shall be omitted.

j. One local control switch, or other control device as required, for control of the speed governor of the generator prime mover for each local generator.

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k. One transfer switch to transfer the governor control of the generator prime mover from local control to the remote control on the master EPCP. It shall also control the indicator light on the master EPCP to indicate when the remote station is in control. This transfer switch shall be provided only on EPCPs for which remote control, item (l), is to be provided on the master EPCP.

l. One remote control switch or other remote control devices as required, and indicator light (white) to control the speed governor on the remote generator prime mover where the unit has control of a remote generator. Control power shall be connected through the transfer switch, item (k), of the remote EPCP, and the indicating light shall be energized when the transfer of control to the remote station is accomplished.

m. One control switch with circuit breaker closed indicator light (blue) for control of each local generator circuit breaker.

n. One control switch with circuit breaker closed indicator light (blue) for control of each bus tie circuit breaker.

o. One control switch with circuit breaker closed indicator light (blue) for control of each electrically-operated distribution circuit breaker.

p. One control switch with space heater on indicator light (white) for control of generator space heater (when provided with the generator) for each local generator when specified (see 6.2).

q. Test receptacle with test plugs for connecting portable instruments to the metering current and potential circuits. Removal of the receptacle protective cover shall automatically short circuit the secondary of metering current transformers, so that only the meters are electrically removed from the secondary circuit.

r. One transfer switch with two indicator lights (white) for connecting the switchboard control bus to the local 450-volt ship's service bus, or the 450-volt emergency supply. The transfer may be accomplished automatically; however, in such cases one source of power shall be the local 450-volt ship's service bus and the second source shall be a 450-volt bus tie supply.

s. One set of three ground indicator lights (clear) with a test switch to indicate ground conditions on the main bus.

t. Dead-front fuse panel for fusing the control circuits. Dead-front fuseholders shall be in accordance with MIL-PRF-19207 or alternative specification approved by the Technical Authority.

u. One mimic bus system (dark type) simulating the generator and bus tie connections shall be provided and shall consist of the following:

(1) Indicator lights (white) for each generator to indicate generator running and voltage available.

(2) Indicator lights (yellow) for each generator and bus tie circuit breaker.

(3) Single-pole, double-throw switches shall be provided for each circuit breaker light. These switches shall be combined with their respective yellow lights and so arranged that when the switch handle indication and the circuit breaker position (open or closed) are the same, the light shall be out. When the circuit breaker position and the switch handle indication are different, the light shall be lit. For indicator light power supply, see 3.6.12.1.

(4) Mimic bus material shall be 1/8 inch thick, and 1/4, 3/4, or 1/2 inch wide, as specified (see 6.2). Mimic bus material shall be laminated plastic in accordance with grades ES-1, ES-2, or ES-3 of ASTM D709; aluminum alloy in accordance with ASTM B209/B209M, alloy 5052; commercial brass in accordance with ASTM B36/B36M and ASTM B121/B121M; CRES in accordance with type 316 of ASTM A240/A240M and ASTM A666; or nickel-copper alloy in accordance with ASTM B127, ASTM B164, and ASTM B564.

v. One ammeter to indicate current in the bus tie.

w. One ammeter for each remote generator for reading one phase of generator current.

x. One shore power circuit breaker control switch for each shore power station to simultaneously operate all circuit breakers for that station. One indicator light (blue) for each shore power circuit breaker to indicate breaker closed. One indicator light (white) per shore power receptacle to indicate the shore power feeder energized. One indicator light (white) to indicate the bus feeder energized. These indicator lights shall be mounted near the shore power circuit breaker control switch.

y. One control switch with one blue and one white indicator light for each electrically-operated circuit breaker feeding an emergency switchboard.

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z. One ammeter for reading the shore power current.

aa. One switchboard type phase sequence indicator with a multiposition selector switch, for use in checking phase rotation and orientation of shore power connections.

bb. One reference switch for use in checking phase orientation of shore power connections. Two indicator lights (green) to indicate phase orientation.

cc. One local/remote control transfer switch for the purpose of transferring control from the local station to the EPCC. Monitoring and indicating circuits shall not be transferred except as noted below. This is required only on ships provided with a centralized machinery control system including an EPCC when specified (see 6.2). Circuits transferred by this switch shall include:

(1) Generator starting and stopping. Remote generator starting shall be transferred by the switch. Remote generator stopping shall be locked out when the switch is in the local position. Local stopping and remote emergency stopping shall be effective regardless of the position of the switch.

(2) Generator output voltage control.

(3) Prime mover governor manual control.

(4) Voltage regulator mode selection.

(5) Governor mode selection.

(6) Circuit breaker control of all circuit breakers which are controlled at the EPCC. This shall include control of the synchronizing control system. Local automatic protection shall remain operative regardless of the position of the local/remote transfer switch.

(7) Elements of the generator green light circuit which are duplicated locally and at the EPCC.

(8) Local/remote transfer switch position indication on the EPCC.

dd. One synchroscope and one set of synchronizing lights (clear) with selector switch for synchronizing with the main bus, the main bus with the bus tie circuit, and the bus tie with shore power for the switchgear group containing shore power connection. Where the switchgear group has control of two generators and two bus tie circuits, the synchroscope and synchronizing lights (clear) shall be connected for synchronizing either generator with the bus and the bus with either bus tie circuit. Note the following:

(1) For synchronizing monitor devices (type I) and automatic paralleling devices (type II) the following applies:

(a) System selector switch for selecting the system to be paralleled. This switch may be combined with the synchroscope selector switch.

(b) An indicator light (white) to indicate that the synchronizing control system is energized.

(2) For synchronizing monitor devices the following applies:

(a) Synchronizing control monitor operating mode selector switch having Operating, Test, and Off positions.

(3) For automatic paralleling devices the following applies:

(a) One operating mode selector switch having Automatic, Manual Permissive, Test, and Off positions.

(b) One pushbutton control switch for initiation of the automatic paralleling operation.

(c) An indicator light (red) to indicate that the synchronizing control equipment is unable to close the circuit breaker when in the automatic mode.

(d) A pushbutton to interrupt the automatic paralleling cycle to completion of paralleling.

3.3.12.5.4 400-Hertz EPCP for solid-state frequency changer systems. The following shall be mounted on the 400-Hertz EPCP:

a. One mimic bus system (dark type) simulating the frequency changer and bus tie connections shall be provided and shall consist of the following:

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(1) Indicator lights (white) for each frequency changer to indicate frequency changer energized and voltage available.

(2) Indicator lights (yellow) for each frequency changer output contactor and bus tie circuit breaker.

(3) Single-pole, double-throw switch for each circuit breaker light. These switches shall be combined with their yellow lights and arranged so that when the switch handle indication and the circuit breaker position (open or closed) are the same, the light shall be out. When the circuit breaker position and the switch handle indication are different, the light shall be on. For indicator light power supply, see 3.6.12.1. Black mimic bus material shall be 1/8 inch thick and 1/4, 3/8, or 1/2 inch wide, as specified (see 6.2). Mimic bus materials shall be laminated plastic in accordance with grades ES-1, ES-2, or ES-3 of ASTM D709; aluminum alloy in accordance with ASTM B209/B209M, alloy 5052; commercial brass in accordance with ASTM B36/B36M and ASTM B121/B121M; CRES in accordance with type 316 of ASTM A240/A240M and ASTM A666; or nickel-copper alloy in accordance with ASTM B127, ASTM B164, and ASTM B564.

b. One current meter for reading one phase of each frequency changer output.

c. One pushbutton switch combination for actuating each frequency changer input contactor. One indicator light (white) to indicate that the unit is energized.

d. One control switch with indicator light (blue) for each frequency changer output contactor.

e. One control switch with indicator light (blue) for control of each 400-Hertz bus tie circuit breaker.

f. One remote control switch with indicator light (blue) for control of each applicable 400-Hertz distribution circuit breaker.

g. One set of three indicator lights (white) with a control switch to indicate ground conditions on the main 400-Hertz bus.

h. Solid-state frequency changer failure mode indicators as required (see 6.2).

3.3.12.6 Power circuit breaker distribution unit. This unit shall provide for the distribution of power from the main bus as specified (see 6.2). The following shall be provided:

a. Circuit breakers having current rating and protective features as required.

b. Indicator lights (blue) mounted on the front panel adjacent to or on the circuit breaker escutcheon to indicate when each of the feeder circuit breakers is closed.

c. Control switches shall be provided for electrically-operated circuit breakers. Where the switchgear group has EPCP or EPCC, or both, these switches shall be located on the EPCP or EPCC, or both.

d. Necessary disconnecting devices for disconnecting cables between switchgear sections and performing other switching functions. NOTE: Where a circuit breaker supplies a feeder which may be energized from another source, such as the emergency switchgear group, an indicator light (white) shall be provided to indicate feeder is energized.

3.3.12.7 Molded case circuit breaker distribution unit. This unit shall provide for the distribution of electric power or lighting supply (450/120 volts) through molded case type circuit breakers as specified (see 6.2). Units receiving two sources of supply shall be provided with interlocked molded case circuit breakers. Where interlocked circuit breakers are installed, an indicator light (white) shall be provided for each power supply to indicate when power is available (see 3.3.13.4).

3.3.12.7.1 Spaces. Spaces for the future installation of circuit breakers shall be blanked off with busing installed for the circuit breakers. Mounting blocks shall also be installed for molded case circuit breakers. Circuit breaker panel blanks shall be securely fastened in place so that the blanks or the fasteners shall not fall into the interior of the switchboard under any anticipated operating conditions. They may be constructed from conducting or nonconducting material.

3.3.12.8 Power-molded case circuit breaker distribution unit. This unit shall provide for distribution of power through both power and molded case types of circuit breakers as specified (see 6.2). It shall contain the devices listed in 3.3.12.6 through 3.3.12.7.1.

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3.3.12.9 Load shedding. Distribution units of ship's service switchgear groups on ships provided with a centralized machinery control system including an EPCP or an EPCC, or both an EPCP and an EPCC, for which load shedding is required, shall contain a load shedding bus or buses segregated from the main bus by electrically-operated non-automatic circuit breakers. Individual electrically-operated ship's service distribution circuit breakers shall also be operated by the load shedding system as specified. Each electrically-operated ship's service distribution circuit breaker not disconnected from the main bus by the load shedding system shall be provided with means for remote control from the EPCC or the EPCP. Relays energized from the EPCC or the EPCP power shall be provided to activate the trip circuit of electrically-operated distribution circuit breakers. Shunt trip circuit of manually operated distribution circuit breakers shall be energized from the EPCC or the EPCP power. The power sensing relay shall be in accordance with MIL-R-24563, MIL-M-24350, or alternative specification approved by the Technical Authority.

3.3.13 Emergency power switchgear groups for ships having AC power. Switchgear groups for controlling larger than 100-kilowatt generators shall employ circuit breaker type automatic bus transfer (ABT) equipment in accordance with 3.6.10.1 and shall be constructed as sectionalized switchgear groups having a separate generator and bus transfer section, and feeder sections. For switchboards controlling emergency generators rated 100 kilowatt and less, contactor type ABT equipment in accordance with 3.6.10.1 shall be used. When the contactor type unit is used, the switchboard shall be assembled with the molded case circuit breaker distribution units to form a single section (see figure 14 and figure 15).

3.3.13.1 AC emergency generator control and circuit breaker type bus transfer section. The following shall be mounted on or within the unit:

a. One circuit breaker type ABT equipment in accordance with 3.6.10.1 having a current rating not less than the continuous rating of the emergency generator and the current rating of the feeders from the ship's service switchboards, as specified. Associated control switches, indicator lights, and relays for panel mounting shall be mounted on one of the front panels.

b. Necessary disconnecting devices for disconnecting cables between switchgear sections and performing other isolation functions.

c. Two potential transformers for metering and relaying.

d. One molded case type circuit breaker (fused or unfused), instantaneous element only, for power supply to the pump supplying cooling (sea) water to the generator diesel as specified (see 6.2) connected to the generator side of the generator circuit breaker. A red warning plate shall be provided stating that the circuit breaker shall always be in the closed position when the diesel is set up for starting.

e. Two current transformers for metering and relaying.

f. One dead-front fuse panel for fusing the potential transformer primary and secondary circuits. Dead-front fuseholders shall be in accordance with MIL-PRF-19207 or alternative specification approved by the Technical Authority.

g. One circuit breaker molded case type (fused or unfused), 250-ampere frame size. Casualty power circuit breaker shall be connected to the generator side of the generator circuit breaker. When the generator capacity is 500 kilowatts or greater, two casualty power circuit breakers shall be furnished.

h. One casualty power terminal, back-connected type in accordance with MIL-DTL-24552 and MIL-DTL-24552/2, connected through the molded case circuit breaker, item (g). When the generator capacity is 500 kilowatts or greater, casualty power terminals, back-connected type, shall be furnished.

3.3.13.1.1 Front panel. The following shall be mounted on the front of the panel:

a. One AC ammeter for reading one phase of generator current.

b. One AC voltmeter for reading one phase of generator potential.

c. One polyphase wattmeter for reading generator output.

d. One frequency meter for reading generator frequency.

e. Instruction plates containing complete operating instructions for the automatic equipment and a single line diagram of the main buses.

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f. One set of three ground indicator lights (clear) with test switch to indicate ground conditions on the 450-volt emergency power bus.

g. One indicator light (green), so connected as to indicate when all manually operated control devices are set up for automatic operation. The applicable following functions and conditions shall be included to complete the green light circuit.

(1) Generator circuit breaker (closed) (when separated from bus transfer).

(2) Voltage regulator transfer switch (automatic position).

(3) Ship's service bus transfer equipment (set for automatic operation).

(4) Emergency bus transfer equipment (set for automatic operation).

(5) Diesel starting control circuits (set for automatic starting).

(6) Diesel starting air shut-off valve (open).

(7) Electrically-operated emergency bus unloader switch (if used, set for automatic operation).

(8) Diesel salt water cooling pump circuit breaker (closed).

(9) Circuit breaker reset switch closed (set for automatic operation).

h. Space, mounting, and wiring for the voltage regulator equipment, including associated current and potential transformers, and for the exciter field rheostat, all equipment to be furnished with the emergency generator. The control switches, rheostats, and similar equipment constructed for panel mounting shall be located for front-of-panel operation as specified (see 6.2).

i. Space, mounting, and wiring for equipment, for the control switches and interlocks, furnished with the bus transfer equipment. The control switches and similar equipment constructed for panel mounting shall be located for front-of-panel operation.

j. One temperature indicator with selector switch and accessories, when temperature detectors are provided with the generator as specified (see 6.2).

3.3.13.2 AC emergency generator control and contactor type bus transfer section. This unit shall be equivalent functionally to the section specified in 3.3.13.1. The devices shall be the same as required in 3.3.13.1 except that the following shall be provided in lieu of 3.3.13.1.a:

a. One generator circuit breaker, molded case type, manually operated.

b. Two contactor type ABT equipments in accordance with 3.6.10.1 having current ratings not less than the current rating of the feeders from the ship's service switchboards, as specified. One equipment shall be connected for selection of the ship's service supply, the second equipment shall be connected for transfer between ship's service supply and the emergency generator supply.

c. Four indicator lights (blue) to indicate closed positions of the ABT contactors. Indicator lights circuit shall be wired through auxiliary contacts provided in the bus transfer equipments.

d. Two indicator lights (white) to indicate that the ship's service power feeders are energized.

e. One non-automatic molded case type circuit breaker, connected to bypass the bus transfer equipment selecting between the ship's service and emergency so that emergency generator power may be supplied to the ship's service switchboards when their generators are secured.

f. Two control switches, one for each bus transfer equipment, for selecting between manual and automatic control of the equipments.

g. When specified (see 6.2), one electrically-operated emergency bus unloader switch with control relay and current transformer shall be provided. This switch shall be connected between the feeder for the steering power switchboard and the emergency bus to disconnect the emergency bus during periods of high current requirements by the steering gear auxiliaries when energized from the emergency generator. Equipment shall function to hold the unloader switch closed unless the demand on the steering power switchboard exceeds 150 percent of the full load current of two steering motors plus 100 percent of the full load current of all other steering gear auxiliaries, in which case, the unloader switch shall open provided the power is being supplied by the emergency generator.

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3.3.13.3 Distribution units. Distribution units of the emergency switchgear group shall be in accordance with 3.3.12.6 through 3.3.12.9.

3.3.13.4 Load center (see 6.3.8) (see distribution sections). These sections shall be switchgear unit type of construction. Units comprising the section shall provide for distribution through molded case or power circuit breakers and shall be constructed in accordance with 3.3.12.6 through 3.3.12.9. Distribution units requiring two sources of supply shall be provided with interlocked molded case (automatic or non-automatic) or power circuit breakers. Where interlocked circuit breakers are installed, an indicator light shall be provided for each power supply to indicate when power is available.

3.3.13.5 Casualty power. Each load center distribution section shall be provided with two molded case type 250-ampere frame size circuit breakers (fused or unfused) for casualty power supply and two casualty power terminals back-connected type in accordance with MIL-DTL-24552 and MIL-DTL-24552/1, connected through the molded case type circuit breakers to the main bus.

3.4 Environmental performance requirements.

3.4.1 Temperature.

3.4.1.1 Ambient temperature. The switchgear shall meet all requirements of this specification when operating in an ambient temperature of 122 °F (50 °C).

3.4.1.2 Temperature rise. Air temperature rise in compartments shall not exceed 45 °F (25 °C).

3.4.2 Shock. Unless specified herein, the complete switchboard section, including all subsidiary components and subassemblies installed, shall be able to withstand grade A, type A, high-impact shock testing in accordance with MIL-DTL-901.

3.4.2.1 Shock test procedures and test results. Test procedures shall be approved by the Technical Authority prior to the start of testing. Test results shall be approved by the Technical Authority.

3.4.3 Vibration. The complete switchboard shall withstand electrically and mechanically the type I vibration tests specified in MIL-STD-167-1.

3.4.3.1 Vibration test procedures and test results. Test procedures shall be approved by the Technical Authority prior to the start of testing. Test results shall be approved by the Technical Authority.

3.4.4 Electromagnetic interference (EMI). The complete switchboard with electronic devices (if applicable) shall meet and demonstrate compliance with the applicable emissions and susceptibility requirements and limits of MIL-STD-461 for surface ship, below deck, metallic hull installations.

3.4.4.1 EMI test procedures and results. EMI test procedures and results shall be in accordance with the Electromagnetic Interference Test Procedures (EMITP) and Electromagnetic Interference Test Report (EMITR) requirements of MIL-STD-461. Test procedures shall be approved by the Technical Authority prior to the start of testing. Test results shall be approved by the Technical Authority.

3.4.5 Water spray. When specified (see 6.2), switchgear shall be designed to meet the water spray test requirements of MIL-STD-108 for drip-proof, 15-degree.

3.4.6 Water mist. When specified (see 6.2), switchgear shall be designed to meet the blowing rain test requirements of MIL-STD-810, Method 506.5, Procedure II. The pressure, flow rate, droplet size range, and duration of the test shall be as specified (see 6.2).

3.4.7 Flame resistance. See 3.3.6.1 and 4.4.3.15.

3.4.8 Noise. When fans are installed, switchboard shall meet airborne and structureborne noise requirements of MIL-STD-1474, Equipment grade E, and MIL-STD-740-2, type IV (see 4.4.3.14).

3.5 Electrical and other performance requirements.

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3.5.1 Rated short-circuit and short-time withstand ratings. Rated short-circuit and short-time withstand ratings as described in IEEE C37.20.1 shall be as specified (see 6.2).

3.5.2 Nominal user voltage. Nominal user voltage shall be one of the values shown in MIL-STD-1399-300-1, as specified (see 6.2).

3.5.3 Dielectric strength. Switchgear shall be constructed to withstand for a period of 1 minute a dielectric test voltage of twice the rated voltage of the equipment plus 1,000 volts rms between separate circuits and to ground.

3.5.4 Voltage spike. The complete switchboard with electronic devices (if applicable) shall withstand voltage spikes on control circuits in accordance with MIL-STD-1399-300-1 (see 4.4.3.10). Voltage spikes shall not cause failure or mal-operation of electronic devices.

3.5.5 Insulation resistance. Insulation resistance from phase to phase and from any phase to the frame/chassis of the switchgear shall not be less than 10 megohms at 77 °F (25 °C).

3.5.6 Grounding.

3.5.6.1 System grounding. Equipment shall operate in an ungrounded system as defined by MIL-STD-1399-300-1.

3.5.6.2 Bond strap. A bond strap in accordance with MIL-STD-1310, class C shall connect the switchboard frame to the hull for grounding purposes.

3.5.6.3 Grounding considerations for powder coat/chemical conversion coating applications. See 3.3.1.3.1.3 and 3.3.1.3.2.

3.5.6.4 Maintenance test points/ground studs. When bus bars are fully insulated, easily accessible test points/ground studs shall be provided to perform initial voltage verification and insulation resistance maintenance checks. Ground studs shall have field replaceable boots that allow maintenance access. Test points shall be provided for every section to be isolated (incoming lines and the bus).

3.5.7 Nominal control voltage. Nominal voltages in the control cabinet/compartment shall not exceed 450 VAC or 125 VDC. DC voltage is preferred.

3.5.8 Phase rotation. For AC, when facing the front of the switchboard or panel, the phase rotation shall be A, B, and C, respectively, from right to left, top to bottom, or front to back.

3.5.9 DC polarity. When facing the front of the switchboard or panel, left hand contacts shall be of negative polarity and right hand contacts shall be of positive polarity.

3.5.10 Maintainability. The switchgear shall facilitate accessibility, troubleshooting, fault isolation, and repair down to the lowest non-repairable part or non-repairable assembly as specified (see 3.3.4, 4.4.3.2, and 6.2).

3.6 Components and devices.

3.6.1 Circuit breakers, stationary elements (cradles), and lifting devices.

3.6.1.1 Circuit breaker type. Circuit breakers shall be molded case type in accordance with UL 489D or power type in accordance with UL 1066A. Power type circuit breakers shall be provided with a cradle for drawout. The frame size rating that the circuit breaker is qualified to in accordance with UL 489D or UL 1066A shall meet or exceed the rated current for the switchboard application in which it will be used. As an example, a power circuit breaker qualified for a frame size rating of 1200 amps in accordance with UL 1066A shall not be used in a switchboard application where the rated current is greater than 1200 amps.

3.6.1.2 Circuit breaker trip coil circuits. Trip coil circuits (both AC and DC) for tripping circuit breakers shall be routed through a normally open “a” contact (open when the circuit breaker is open) of the circuit breaker auxiliary switch so that the trip coil cannot remain energized after the circuit breaker is open.

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3.6.1.3 Power circuit breaker lifting device. Maintenance equipment capable of removing power type circuit breakers from all applicable compartments shall be required (see 6.2). Details of the required aisle space and storage requirements shall be as specified (see 6.2).

3.6.1.4 Tightening jam nut assembly (jam nut and regular nut) connections to molded case circuit breaker or molded case switch mounting base cylindrical studs. For jam nut connections made to molded case circuit breaker or molded case switch mounting base cylindrical studs, the torque requirements of table VI shall apply. Jam nuts shall be torqued to 100 percent of applicable table IV values and then the regular nuts installed. The regular nut shall be torqued to 110 to 115 percent of applicable table IV values while the jam nut is held stationary. See figure 16 for typical connections for molded case circuit breaker or molded case switch mounting bases with cylindrical studs.3

TABLE VI. Torque requirements for molded case circuit breaker/molded case switch mounting base cylindrical stud connections.

Copper stud size	Steel cap screw size	Torque	
		Min.-max. lubricated	Min.-max. non-lubricated
3/8 - 16	----	7-8 foot-pounds (84-96 inch-pounds)	10-12 foot-pounds (120-144 inch-pounds)
1/2 - 13	----	15-17 foot-pounds (180-204 inch-pounds)	22-25 foot-pounds (264-300 inch-pounds)
3/4 - 16	----	25-28 foot-pounds (300-336 inch-pounds)	37-42 foot-pounds (444-504 inch-pounds)
1-1/8 - 12	----	40-44 foot-pounds (480-528 inch-pounds)	60-66 foot-pounds (720-792 inch-pounds)
----	5/8 - 11	50-55 foot-pounds (600-660 inch-pounds)	75-80 foot-pounds (900-960 inch-pounds)
----	1 - 8	130-145 foot-pounds (1560-1740 inch-pounds)	195-215 foot-pounds (2340-2580 inch-pounds)
5/16 - 24	----	5-6 foot-pounds (60-72 inch-pounds)	7-9 foot-pounds (84-108 inch-pounds)

3.6.2 Arc fault detectors. When required (see 6.2), arc fault detectors (AFD) that have been previously approved for Navy use shall meet the requirements of MIL-DTL-32656.

3.6.3 Synchronizing control circuits. The synchroscope shall be supplied by the phase A-B transformers, and the synchronizing lights shall be supplied by the phase B-C transformers provided for metering, synchronizing, and relaying potential as shown on figure 17. The loss of one transformer shall not disrupt both synchronizing facilities. Closing control circuit of electrically-operated circuit breakers (AC generators, bus ties, and shore power) shall be connected through contacts of the corresponding synchronizing switch. Synchronizing control equipment shall be provided as specified in MIL-S-24188 or alternative specification approved by the Technical Authority to electrically prevent closing of the generator circuit breakers, bus tie circuit breakers, or shore power circuit breakers, as applicable, unless the energized systems to be connected are properly synchronized. Control circuitry external to the synchronizing protective equipment shall be arranged so synchronizing and paralleling can be accomplished after failure of the synchronizing protective equipment. Automatic paralleling devices shall be as specified (see 6.2).

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3.6.4 Generator space heater circuit. The space heater circuit shall be wired through a normally closed contact (closed when circuit breaker is open) of the associated generator circuit breaker auxiliary switch. The space heater circuit shall monitor current flow to the heater element to account for open circuited heater element or loss of heater power supply. An indicator light (white) shall show when generator heaters are energized.

3.6.5 Metering and control options. The level of metering and control that is required for each switchboard section in the system shall be as specified (see 6.2).

3.6.5.1 Human machine interface (HMI). Note the following:

a. MIL-STD-1472 shall be used as a guideline for the HMI design.

b. The HMI panel shall be mounted in the front of the control cabinet.

c. The top of the HMI panel display screen shall be no higher than 72 inches above the deck (total height includes the switchboard foundation).

d. The HMI shall be the local operational interface for all features and functions available in the switchboard or switchboard group in which it is installed.

e. The HMI screen shall normally be blank. The display shall become automatically active upon alarm, warning, or manual initiation. The manual initiation shall not result in the initiation of a command to any devices; the HMI screen shall simply turn on.

3.6.5.1.1 Display pages. The HMI screen shall have several display pages including, but not limited to:

a. One-line diagram page. This shall be a mimic one-line diagram of the switchboard, showing the status of all circuit breakers, and the current through each breaker, and bus voltage.

b. Circuit breaker page. A page for each circuit breaker showing positions as applicable (open/closed/test/removed), currents for all three phases, controls for operating each circuit breaker.

c. Alerts page. The alarms, warnings, and notifications associated with the class switchboard shall be provided on the alert page. Local alarm silence shall be provided. When an alarm is cleared at the Electric Plant Control Panel (EPCP), it shall also clear the alarm at the HMI. Audio and visual indication shall be provided for alarms; visual indication only shall be required for notices and warnings. This feature shall be selectable. Circuit breaker trip information shall be provided on the alert page as applicable.

d. Generator page. This applies only to the ship service and emergency switchboards. The generator page shall provide indications necessary for operations such as, but not limited to, paralleling, start-up, and securing of generators (electrically).

3.6.5.1.1.1 Color conventions. The HMI screens shall use the following colors to display unit operating conditions. Color conventions for other operating conditions not covered below shall be as specified (see 6.2).

a. Green. To indicate that a component is operating within its normally acceptable conditions.

b. Yellow. To indicate that a component is operating outside its normally acceptable conditions and is in a pre-alarm state.

c. Red. To indicate that a device has entered an alarm condition.

3.6.5.1.2 Touch screen. The HMI control interface shall be touch screen. The HMI display shall be flush mounted to the front of the control cabinet.

3.6.5.1.2.1 Viewing requirements.

a. Minimum screen diagonal shall be 18 inches.

b. Maximum screen diagonal shall be 23 inches.

c. Minimum resolution shall be 1280 x 1024.

d. Minimum luminance shall be 150 candles/square meter.

e. Minimum contrast ratio shall be 100:1.

f. Minimum viewing area shall be ±45 degrees side to side, 30 degrees up, and 10 degrees down.

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3.6.5.1.3 Software. A graphical user interface (GUI) with software shall be provided. Instructions shall be provided for software and GUI use. GUI software program(s) shall be provided to communicate with external components. This software shall provide functionality as required by ships specifications.

3.6.5.1.4 Redundancy. Provisions shall be made for a backup system for all control and metering functions to be available to the user in the event of an HMI failure. Control shall be easily available to the user through the use of a back-up to the HMI or through other switches and other lights. A dedicated ruggedized personal computer (PC) or similar acceptable for naval shipboard use may be used as a backup system to an HMI through an access port. All displays and functions on the HMI shall be accessible on the backup system.

3.6.5.2 Discrete meters. Electrical measuring instruments shall be in accordance with MIL-I-1361, ASTM F2362, MIL-DTL-16034, MIL-S-16104, MIL-M-16125, MIL-W-19088, MIL-V-23151, MIL-M-23167, or alternative specification approved by the Technical Authority. Electrical measuring instruments shall have a 4-1/2-inch, rectangular case with a 250-degree scale. Instrument scales shall be white with black markings and lettering, and shall be marked to indicate the full load and normal voltage calibration points.

3.6.5.3 Digital indicators. Digital readout indicators shall be segment readouts in accordance with MIL-DTL-28803 or alternative specification approved by the Technical Authority. The height of the indicator characters shall be a minimum of 0.45 inch. Display colors shall be selected in accordance with 3.6.12, except that red shall be used for alpha-numeric displays.

3.6.5.3.1 Digital readout circuitry. The need for digital indicators may require that some circuitry be mounted with the digital indicators to reduce the number of connections necessary between the equipment and the digital indicators.

Indicators shall be constructed according to the following guides with the order of importance being the same as the order of the guides:

a. Minimize the complexity of the indicator component.

b. Minimize susceptibility.

c. Minimize the number of connections between the equipment and the digital indicators

3.6.5.3.2 Digital indicator flicker. The digital indicators that display the parameter measurements in engineering units shall reduce flickering of the readout equipment. The least significant digit will inherently change from one digit to another at the transfer point. Every effort shall be made to reduce objectionable flicker in the least significant digit.

3.6.5.3.3 Isolation. Isolation shall be used among the various instrument outputs so that shorting or opening any digital output circuit shall not produce a change in any other output in excess of the specified accuracy requirements.

3.6.5.4 Controls. The following controls shall be provided:

a. Local HMI (when required). Equipment shall be capable of operation directly through the HMI interface (normal operation).

b. Remote. Equipment shall be capable of operation through offsite interfaces.

c. Local manual. Equipment shall be capable of operation directly.

3.6.5.4.1 Controls operations in each control mode. A detailed list of required controls applicable to Local HMI, Remote, and Local Manual shall be provided as specified (see 6.2).

3.6.6 Transformers. Transformers shall meet the requirements of MIL-T-15108 and MIL-T-16315, as applicable, or alternative specification approved by the Technical Authority.

3.6.7 Resistors and rheostats. Resistors and rheostats shall be in accordance with MIL-DTL-15109 or alternative specification approved by the Technical Authority. Power dissipation shall not exceed 50 percent of the rated value after ambient temperature and other applicable derating factors are applied in accordance with the part specification.

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3.6.8 Fuses and terminal blocks. Fuses shall be in accordance with MIL-PRF-15160 or alternative specification approved by the Technical Authority. Fuses, fuseholders, and fuseclips shall be silver plated in accordance with MIL-PRF-15160 or alternative specification approved by the Technical Authority. Fuseholders shall be dead-front type with blown fuse indicator in accordance with MIL-PRF-19207 or alternative specification approved by the Technical Authority. Dead-front fuse panel for fusing of the potential and control transformer primary and secondary circuits and any control circuits extending outside of the section shall be mounted on or within the unit.

3.6.8.1 Fuse mounting. Fuses mounted on switchboards shall be accessibly located for replacement. External mounting of fuses is preferred. Fuses shall be grouped on subpanels insofar as practicable, but the length of leads between fuses and the power circuit taps shall be kept to a minimum.

3.6.8.1.1 Dead-front fuseholders. Dead-front fuseholders shall be of the types in table VII in accordance with MIL-PRF-19207 or shall be equivalent types in accordance with alternative specification approved by the Technical Authority and shall not be installed for any distribution circuit fuse on power or lighting distribution switchboards or panels:

TABLE VII. Dead-front fuseholder types.

Type	Specification
FHL10U	MIL-PRF-19207/1
FHL11U	MIL-PRF-19207/2
FHL12U	MIL-PRF-19207/3
FHL14G	MIL-PRF-19207/5
FHL32W	MIL-PRF-19207/21
FHL33W	MIL-PRF-19207/22
FHL34W	MIL-PRF-19207/23
FHL35W	MIL-PRF-19207/24

3.6.9 Terminal boards. To minimize the possibility of breakage of the terminal boards caused by distortion of the mounting support under shock, the length of the terminal board shall not exceed 12 terminal points. Terminal boards shall be in accordance with A-A-59125 and A-A-59125/2 or A-A-59125/9, or the equivalent.

3.6.10 Switches, disconnecting devices, and transfer switches. Control and instrument switches shall be in accordance with specification approved by the Technical Authority. For applications below 125 volts, style JR switches as specified in MIL-DTL-21604 and MIL-DTL-21604/5 or alternative specification approved by the Technical Authority may be used. Control and instrument switches shall have gray or natural light tan escutcheon plates with black lettering. Switches shall be provided with four styles of handles to identify the function for which the switch is provided. The handle styles and corresponding function shall be as follows:

a. Round - governor control (spring return)

b. Oval - synchronizing and voltage regulator, synchronizing monitor, and field flashing

c. Round (knurled) - instruments (voltmeter, ammeter, frequency meter, temperature indicator)
- test (emergency switchboard, spring return)

d. Pistol-grip - circuit breaker and ground lamps (spring control)
- control bus selector

Control switches installed within the enclosure (such as relay defeater switches) shall be sufficiently close to the access opening to minimize personnel exposure to energized components.

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3.6.10.1 ABT equipment. As specified herein, ABT equipment shall be circuit breaker or contactor type, part or identifying number (PIN) M17773-4A29 in accordance with MIL-PRF-17773 or equipment in accordance with alternative specification approved by the Technical Authority.

3.6.11 Fans and fan monitoring system. Thermostatic fans may be added to the switchboard. Fans shall be used for switchboard compartment cooling or ambient air stratification. Fans shall not be used to cool a specific component. When specified (see 6.2), fans and fan monitoring system shall meet the requirements of 3.6.11.1, 3.6.11.2, 3.6.11.3, and 3.6.11.4. When fans are installed, the noise requirements of 3.4.8 shall be met. Fans shall be activated during selected first article tests as specified herein.

3.6.11.1 Thermostatic activation and control. Fans shall activate only when the internal ambient temperature in a compartment exceeds 140 °F (60 °C). All fans in a switchboard section shall be controlled by the same control/sensor.

3.6.11.2 Fan redundancy. Fans shall incorporate redundant capability. Only one fan in a redundant pair shall be operating at any given time. Each redundant fan shall be powered and fused separately from its associated primary fan. Provision shall be made to manually switch to redundant fan upon primary fan failure. Switch can be installed on switchboard or fan housing.

3.6.11.3 Fan filters. Easily accessible, washable air filters shall be provided for intake fans.

3.6.11.4 Maintenance switch and alarms. Fans shall be disabled by operation of an appropriate maintenance switch. Remote and local audible and visual alarms shall be provided to indicate fan failure and over-temperature conditions. The local audible alarm shall have a pushbutton that shall silence an active alarm. The audible and visual alarms shall remain active until the internal switchboard temperature drops below 149 °F (65 °C).

3.6.12 Indicator lights. Whenever possible, indicator lights shall be Light Emitting Diode (LED) type. If an LED type indicator light is not available for a particular application, an incandescent type indicator light may be used. LED indicator lights shall be in accordance with A-A-59781. Incandescent indicator lights shall be in accordance with MIL-DTL-3661, MIL-L-3661/38, MIL-L-3661/62, MIL-L-3661/63, MIL-L-3661/64, and MIL-DTL-3661/65, as applicable or alternative specification approved by the Technical Authority. Indicator lights shall be furnished with lamps. For position indication of circuit breakers, blue indicator lights shall be used to show that the circuit breaker is closed. Other indicator lamps have color caps as specified for the particular application. The color code shall be as follows:

a. Red - danger or emergency condition requiring immediate attention or corrective action.
b. Green - normal condition.
c. White - power available or power on.
d. Blue - closed, advisory.
e. Clear (not etched) - synchronizing or ground detector lights.
f. Yellow - abnormal, but not requiring immediate attention.

3.6.12.1 Indicator light power. On surface ships, the indicator lights of the mimic bus shall be supplied from a local lighting system that has available both normal and emergency power. On submarines, power for the indicator lights shall be taken from the AC or DC control power buses. All LED type indicator lights shall be of the same operating voltage.

3.6.12.2 Ground detector lights. Ground detector lights shall be connected as shown on figure 18.

3.6.13 Non-contact thermal sensors. When specified (see 6.2), fiber optic temperature sensors or other technologies shall be provided to determine temperatures of selected current carrying joints or expected hot spots (see 4.4.3.8.7). Temperature sensors shall be permanently mounted and be provided with dedicated maintenance test ports. The test equipment used shall be capable of operation in the field for maintenance. Functionality shall be verified by temperature rise testing when required.

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3.6.14 Voltage regulator equipment. Where voltage regulator equipment in accordance with MIL-R-2729 or alternative specification approved by the Technical Authority is required to be installed in the switchboard, the space and arrangement of this equipment shall allow for accessibility for maintenance and provide the electrical insulation distances as specified (see 6.2).

3.6.15 Tool storage. When specified (see 6.2), maintenance equipment storage shall be provided for all recommended on-board maintenance tools.

3.6.16 Charging motor protection. When specified (see 6.2), a device that protects the circuit breaker charging motor from excessive current and extended operation shall be provided.

3.7 Cybersecurity. Low voltage switchgear cybersecurity (see 6.3.6) design shall follow requirements for a secure industrial control system (ICS) in accordance with NIST Special Publications 800-53 and 800-82 and CNSSI No. 1253. ICS includes supervisory control and data acquisition (SCADA) systems, distributed control systems (DCS), and other control system configurations such as Programmable Logic Controllers (PLCs) that are often found in the industrial control sectors. Cybersecurity shall be in accordance with Appendix A.

3.7.1 Cybersecurity strategy. A low voltage switchgear cybersecurity strategy shall be as specified (see 6.2 and A.3) and approved by the Technical Authority.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection samples shall be subjected to the examination and tests in the order specified in table VIII. Prior to first article inspection, any exceptions or changes to the order specified in table VIII shall be approved by the Technical Authority.

TABLE VIII. First article and conformance inspection.

Order	Inspection	First article	Conformance	Requirement paragraph(s)	Inspection paragraph(s)
1	General examination	X	X	----	4.4.3.1
2	Maintainability (if required)	X	----	3.5.10	4.4.3.2
3	Dielectric strength	X	X	3.5.3	4.4.3.5
4	Insulation resistance	X	X	3.5.5	4.4.3.13
5	EMI	X	----	3.4.4	4.4.3.9
6	Voltage spike	X	----	3.5.4	4.4.3.10
7	Ambient temperature and temperature rise	X	----	3.4.1	4.4.3.8
7A	Thermal imaging window and non-contact thermal sensor data check (if required)	X	----	3.3.5.4 3.6.13	4.4.3.8.7
8	Short-circuit withstand	X	----	3.5.1	4.4.3.11
9	Short-time withstand	X	----	3.5.1	4.4.3.12
10	Shock	X	----	3.4.2	4.4.3.3
11	Dielectric strength (65% reduced value)	X	----	3.5.3	4.4.3.5
12	Insulation resistance	X	----	3.5.5	4.4.3.13

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TABLE VIII. First article and conformance inspection – Continued.

Order	Inspection	First article	Conformance	Requirement paragraph(s)	Inspection paragraph(s)
13	Vibration	X	----	3.4.3	4.4.3.4
14	Dielectric strength (65% reduced value)	X	----	3.5.3	4.4.3.5
15	Insulation resistance	X	----	3.5.5	4.4.3.13
16	Noise (if required)	X	----	3.4.8	4.4.3.14
17	Water spray (if required)	X	----	3.4.5	4.4.3.6
18	Dielectric strength (65% reduced value) (if water spray required)	X	----	3.5.3	4.4.3.5
19	Insulation resistance (if water spray required)	X	----	3.5.5	4.4.3.13
20	Water mist (if required)	X	----	3.4.6	4.4.3.7
21	Dielectric strength (65% reduced value) (if water mist required)	X	----	3.5.3	4.4.3.5
22	Insulation resistance (if water mist required)	X	----	3.5.5	4.4.3.13
23	Flame test	X	----	3.4.7	4.4.3.15

4.3 Conformance inspection. Conformance inspection samples shall be subjected to the examination and tests in the order specified in table VIII. Prior to conformance inspection, any exceptions or changes to the order specified in table VIII shall be approved by the Technical Authority.

4.3.1 Sampling for conformance inspection.

4.3.1.1 Inspection lot. Equipment of the same type and size identified for inspection at one time shall be considered a lot for conformance inspection.

4.3.2 Classification of defects. Classification of defects shall be as listed in table IX. In addition, failure to satisfy any test of 4.4 shall be considered a defect. Defects resulting from failure to satisfy tests of 4.4.3 shall be classified as major.

TABLE IX. Classification of defects.

Classification	Defects
Critical:	
1	Enclosures, panels, compartmentation not provided to prevent injury from electric shock; rear enclosure perforations, ventilating louvers exceed size limit.
2	Grab rails and guard rails not located or mounted as specified to ensure safe operations.
3	Cybersecurity not approved or implemented in accordance with applicable standards.
Major:	
101	Type, number, or section arrangement nonconforming.
102	Section unit not complete, devices or component parts missing.

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TABLE IX. Classification of defects – Continued.

Classification	Defects
103	Evidence of use of unauthorized material.
104	Parts broken, cracked, distorted, or chipped.
105	Control benchboard (if required) dimensions, arrangement of devices nonconforming.
106	Structure not in accordance with drawing; not framed, bolted, welded, or dimensioned as required; gusset plates nonconforming.
107	Framework bracing not as specified; bolt size less than minimum; unauthorized use of rivets.
108	Standard units or special sections not bolted to foundation channel in accordance with drawing; number and size of bolts nonconforming.
109	Welding to frame nonconforming, bus bar cutouts not located in accordance with applicable drawing.
110	Compartmentation not as specified; circuit breakers not isolated; insulation distance not as specified.
111	Ventilation not adequate; openings not sized for circuit breaker load; openings not screened.
112	Locking devices not of acceptable type; not provided where required.
113	Panel thickness, size, forming, stiffening, or location nonconforming.
114	Panel not hinged as specified, when required.
115	Equipment brackets, bolting, or arrangement nonconforming; switches not mounted in specified plane; phases not disposed as required.
116	Casualty power connection, if required, not accessible; mounting nonconforming; casualty power circuit breaker not as specified.
117	Voltage regulator mounting nonconforming.
118	Bus bar not sized properly for amp rating; not silver surfaced formed as specified; ends not neatly finished; bolt hole contact areas not flat; bolt size and number not as specified.
119	Cable connectors not specified type and size for required load and use.
120	Bus bar assemblies not marked with specified warning.
121	Evidence that bus bar supports not spaced in accordance with short circuit current available, high-impact shock and vibration requirements.
122	Evidence that bus bar capacity nonconforming.
123	Cable connectors not of specified type; solder, if authorized, not as specified.
124	Means not provided for connecting cables from other units or sections; cable connection arrangement restricts access.
125	Instrument and control wiring nonconforming; not connected as specified; not grouped and strapped as specified.
126	Terminal boards not provided where required.
127	Synchronizing circuits, where required, not connected in specified manner; synchronizing control equipment not as specified.
128	Indicator lights, space heater circuits, transformers, voltage regulators, and trip circuits for power type circuit breakers not connected or arranged in accordance with applicable specifications or drawings.
129	Wiring not marked or tagged.

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TABLE IX. Classification of defects – Continued.

Classification	Defects
130	Circuit breakers not of type specified; location and mounting nonconforming; drawout mechanism (for power type circuit breakers only) not operable as specified.
131	Switches not in accordance with specification; instrument control switches not type or not marked as specified; handle not required shape.
132	Disconnect devices not of specified type; not mounted as required; tools, if required, not located properly.
133	Fuses not in accordance with specification; location identifications and mounting nonconforming.
134	Fuseholders, where required, not enclosed or not in accordance with drawing.
135	Terminal boards exceed allowable length.
136	Instruments not in accordance with specification; case, scale, and mounting nonconforming.
137	Transformers not in accordance with applicable specifications; not provided; not connected as specified; not protected where required.
138	Relays not as specified; not accepted type.
139	Indicator lights nonconforming; not color required by code; resistors not as specified; lamps not provided.
140	Rear and top enclosures not secured firmly to frame; will not clear live bus work if buckled; top flanges do not prevent water from seeping into the enclosure; panel fastening not as specified method employed for making thumbscrews captive inadequate.
141	Top enclosure shape, form, and flanges nonconforming; corners not welded watertight.
142	Buses not marked to indicate polarity or phase or buses not in the correct orientation.
143	Devices, where so required by applicable specification, not qualified for listing on Qualified Products Lists (QPLs).
144	When specified (see 6.2), user name and password control not provided as specified.
Minor:	
201	Painting procedure specified not followed; primer not applied as specified; enamel not of specified type.
202	Identification and information plates not of specified shape, size, thickness, location, letter, and fastening.
203	Special tools, if required, not furnished.
204	Switchboard not cleaned following construction; loose parts, dirt, or metal shavings not removed from unit.

4.4 Examination and test methods.

4.4.1 Test equipment. Test equipment shall be as specified in each individual test herein. Test equipment requiring calibration shall be certified to be acceptable at the time of testing.

4.4.2 Test conditions. Unless otherwise specified herein, the inspection and tests of 4.4 shall be performed under the following conditions. Unless otherwise specified herein, ambient conditions within the specified range need not be controlled.

a. Ambient temperature from 50 to 122 °F (10 to 50 °C).

b. Attitude: Normal operation position.

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4.4.3 Detailed inspection.

4.4.3.1 General exam. Switchgear shall be subjected to a thorough examination to determine that the material, workmanship, safety to operating personnel, design, and construction are in accordance with this specification and applicable drawings. Examination shall include, but need not be limited to, the following:

a. Prohibited materials shall be verified as specified (see 3.3.1.1 and 6.2).

b. Non-preferred component materials are not used without prior approval by the Technical Authority (see 3.3.1.2).

c. Electrical connections and wiring in accordance with applicable documents. An operational (manual, local, remote) check of all low voltage switchgear components and with a ring out of wired connections.

d. Outline and mounting dimensions in accordance with applicable documents (not needed for conformance testing).

e. Designation and marking, including safety warnings, are as specified.

f. Weights and centers of gravity are in accordance with applicable documents (not needed for conformance testing). Centers of gravity may be determined by actual measurement or through the use of three dimensional models.

g. Check of workmanship, fastener integrity, damage to platings or components.

h. The 100-hour burn-in test shall be conducted by energizing control voltage to the equipment for 100 hours at nominal voltage and frequency, ambient temperature and with all inputs and outputs connected to effect maximum rated loading. The 100-hour burn-in test may be accomplished at the end of first article testing.

i. Verification of arc fault equipment functionality (as applicable).

j. Verification of UPS hold up time.

k. Ground connections shall be inspected for resistance in accordance with MIL-STD-1310.

l. Compartmentation gaps are as specified.

m. Louver openings are as specified.

n. Verification that power circuit breaker lifting device is capable of removing power type circuit breakers from all applicable compartments.

o. Cybersecurity approved or implemented in accordance with applicable standards (see 3.7 and Appendix A).

p. Verification of charging motor protection. If equipped, a device that protects the circuit breaker charging motor from excessive current (locked rotor) and extended time of operation (failed cut off) shall be verified to be operational and protect the circuit breaker charging motor from damage during mal-operation.

4.4.3.2 Maintainability. When specified (see 6.2), compliance shall be verified through a maintenance demonstration procedure, maintenance task selection, and maintenance task performance. For guidelines, refer to test Method 1 of MIL-HDBK-470.

4.4.3.3 High-impact shock. The complete switchboard shall be subjected to grade A, type A, high-impact shock test in accordance with MIL-DTL-901. This equipment shall be tested as a principal unit. The complete switchboard section, including all subsidiary components and subassemblies installed within the switchboard during the test, shall meet the shock test acceptance criteria.

4.4.3.3.1 Test setup. Test setup shall include the following:

a. All equipment, including spare circuit breakers, shall be installed in the switchboard section in accordance with the shipboard configuration during testing.

b. All lengths of ship's cabling (except for spare and blank circuit breakers) and buswork shall be installed in the switchboard section in accordance with the shipboard configuration during testing.

c. All lengths of ship's cabling (except for spare and blank circuit breakers) that extend outside the foundation of the switchboard section shall be stubbed to 12 to 18 inches with the bitter ends insulated.

d. Cable, cable terminations, and cable restraints required for shipboard shall be included in shock testing.

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e. The test shall be conducted with all equipment energized at rated voltage with no load.

f. Switchboard section mounting location, mounting plane, and mounting orientation for the shock test shall simulate the worst case shipboard installation as specified (see 6.2).

g. For each compartment of the switchboard section with circuit breakers installed, a minimum of one circuit breaker of each type shall be monitored for contact bounce during testing. Monitoring shall include all main contacts and at least one normally open and one normally closed auxiliary contact.

h. Fans when required shall be energized during this test.

i. Foundations used in shock testing shall simulate shipboard configuration.

j. The switchboard shall be instrumented with a sufficient number of accelerometers to support future shock analysis of components.

4.4.3.3.2 Principal function pass/fail criteria. The following criteria shall be used to determine switchgear component pass or failure acceptance criteria:

a. Hair line cracks are permitted in the framework, but no separation of parts shall result.

b. Any resulting distortion shall not be of sufficient magnitude to impair the operation of the equipment contained within or prevent the circuit breakers from being withdrawn.

c. The enclosure shall not distort to an extent which might result in contact with any electrically conducting part within the section.

d. The bus work shall not be mechanically damaged nor the bus clearance distances reduced below the minimum requirements.

e. Insulation resistance shall not be reduced below 10 megohms at 77 °F (25 °C).

f. The section shall pass the dielectric strength test at 65 percent of the specified voltage.

g. Circuit breakers shall not change position/state from “on” to “off”, from “off” to “on”, or from “on” to “trip” during the test. An “off” to “trip” position/state change is permitted.

h. Any closing of circuit breaker main and auxiliary contacts from the open position shall not be acceptable.

i. Momentary opening of circuit breaker main and auxiliary contacts shall not be in excess of 0.020 second during a single shock event.

j. Circuit breaker control relays or protective relays shall not cause the circuit breakers to change position/state.

k. Protective relay adjustable settings shall not change position.

l. All circuit breakers shall complete an open/close operation after each shock blow.

m. All circuit breakers with racking capability shall complete a racking cycle after each shock blow. Racking shall be accomplished using the racking mechanism and standard racking tool without removal of circuit breaker front cover.

n. No components or pieces of components shall become adrift. Cracks in thermal imaging windows are permitted.

o. Bus and cable fasteners shall not have loosened.

4.4.3.3.3 Disposal of shock-tested equipment. Equipment which has been subjected to high-impact shock may be accepted as a production unit of the contract or order only if each of the following conditions are met:

a. The vibration test is conducted after the completion of all shock tests.

b. Damaged parts are replaced.

c. Damaged structural members are repaired or replaced.

d. Post shock and vibration electrical tests and inspections show conformance to specified performance.

e. The unit is covered by the same guarantee by the manufacturer as other production units.

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f. The shock-tested unit is free from any distortion of the frame members or enclosure that would prevent its being bolted to the deck or to other switchgear units to form a switchboard section. Measurements shall be made to detect any misalignment of bolt holes or bent framework.

4.4.3.3.4 Extension of shock tests. Extension of satisfactory shock test results to other sections shall be in accordance with the requirements of MIL-DTL-901. Requests for test extension shall include, but shall not be limited to, the following (see 6.2):

a. Change in weight and mass.

b. Change in center-of-gravity.

c. Change in moments-of-inertia.

d. Changes in dimensions (i.e., height and footprint).

e. Changes in circuit breaker frame sizes, locations, and quantities.

f. Change in materials or size and location of structural members.

g. Changes in method of circuit breaker mounting.

h. Changes in bus bar arrangement or method of support.

i. Changes in ship mounting classification (i.e., deck or hull mounting).

j. Changes in mounting orientation (i.e., athwartship or fore/aft).

4.4.3.4 Vibration test. The switchgear shall be subjected to type I tests of MIL-STD-167-1. Maximum frequency of vibration shall be as specified (see 6.2). Test setup, pass/fail criteria, and extensions shall be the same as for shock testing (see 4.4.3.3.1, 4.4.3.3.2, 4.4.3.3.4) with the following exceptions:

a. No circuit breaker contact bounce shall be acceptable.

b. All circuit breakers shall complete an open/close operation between endurance axis change.

c. All circuit breakers with racking capability shall complete a racking cycle between endurance axis change. Racking shall be accomplished using the racking mechanism and standard racking tool without removal of circuit breaker front cover.

d. Extension of vibration test results to other sections shall meet the requirements as specified (see 6.2).

4.4.3.5 Dielectric strength. Equipment shall withstand, for a period of 1 minute, a dielectric test voltage of twice the rated voltage of the equipment plus 1,000 volts rms value. The frequency of the test voltage shall be not less than 60 Hertz. It shall approximate a true sine wave. The source of the test potential shall have a rating of at least 1 kilowatt.

4.4.3.6 Water spray. Switchgear shall be subjected to the water spray test requirements of MIL-STD-108 for drip-proof, 15-degree protected enclosure except that visual exam shall not be permissible to determine acceptability of drip-proof design. Fans shall be energized during this test.

4.4.3.6.1 Pass/fail criteria. Within 30 minutes of the completion of water spray testing, the insulation resistance and dielectric tests as shown in table VIII shall be successfully conducted.

4.4.3.7 Water mist. Switchgear shall be subjected to the blowing rain test requirements of MIL-STD-810, Method 506.5, Procedure II. The pressure, flow rate, droplet size range, and duration of the test shall be as specified (see 6.2). Fans shall be activated before water mist testing begins.

4.4.3.7.1 Pass/fail criteria. The following criteria shall be met:

a. Fans shall automatically shut down when a water mist event occurs.

b. Within 30 minutes of the completion of water mist testing, the insulation resistance and dielectric tests as shown in table VIII shall be successfully conducted.

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4.4.3.8 Temperature rise.

4.4.3.8.1 Switchboard sample(s). The test shall be conducted on a complete switchboard section for each sample or samples chosen. All equipment (including spare circuit breakers), cabling, and buswork shall be installed in accordance with the shipboard installation. All circuit breakers (except spares) and all disconnects shall be fully cabled including those that are not used as a load during testing.

4.4.3.8.2 Pass/fail criteria. Temperature rise tests shall be conducted to verify the following when the switchboard section is operating within an ambient temperature of 122 °F (50 °C).

a. The air temperature within any compartment in the switchboard section shall not exceed 167 °F (75 °C).

b. Insulation, components, and wiring required to be installed in the units shall operate within rated insulation thermal limits or rated limits. A functional check of operating controls and circuit functions shall be performed at temperature stabilization.

c. Circuit breaker terminals shall not exceed a temperature rise limit of 126 °F (70 °C).

d. The temperature rise limit of power cables shall be based on the power cable chosen as specified (see 6.2).

e. When specified (see 6.2), fans shall activate only when the internal ambient temperature in a compartment exceeds 140 °F (60 °C).

f. Bus bars not in accordance with the requirements of table II shall not exceed a temperature rise of 144 °F (80 °C).

4.4.3.8.3 Test chamber. Testing shall be conducted without recirculation of air into the test chamber. Fans may be used to maintain ambient temperature and to prevent air stratification within the test chamber. These fans shall be directed away from the switchboard towards the sides of the test chamber enclosure. Air from these fans shall not blow directly onto or be deflected onto any part of the switchboard section during testing.

4.4.3.8.4 Sensors and sensor locations. Temperature sensors (thermocouples or other methods approved by the Technical Authority) shall be installed to determine ambient temperature within the test chamber (but outside the switchboard section) and temperature rise in each compartment of the switchboard section.

4.4.3.8.4.1 Ambient temperature sensors. Six sensors shall be installed in one of the following configurations:

a. Three sensors shall be installed 3 feet away from both the left and right sides of the switchboard section at half the depth and at the following locations:

(1) One sensor shall be 6 inches from the bottom of the switchboard section.

(2) One sensor shall be 6 inches from the top of the switchboard section.

(3) One sensor shall be at half the height of the switchboard section.

b. Three sensors shall be installed 3 feet away from both the front and back surfaces of the switchboard section at half the section width and at the following locations:

(1) One sensor shall be 6 inches from the bottom of the switchboard section.

(2) One sensor shall be 6 inches from the top of the switchboard section.

(3) One sensor shall be at half the height of the switchboard section.

4.4.3.8.4.2 Compartment temperature sensors.

4.4.3.8.4.2.1 Compartments other than bus compartments. Each switchboard section compartment other than a bus compartment shall have two temperature sensors at the following locations:

a. One sensor shall be 2 inches from the top at the compartment midway point from front-to-back and as close to the center of the compartment as possible.

b. One sensor shall be 2 inches from the bottom at the compartment midway point from front-to-back and as close to the center of the compartment as possible.

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4.4.3.8.4.2.2 Bus compartments. There shall be three temperature sensors per switchboard unit in the bus compartment (i.e., a three-unit section shall require nine separate sensors) at the following locations:

a. One sensor shall be 2 inches from the top at the compartment midway point from front-to-back and as close to the midway point of the unit width as possible.

b. One sensor shall be 2 inches from the bottom at the compartment midway point from front-to-back and as close to the midway point of the unit width as possible.

c. One sensor shall be at the halfway point from the top to bottom of the compartment at the compartment midway point from front-to-back and as close to the center of the unit width as possible.

d. Sensors shall be mounted to the bus (if required) to validate the bus insulation's temperature rating for non-standard bus geometry.

e. Sensors shall be mounted to each phase of the circuit breaker stationary element line and load terminals.

f. Sensors shall be mounted to each phase of the primary power cable connections.

4.4.3.8.5 Temperature stabilization and temperature rise determination. All temperature readings for each test shall be recorded during the same time frame and shall be of sufficient duration to allow stabilization of the measured temperatures to a rate of rise within 3.6 °F (2 °C) per hour. Ambient temperature at each ambient sensor location shall be maintained at 122±3.6 °F (50±2 °C). An average ambient temperature shall be determined by averaging together the ambient sensor readings. The average ambient temperature shall be used to determine the temperature rise of the switchboard compartments.

4.4.3.8.6 Loading and equipment energization.

4.4.3.8.6.1 One-unit switchboard section. Current equivalent to the rating of the main bus shall be applied at any convenient voltage. All circuit breakers shall be loaded in accordance with 4.4.3.8.6.3. If applicable, any remaining load shall be divided among as many different circuit breaker frame sizes as possible. If applicable, disconnects may be used for any remaining load after circuit breakers are loaded in accordance with 4.4.3.8.6.3. Total loading of the unit shall not exceed the rating of the unit.

4.4.3.8.6.2 Multiple-unit switchboard section. Separate temperature rise tests equaling the number of units in the section shall be conducted (i.e., a three-unit section shall require three separate tests). Each unit shall be loaded in accordance with 4.4.3.8.6.1. If applicable, any remaining load shall be divided among the other units as necessary. Total loading of a particular unit shall not exceed the rating of that unit. If the unit being fully loaded receives power from a circuit breaker in another unit of the same switchboard section, that circuit breaker shall be used to provide power to the fully loaded unit.

4.4.3.8.6.3 Circuit breaker loads. Circuit breakers shall be loaded to their designed full load (element rating, continuous current setting, or rating plug value as applicable) at any convenient voltage to simulate the heat dissipation characteristics the switchboard will have under actual operating conditions. All transformers and equipment that utilizes internal transformers shall be energized.

4.4.3.8.6.4 Voltage regulator loads. Dummy-type resistive loads shall be used to simulate voltage regulator components in the spaces allocated to these. These dummy loads shall be equal to the maximum heat dissipation ratings of the regulator components.

4.4.3.8.7 Thermal imaging windows and non-contact thermal sensor data check. During temperature rise testing, thermal imaging shall be conducted through the thermal imaging windows, if applicable, and data shall be gathered from non-contact thermal sensors. To verify accuracy, data shall be compared to readings taken from thermocouples during temperature rise testing. Temperature data may be used to create labels located adjacent to the window to aid thermographers for maintenance.

4.4.3.9 EMI test. The EMI tests of MIL-STD-461 for surface ship, below deck, metallic hull installations shall be conducted on switchboards with electronic devices. Testing shall not cause false tripping of circuit breakers or failure of electronic devices. Fans shall be energized during EMI testing.

4.4.3.10 Voltage spike. The control circuits shall be tested in accordance with the voltage spike testing requirements of MIL-STD-1399-300-1.

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4.4.3.11 Short-circuit withstand. Short-circuit withstand testing shall be conducted in accordance with IEEE C37.20.1. The maximum rms asymmetrical short circuit current shall be as specified (see 6.2).

4.4.3.12 Short-time withstand. Short-time withstand testing shall be conducted in accordance with IEEE C37.20.1. The maximum rms asymmetrical short circuit current shall be as specified (see 6.2).

4.4.3.13 Insulation resistance.

4.4.3.13.1 Test setup. Insulation resistance tests shall be conducted in accordance with Method 302 of MIL-STD-202. Test conditions shall be as follows:

a. Test potential: 500 V ±10 percent for operating potentials of 450 VAC or less.

b. Points of measure: Between each electrically isolated circuits and all other circuits connected together to ground (frame, chassis, or enclosure as applicable).

c. Application time of test voltage shall be sufficient to take resistance readings.

d. Temperature at time of test: Temperature of parts to be tested shall be measured and recorded.

4.4.3.13.2 Pass/fail criteria. Insulation resistance shall not be less than 10 megohms corrected to 77 °F (25 °C) for any readings before and after shock testing. Correction shall be made on the basis of insulation resistance doubling for each 27 °F (15 °C) decrease in temperature.

4.4.3.14 Noise. When fans are installed, airborne and structureborne noise testing in accordance with MIL-STD-1474, Equipment grade E, and MIL-STD-740-2, type IV shall be conducted. Fans shall be activated during testing.

4.4.3.15 Flame test. Primary conductor insulating material not in accordance with MIL-DTL-917 requirements shall be subjected to flame testing in accordance with IEEE C37.20.1.

4.4.3.15.1 Extension of flame test. Requests for flame test extension shall include information from previous testing performed in accordance with 4.4.3.15.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Switchgears covered by this specification are intended to provide protection, control, and distribution of electrical power on naval ships.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification

b. When first article inspection is required (see 3.1).

c. Door interlocks required (see 3.2.1.2).

d. Requirement for prohibited materials verification (see 3.3.1.1 and 4.4.3.1.a).

e. Non-preferred component coatings allowed (see 3.3.1.3.3).

f. Thermal imaging windows required (see 3.3.5.4).

g. Removable guard rail sections required (see 3.3.5.10.1).

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h. Location of cable entrance, cable restraints, and type of cable terminations (see 3.3.9).

i. Bus bar capacity (see 3.3.10.3).

j. Bus bar insulation (see 3.3.10.5).

k. Maximum available short circuit current (see 3.3.10.7).

l. Type, number, and arrangement of units comprising the switchgear sections (see 3.3.12 to 3.3.13, inclusive).

m. Sections comprising a switchgear group (see 3.3.12 to 3.3.13, inclusive).

n. Power, current, and voltage rating of generators controlled and quantity of casualty power equipment (see 3.3.12.1, 3.3.12.2, and 3.3.13.1).

o. Space, mounting, and wiring requirements for the voltage regulator equipment including associated current and potential transformers (see 3.3.12.1.g, 3.3.12.1.1.g, 3.3.12.1.1.1.j, 3.3.12.2.1.i, 3.3.12.5.3.a, 3.3.13.1.1.h, and 3.6.14).

p. Space, mounting, and wiring and associated control equipment for electric governor equipment (see 3.3.12.1.m, 3.3.12.1.1.o, and 3.3.12.2.1.f).

q. If load sensing relays are provided (see 3.3.12.1.1.i).

r. When automatic paralleling devices are required (see 3.3.12.1.1.q, 3.3.12.2.1.m, and 3.6.3).

s. Control switches, rheostats, and similar devices required in conjunction with speed regulating equipment (see 3.3.12.1.1.1.f).

t. If generator space heaters and temperature detectors are provided with the generators (see 3.3.12.1.1.1.i and n, 3.3.12.2.1.h and o, 3.3.12.5.3.g and p, and 3.3.13.1.1.j).

u. Disconnect devices (see 3.3.12.1.2.c and 3.3.12.3.c).

v. When a chair for EPCP is required (see 3.3.12.5.1).

w. Whether desk type or vertical type EPCP is required (see 3.3.12.5.1 and 3.3.12.5.2).

x. Space for internal communications (IC) equipment (see 3.3.12.5.1.1).

y. Whether or not the ship has a centralized machinery control system including EPCC (see 3.3.12.5.3.cc).

z. When mimic bus material is required (see 3.3.12.5.3.u(4) and 3.3.12.5.4.a(3)).

aa. Material to be used in construction of switchgear structure and switchgear buses (see 3.3.12.5.4.a(3)).

bb. Solid-state frequency changer failure mode indicators (see 3.3.12.5.4.h).

cc. Feeders, breakers, and switches (see 3.3.12.6, 3.3.12.7, and 3.3.12.8).

dd. Requirement for molded case type circuit breaker on emergency switchgear for diesel cooling water pump (see 3.3.13.1.d).

ee. One electrically-operated emergency bus unloader switch with control relay and current transformer (see 3.3.13.2.g).

ff. Water spray required (see 3.4.5).

gg. MIL-STD-108 degree of enclosure (see 3.4.5).

hh. Water mist required. Fan shut off permissible. Water spray pressure, flow rate, droplet size range, and duration of the test (see 3.4.6 and 4.4.3.7).

ii. Short-circuit and short-time withstand ratings (see 3.5.1, 4.4.3.11, and 4.4.3.12).

jj. Rated maximum voltage (see 3.5.2).

kk. Whether a maintainability demonstration is required (see 3.5.10 and 4.4.3.2).

ll. Circuit breaker lift device (see 3.6.1.3).

mm. Arc fault protection required (see 3.6.2).

nn. Synchronizing control circuits required (see 3.6.3).

oo. Layout of control switches, metering, and similar devices required (see 3.6.5).

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pp. Centralized machinery control system requirements of equipment, controls applicable to local HMI, remote, and local manual (see 3.6.5.4.1).

qq. Fans allowed (see 3.6.11).

rr. Non-contact thermal sensors required (see 3.6.13).

ss. Tool storage location (see 3.6.15).

tt. When a device that protects the circuit breaker charging motor from excessive current and extended operation is required (see 3.6.16).

uu. Cybersecurity strategy (see 3.7.1).

vv. Switchboard section shipboard mounting location, mounting plane, and mounting orientation, additional shock test requirements (see 4.4.3.3.1.f).

ww. Extension of shock and vibration tests (see 4.4.3.3.4 and 4.4.3.4.d).

xx. Maximum frequency of vibration (see 4.4.3.4).

yy. Temperature rise limit of power cables (see 4.4.3.8.2).

zz. Packaging requirements (see 5.1).

aaa. System categorization results (see A.3.1).

bbb. Vulnerability analysis (see A.3.2).

ccc. When intrusion detection is required (see A.3.3.h(2)).

ddd. Additional cybersecurity requirements (see A.4.2).

6.3 Definitions.

6.3.1 Branch bus taps. Branch bus taps are bus bars connected to the main bus taps.

6.3.2 Compartment. An internal switchboard section or unit subdivision formed by barriers of sheet metal, nonconductive material, or a combination of both.

6.3.3 Cybersecurity. Prevention of damage to, protection of, and restoration of computers, electronic communication systems, electronic communication services, wire communication, and electronic communication, including information contained therein, to ensure its availability, integrity, authentication, confidentiality, and non-repudiation.

6.3.4 Desk type EPCP. A desk type EPCP is a switchgear section consisting of switchgear type units and a control desk combined in a single structure with the top of the control desk, either horizontal or inclined, attached to the front switchgear section panels. It is provided with the necessary instruments and controls for centralized operation, monitoring, and control of the electric plant including control of generator, bus tie, and feeder circuits (including battery on submarines) of one or more switchgear groups.

6.3.5 Distribution unit. This unit provides for distribution of power through both molded case and power type circuit breakers as applicable.

6.3.6 Electronic devices. Electronic devices are those that have active electrical components such as transistors, diodes and integrated circuits, and associated passive interconnection technologies that would typically be found on a printed circuit board.

6.3.7 EPCC. A desk type EPCC is a console consisting of control desks provided with the necessary instruments and controls for centralized operation, monitoring, and control of the electric plant including control of generator prime movers, generators, bus ties, and feeder circuits of one or more switchboard groups.

6.3.8 Load center. A load center distribution section is a section consisting of one or more distribution units. It serves as a power distribution center in locations requiring power supplies exceeding the capacity of distribution panels.

6.3.9 Low voltage. Low voltage refers to voltages up to and including 1 kilovolt.

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6.3.10 Main bus taps. Main bus taps are bus bars connected to the main bus.

6.3.11 Section. A switchgear section is a self-contained assembly of switchgear units consisting of a complete framework with all required enclosures and with a common base mounting.

6.3.12 Switchboard. A switchboard is a large single panel, frame, or assembly of panels on which are mounted on the face, the back, or both, switches, overcurrent and other protective devices, buses, and instruments.

6.3.13 Switchgear group. A switchgear group is synonymous with the term switchboard; however, switchboard groups may consist of two or more individual sections rather than all units being assembled together into a single structure.

6.3.14 Technical Authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05S may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

6.3.15 Unit. A standardized switchgear unit is a vertical subdivision of a section complete with separate front panels and all required devices.

6.3.16 Vertical EPCP. A vertical EPCP is functionally similar to the desk type EPCP except that all necessary instruments, controls, and indicating apparatus are mounted on vertical panels.

6.4 Subject term (key word) listing.

Bus bars

Compartmentation

Cybersecurity

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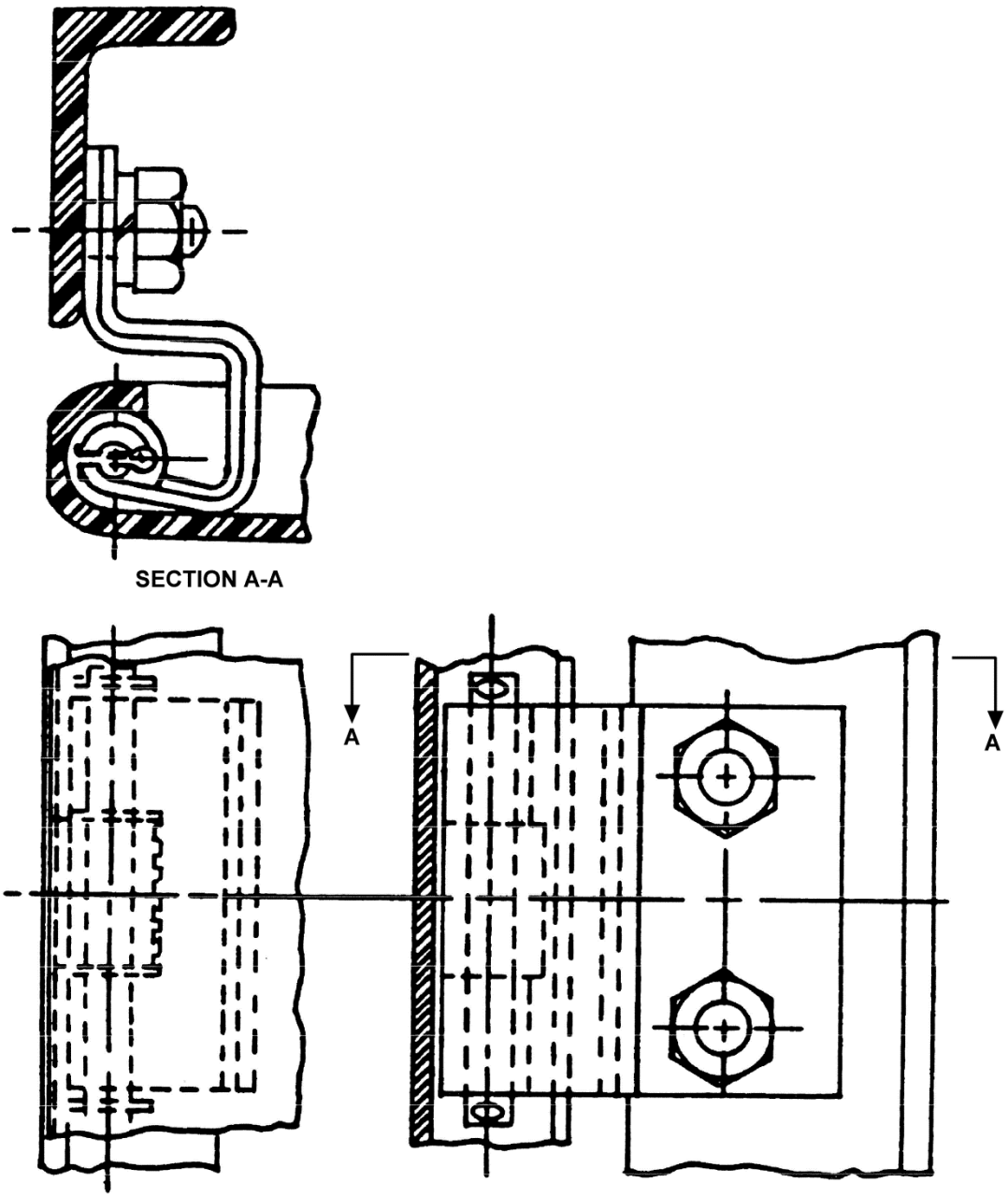


FIGURE 1. Typical panel hinge assembly.

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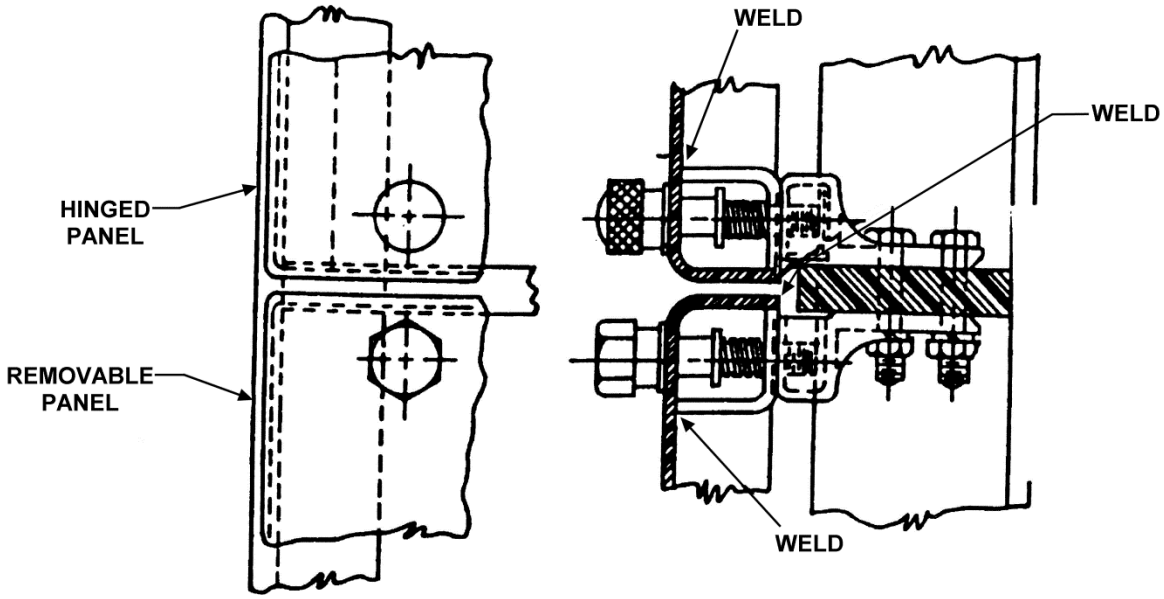


FIGURE 2. Typical panel latch bolt for hinged and removable panel.

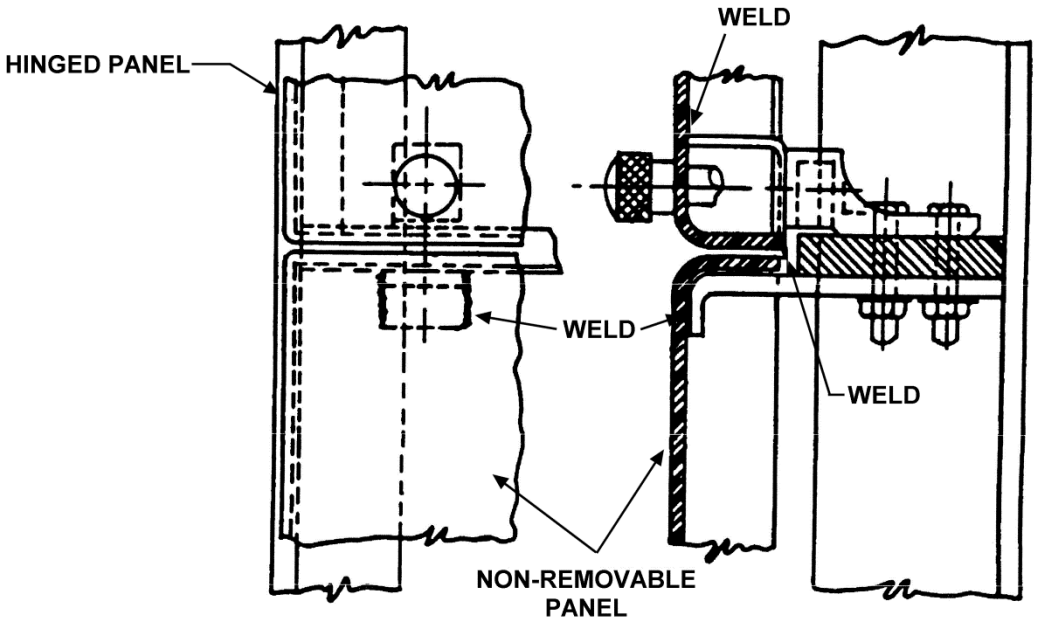


FIGURE 3. Typical panel latch bolt for hinged panel and method of securing stationary panel.

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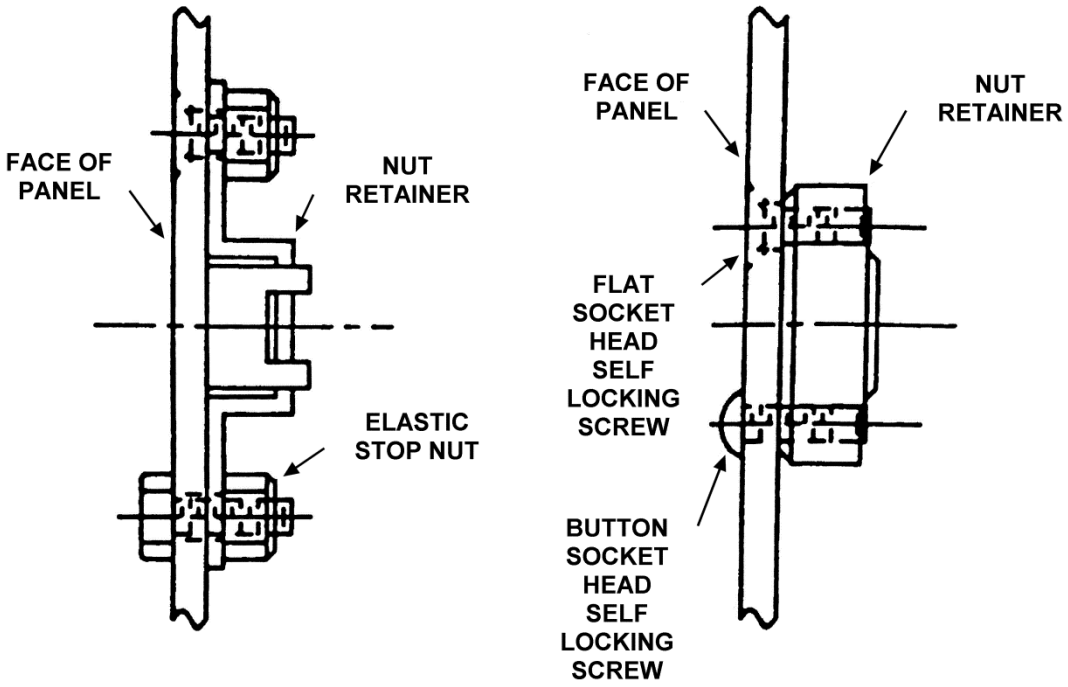


FIGURE 4. Anchor devices for thumbscrews.

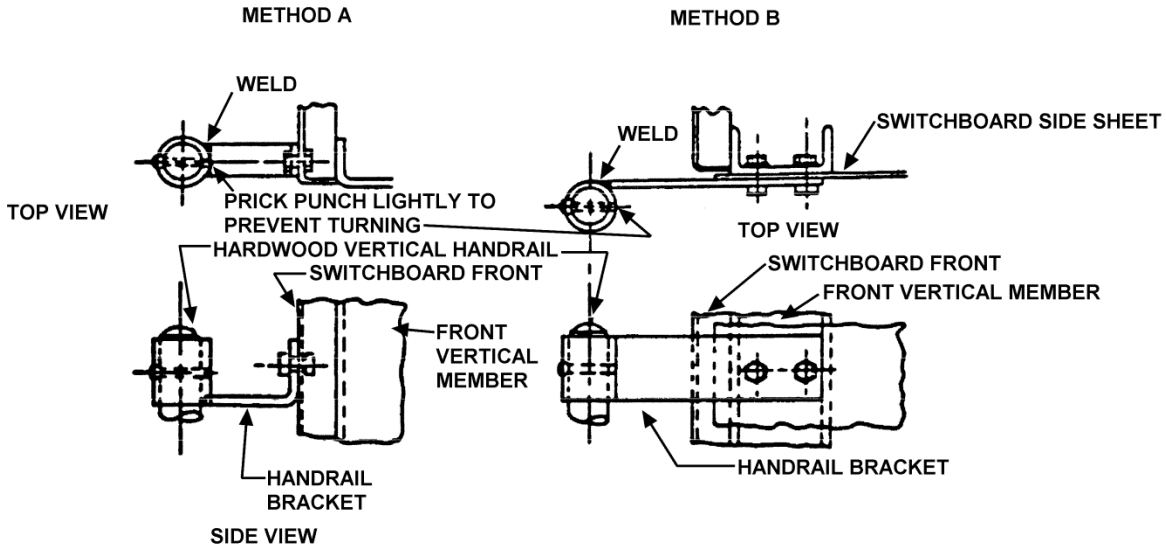


FIGURE 5. Typical vertical guard rail mounting.

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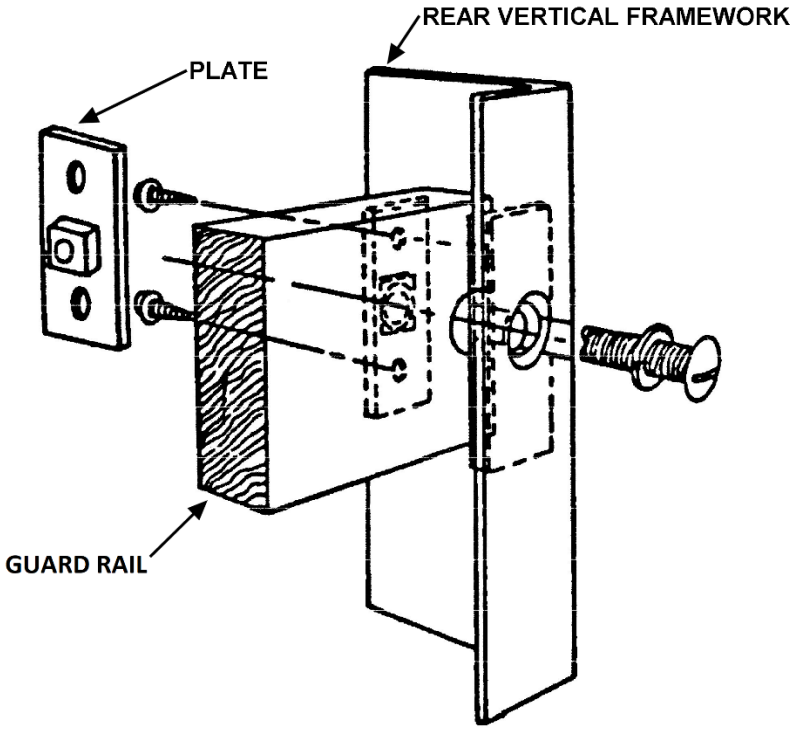
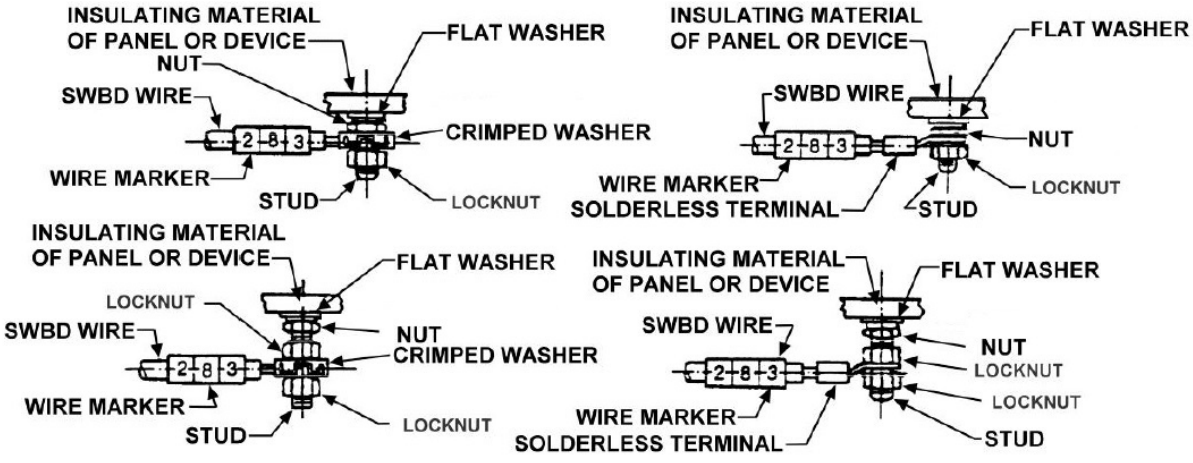


FIGURE 6. Typical removable guard rail mounting.



NOTE: Markings may be stamped on wire terminals or, if terminals are too small to permit stamping, synthetic resin tubing of MIL-I-631, type F, grade A, or fiber tags shall be used.

FIGURE 7. Typical cable marking and connections.

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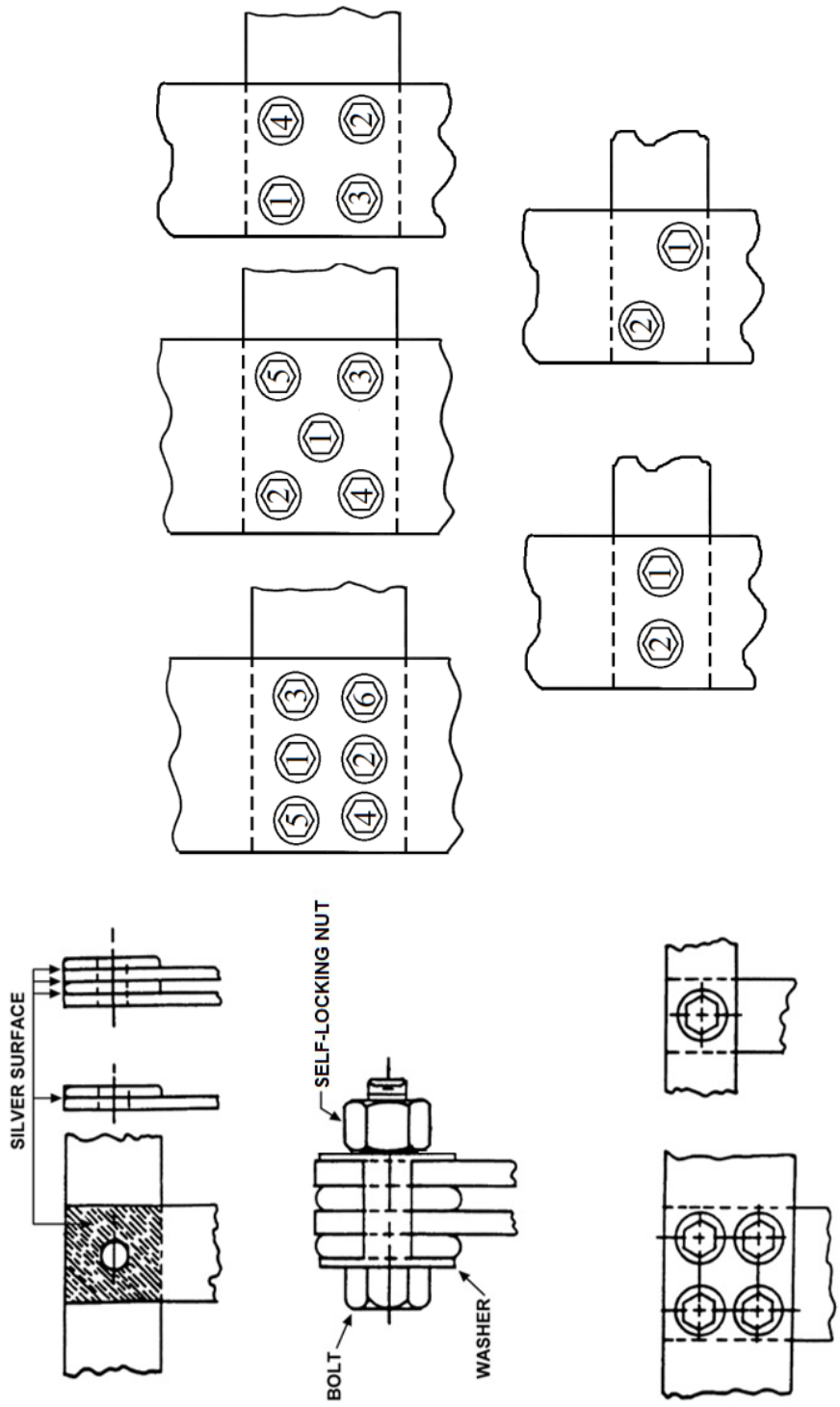


FIGURE 8. Tightening sequence and typical arrangement for bolting of bus connections.
(for combination of bolt and bus sizes, see table III.)

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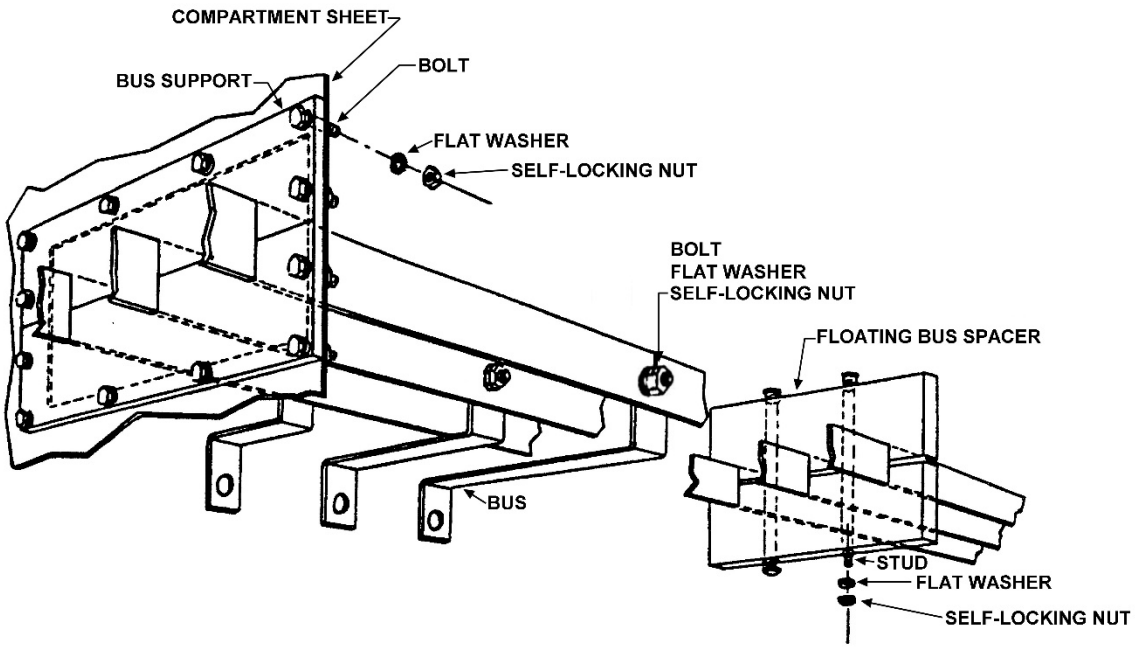


FIGURE 9. Typical bus bar mounting.

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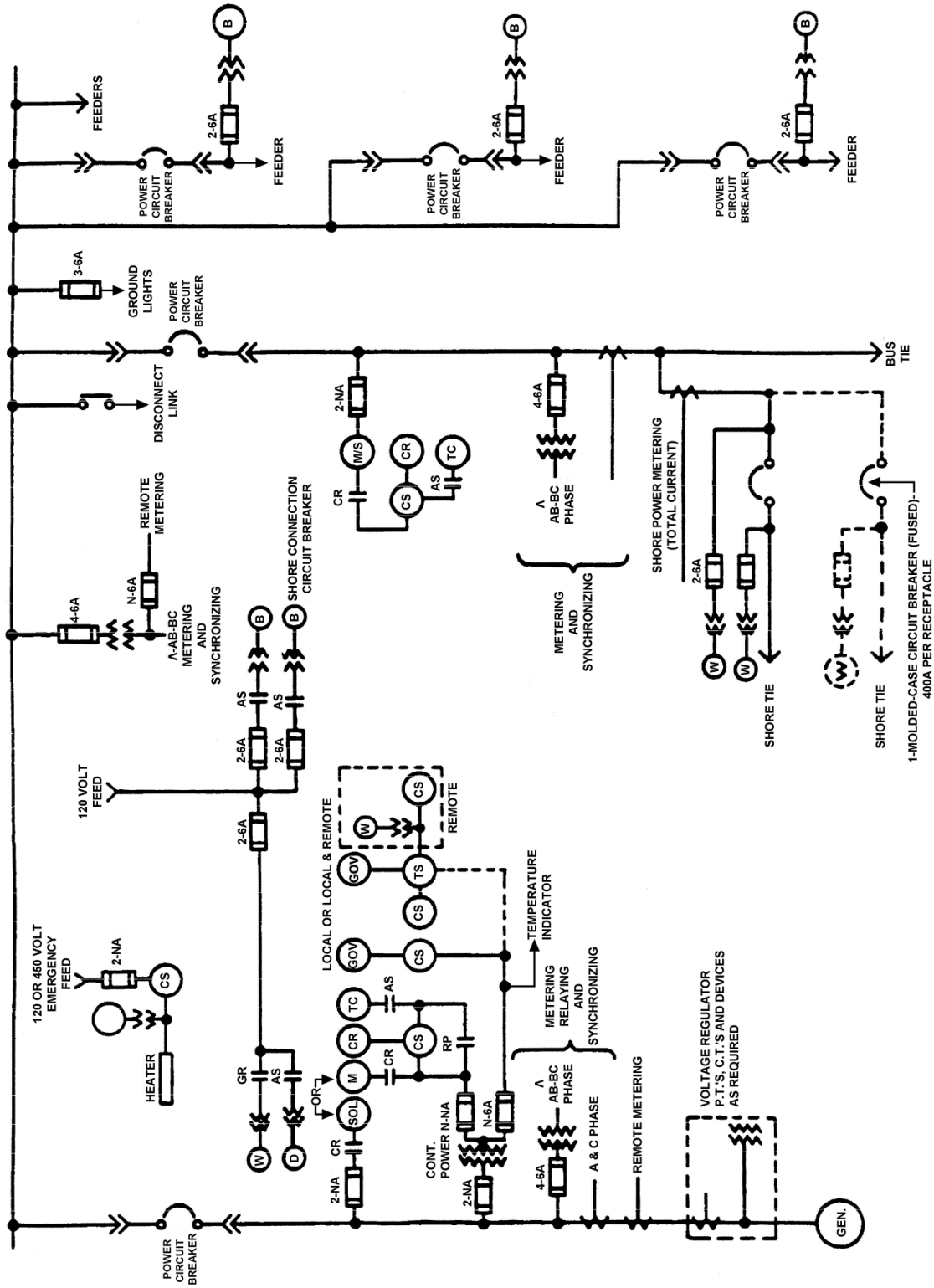


FIGURE 10. Typical one-line control diagram for ship's service AC power switchgear group having an EPCP. (see figure 12 for legend)

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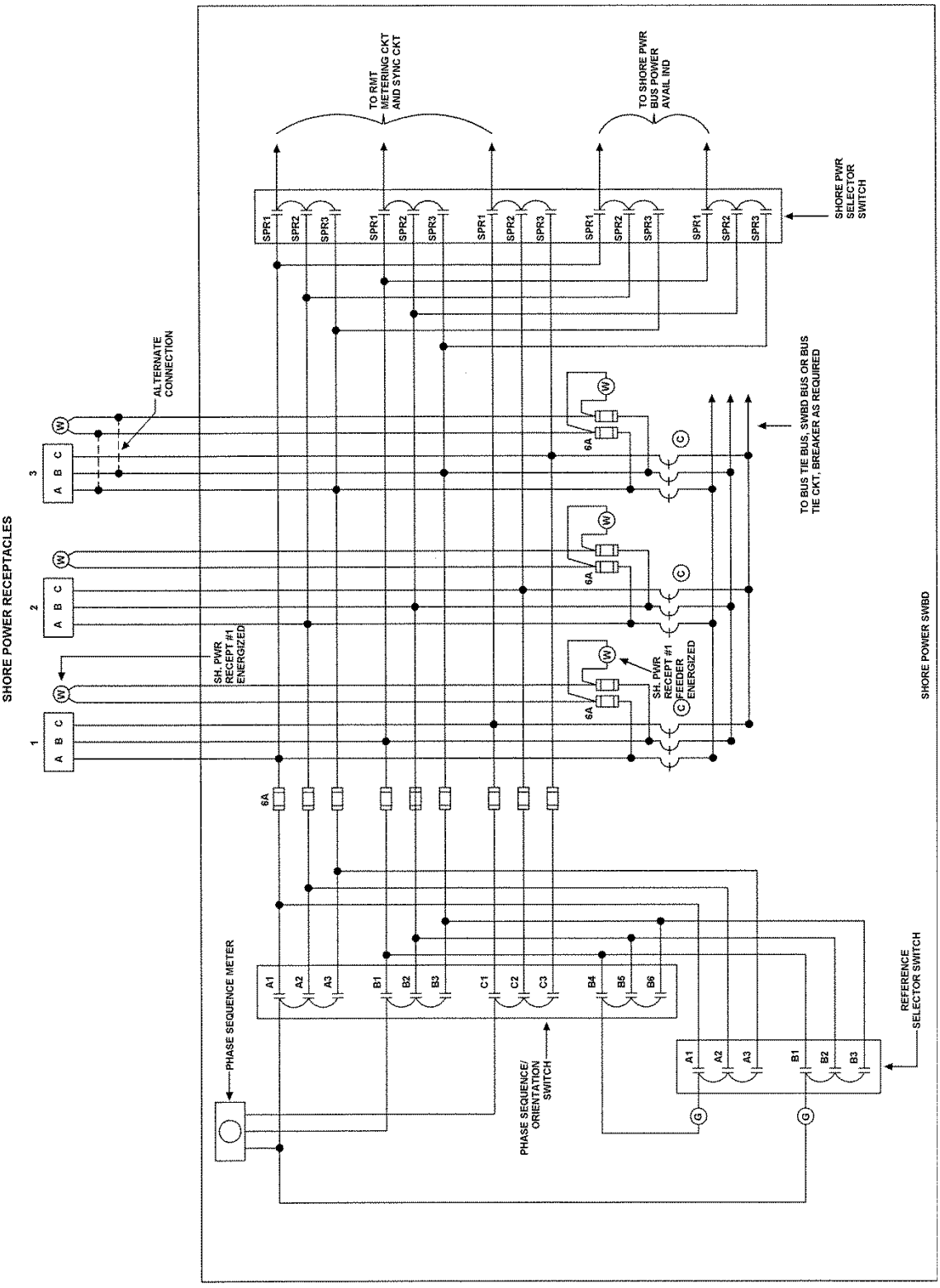


FIGURE 11. Typical shore power connection diagram (sheet 1 of 3).

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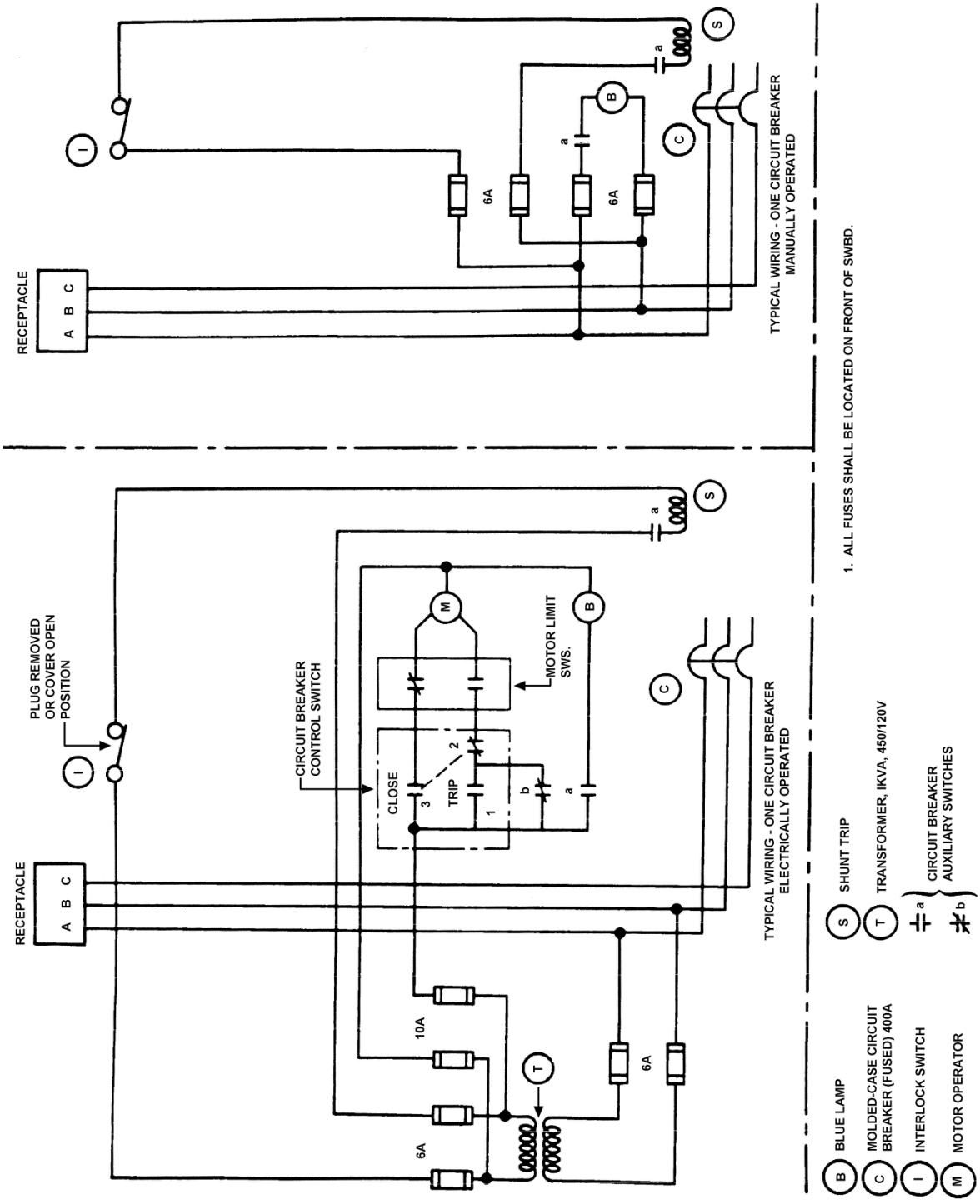


FIGURE 11. Typical shore power connection diagram (sheet 2 of 3) – Continued.

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TYPICAL SWITCH CONTACT ARRANGEMENTS

PHASE SEQUENCE/ORIENTATION TEST SWITCH

CONTACT	POSITION			
	OFF	1	2	3
A1		X		
B1		X		
C1		X		
A2			X	
B2			X	
C2			X	
A3				X
B3				X
C3				X
B4		X		
B5			X	
B6				X

1. CONTACT ARRGTS. APPLY TO THE 3 RECEPTACLE CIRCUITS SHOWN ON SHT. NO. 1 OF 3

2. ADDITIONAL CONTACTS SHALL BE PROVIDED ON THE REFERENCE SELECTOR SWITCH TO SWITCH METERING AND SYNCH. EQUIPT. TO THE RECEPTACLE SELECTED AS THE REFERENCE, AS REQUIRED

REFERENCE SELECTOR SWITCH

CONTACT	POSITION			
	OFF	1	2	3
A1		X		
A2			X	
A3				X
B1		X		
B2			X	
B3				X

CIRCUIT BREAKER CONTROL SWITCH

CONTACT	POSITION		
	TRIP	NORMAL	CLOSE
1	X		
2	X	X	
3			X
4	X		
5	X	X	
6			X
7	X		
8	X	X	
9			X

FIGURE 11. Typical shore power connection diagram (sheet 3 of 3) – Continued.

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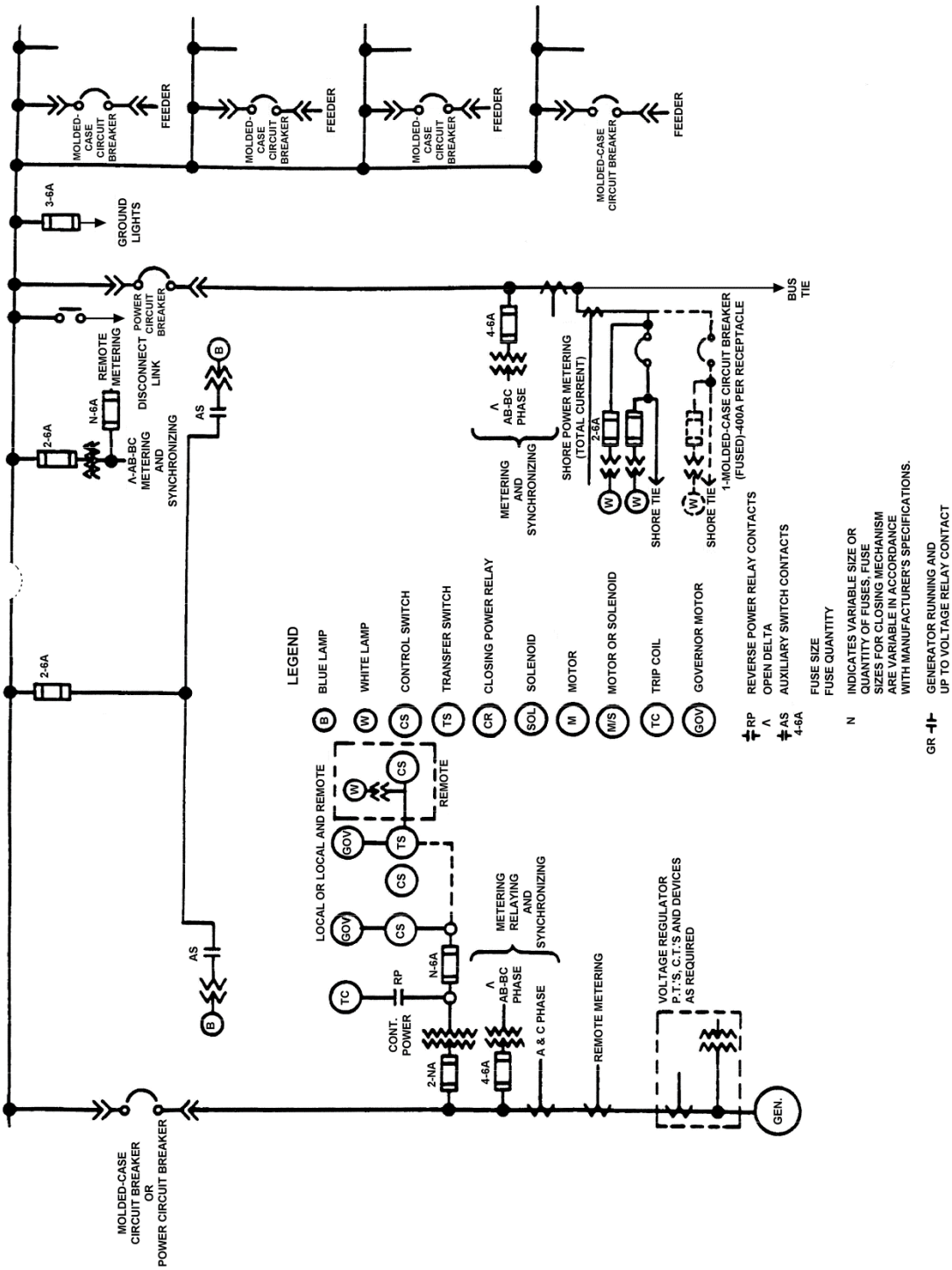


FIGURE 12. Typical one-line control diagram for ship's service AC power switchgear group not having EPCP or EPCC.

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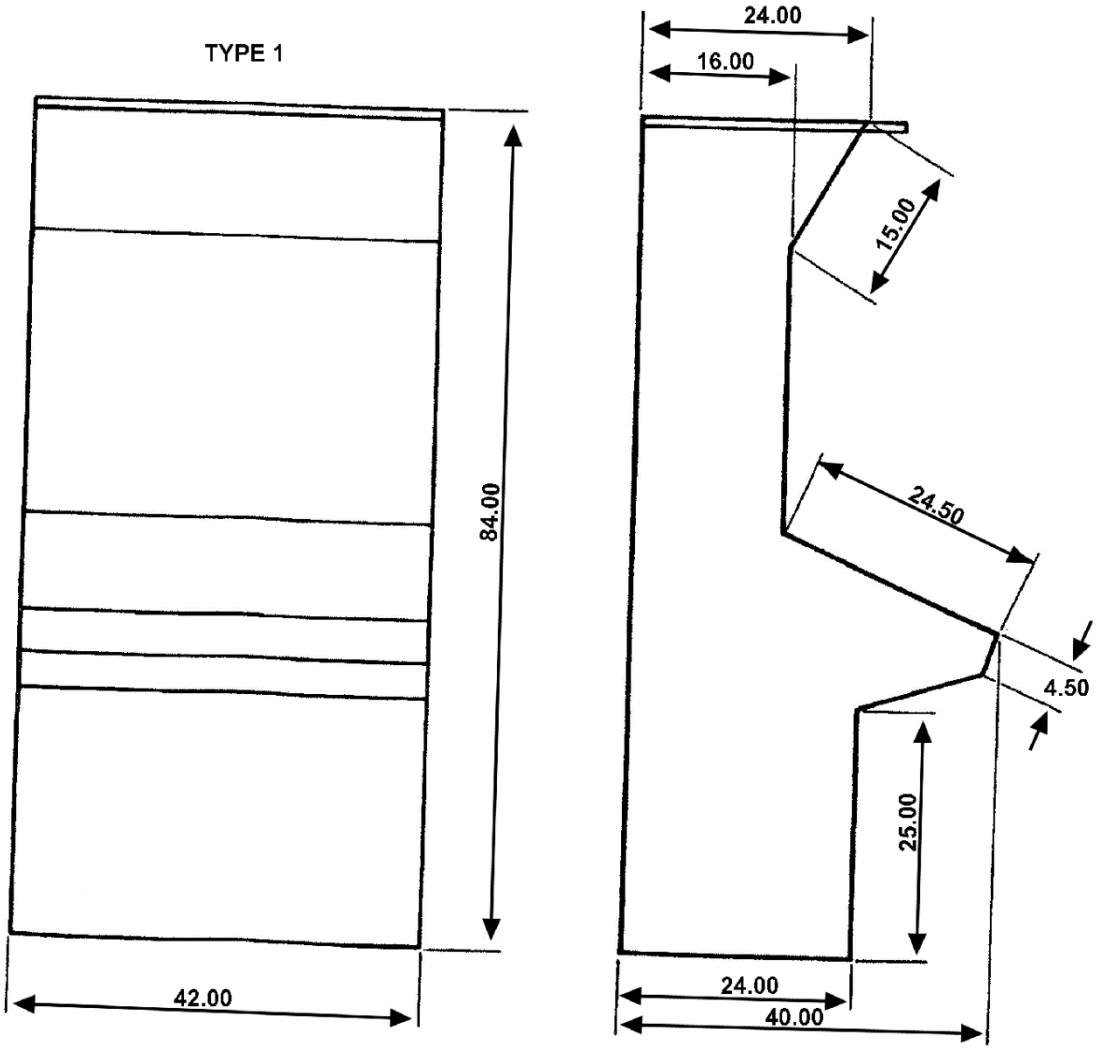


FIGURE 13. Desk types EPCP (sheet 1 of 2).

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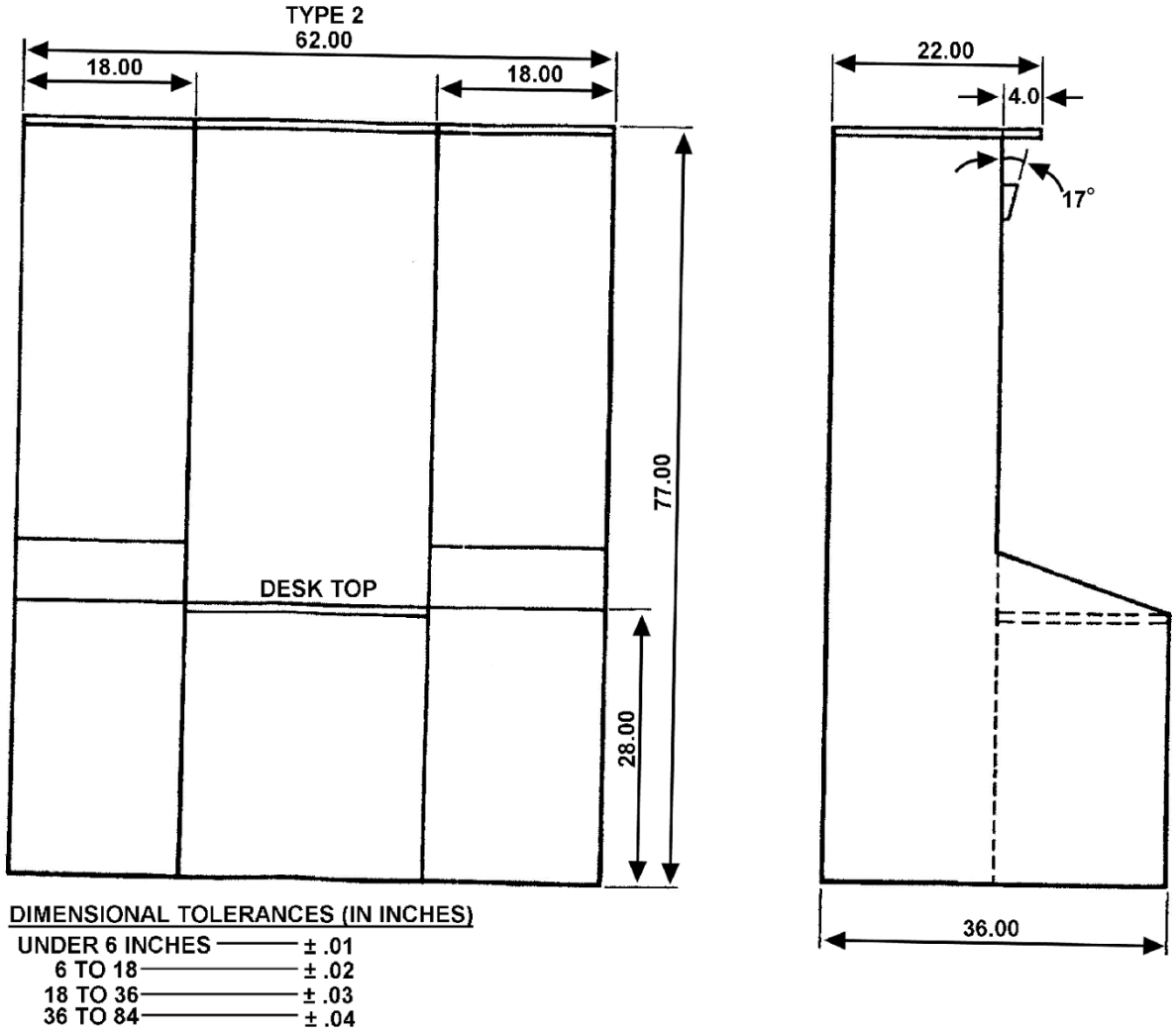


FIGURE 13. Desk types EPCP (sheet 2 of 2) – Continued.

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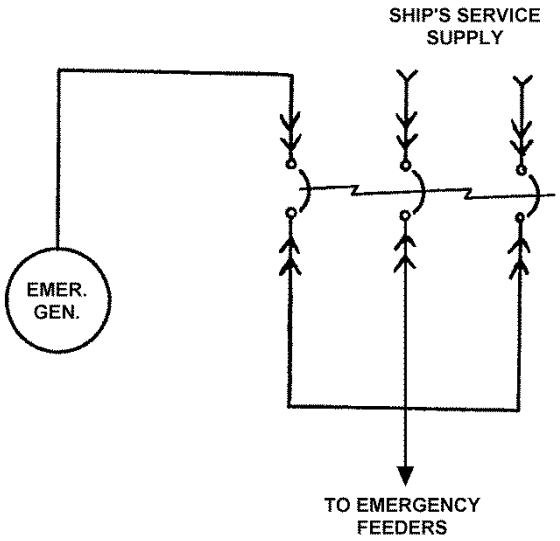


FIGURE 14. ABT, circuit breaker type, PIN M17773-4A29.

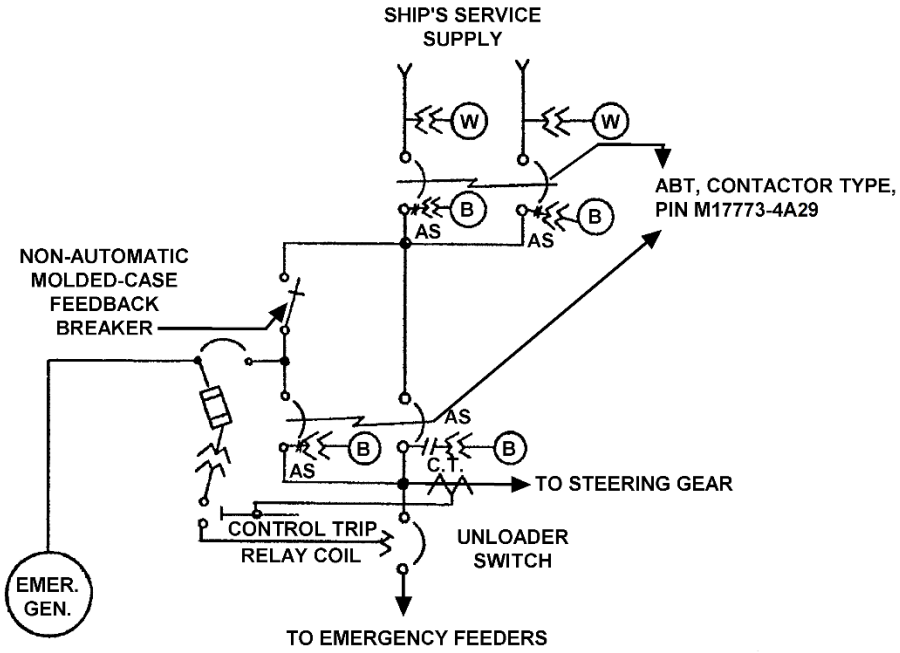


FIGURE 15. One-line diagram of emergency power system.

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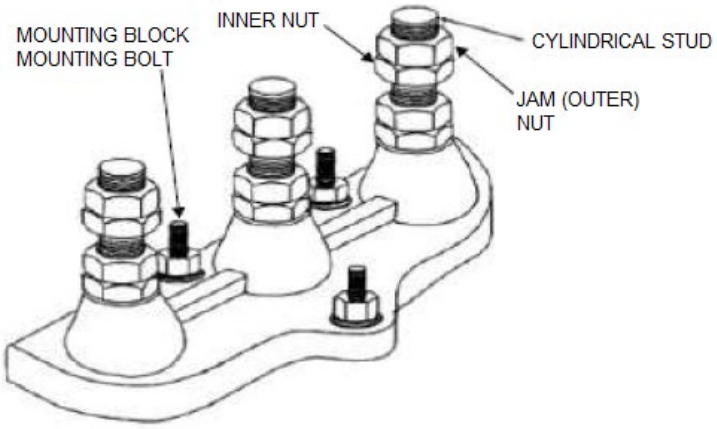
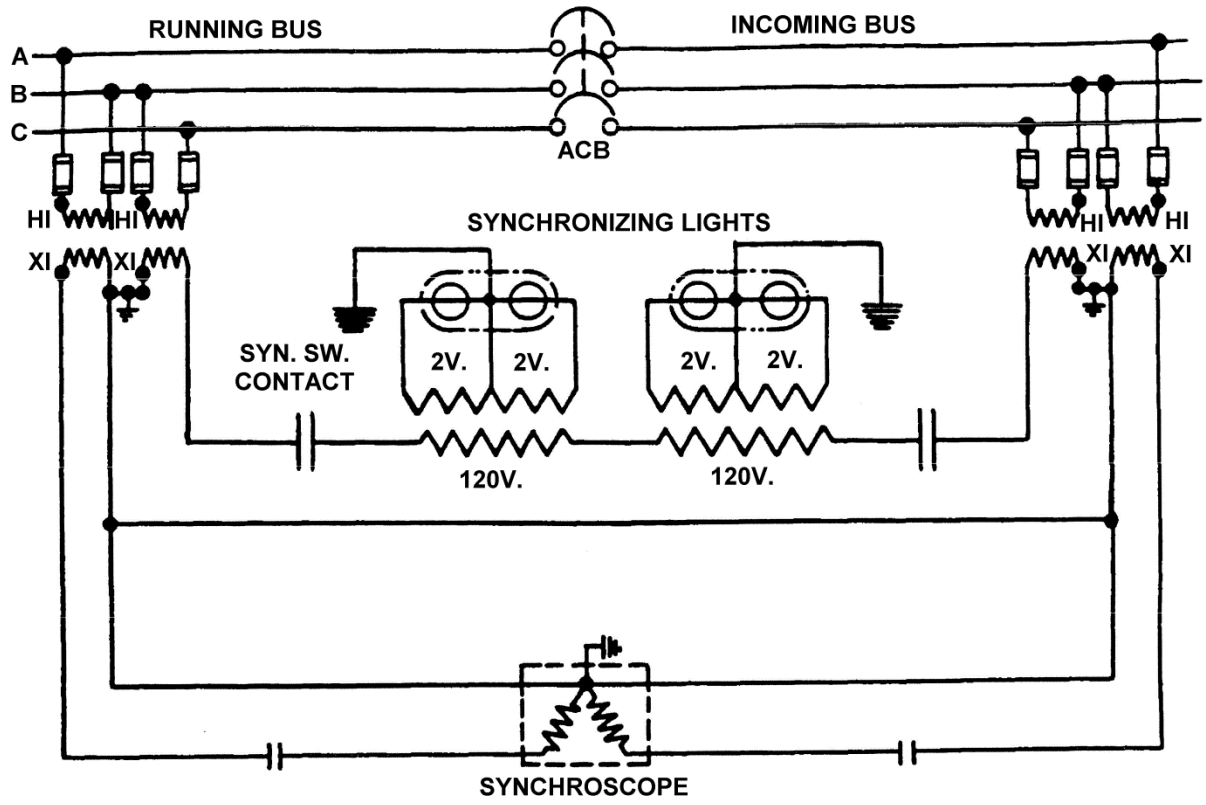


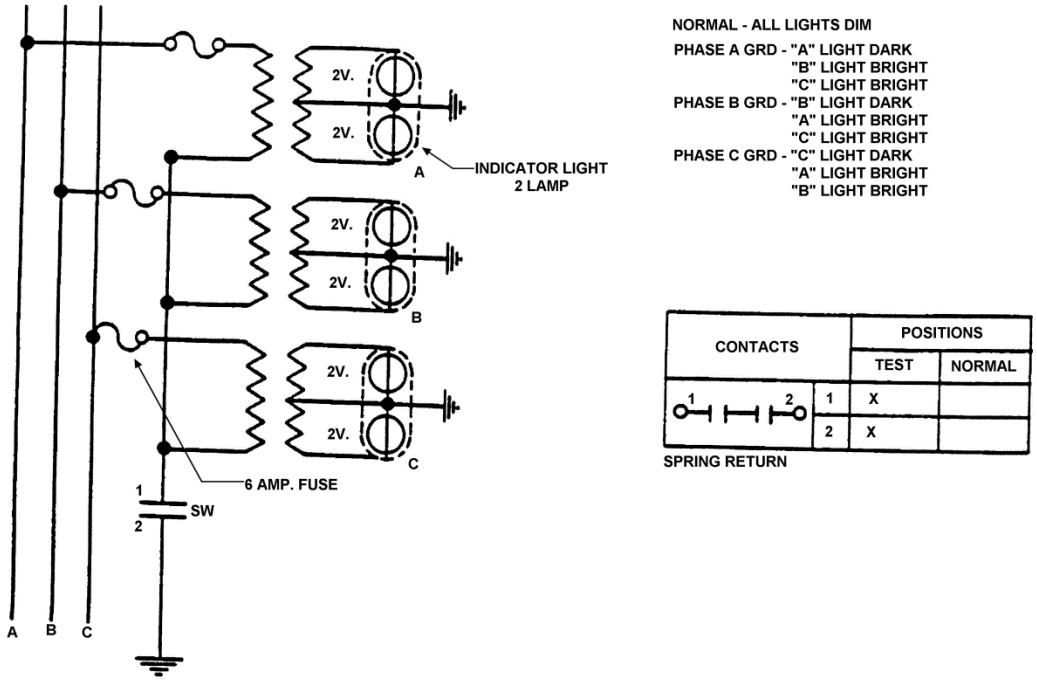
FIGURE 16. Typical connections for molded case circuit breaker or molded case switch mounting bases with cylindrical studs.



NOTE: This diagram is typical only, representing a reference bus and a second bus to be synchronized with it. Where provision should be made for a third bus to be synchronized with either of these buses, additional contacts in the synchronizing transfer switch shall be provided as required.

FIGURE 17. Typical synchronizing connections.

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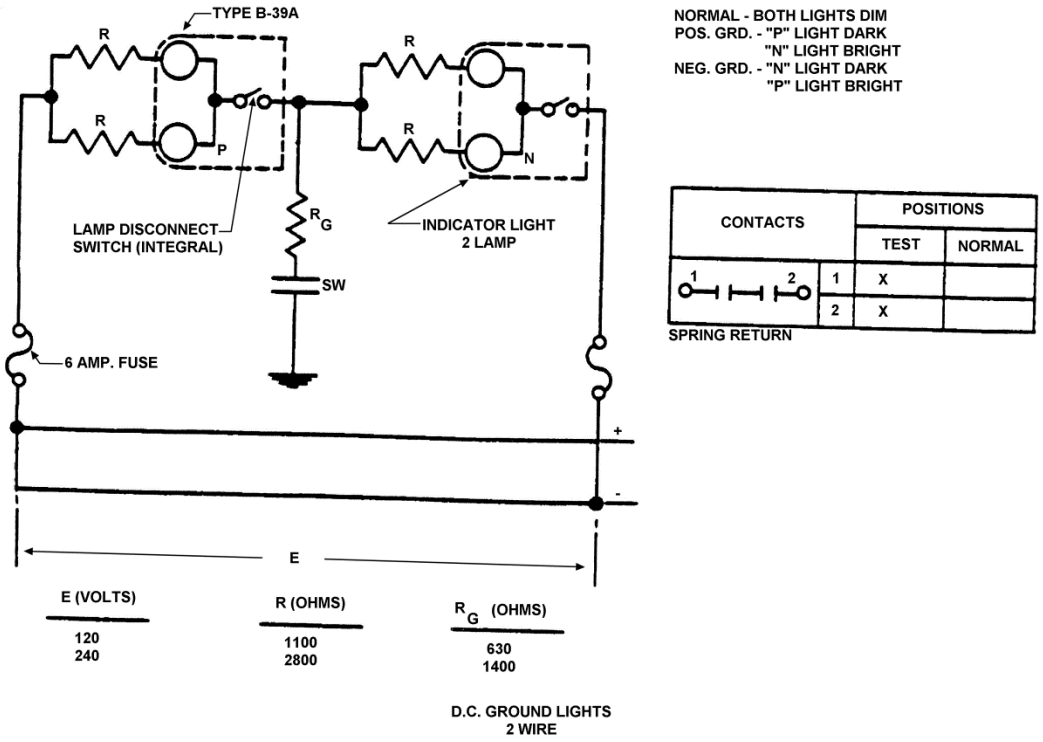


FIGURE 18. Ground light schemes.

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LOW VOLTAGE SWITCHGEAR CYBERSECURITY REQUIREMENTS

A.1 SCOPE

A.1.1 Scope. This appendix covers low voltage switchgear cybersecurity requirements. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

A.2 APPLICABLE DOCUMENTS

A.2.1 General. The documents listed in this section are specified in Appendix A of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in Appendix A of this specification, whether or not they are listed.

A.2.2 Government documents.

A.2.2.1 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMITTEE ON NATIONAL SECURITY SYSTEMS INSTRUCTION (CNSSI)

CNSSI No. 1253 - Security Categorization and Control Selection for National Security Systems

(Copies of this document are available online at www.cnss.gov/CNSS/issuances/Instructions.cfm.)

DEPARTMENT OF DEFENSE ISSUANCES

DoDI 8500.01 - Cybersecurity

DoDI 8510.01 - Risk Management Framework for DoD Systems

(Copies of these documents are available online at www.esd.whs.mil/DD/.)

FEDERAL INFORMATION PROCESSING STANDARDS (FIPS) PUBLICATION

FIPS Pub 199 - Standards for Security Categorization of Federal Information and Information Systems

(Copies of this document are available online at https://csrc.nist.gov/publications/PubsFIPS.html.)

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY (NIST)

NIST SP 800-53 - Security and Privacy Controls for Information Systems and Organizations

NIST SP 800-53A - Assessing Security and Privacy Controls in Federal Information Systems and Organizations: Building Effective Assessment Plans

(Copies of these documents are available online at www.nist.gov.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) MANUAL

NAVSEA 9400.2-M - NAVSEA PIT-Control System Cybersecurity Implementation Manual

(Copies of this document are available online at www.navsea.navy.mil.)

A.2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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A.3 CYBERSECURITY REQUIREMENTS

A.3.1 System categorization (SC). The generalized format for expressing the system category is confidentiality, integrity, availability, and specific impact where the acceptable values for potential impact are low, moderate, or high. In accordance with FIPS Pub 199, the security category of the low voltage switchgear is determined as:

SC low voltage switchgear = {(confidentiality, LOW), (integrity, HIGH), (availability, HIGH)}

When specified (see 6.2), if a specific low voltage switchgear design is determined to have different impact values, the new impact values shall be reviewed and approved by the Technical Authority.

A.3.2 Vulnerability analysis. When specified (see 6.2), an analysis of vulnerabilities shall drive implementation details of the low voltage switchgear, both physically and with respect to cybersecurity, to mitigate those vulnerabilities and the impacts of attacks. The analysis of vulnerabilities shall identify the following and be documented:

a. Low voltage switchgear assets (e.g., control consoles, human-machine interfaces [HMIs], backbone networks, control processors, communication interfaces, input/output [I/O], networks, and other such assets)

b. Low voltage switchgear vulnerabilities, both accidental and malicious (e.g., advanced persistent threats [APTs] and vulnerabilities of system availability, integrity, and authenticity)

c. Risk to operations and assets, individuals, other organizations, and the Nation posed by the vulnerabilities

d. Mitigation considerations

A.3.3 Security controls. The security controls from the baseline, based on system categorization in A.3.1, shall be tailored from NIST SP 800-53 using CNSSI No. 1253. The tailored-out security controls shall be provided with technical justification. The security control selection process shall be fully executed in accordance with DoDI 8510.01 by considering the vulnerabilities and risks identified in A.3.2. Special emphasis shall be applied in the following areas:

a. Access control

(1) The principle of least privilege

(2) Automatic account lockout after unsuccessful logon attempts

(3) Remote access capability

(4) Prohibition of wireless access

b. Audit and accountability

(1) Generation of time stamps for audit records

(2) Protection of audit information from unauthorized access, modification, and deletion

c. Configuration management

(1) Access restrictions for any changes including upgrades and modifications to hardware, software, or firmware components

(2) Documentation of configuration settings that can be changed in hardware, software, or firmware components

(3) Disabling of unused or unnecessary physical and logical ports and protocols.

(4) Software whitelisting of a deny-all, permit-by-exception policy to allow the execution of authorized software

d. Identification and authentication

(1) User identification and authentication.

(2) Password-based authentication including minimum password complexity, lifetime of password, and prohibition of password reuse

(3) Hardware token-based authentication such as PKI-based tokens (CAC)

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(4) Change of vendor default authenticator prior to delivery/installation

(5) Device identification and authentication using Media Access Control

e. Physical and environmental protection

(1) Use of alarms to monitor physical access point

(2) Use of lockable physical enclosures

f. System and services acquisition

(1) Structure hardware, software, and firmware to control access with least privilege

g. System and communications protection

(1) Failure in a known safe state

h. System and information integrity

(1) Malicious code protection mechanisms including anti-virus signature definitions and secure coding practices

(2) Intrusion detection for network connections

(3) Integrity checks of software, firmware, and information at system startup

(4) Code authentication of software or firmware prior to installation

(5) Check of the valid syntax and semantics of inputs

A.3.4 Implementation. The selected security controls for the low voltage switchgear design from A.3.3 shall be implemented in accordance with DoDI 8500.01, DoDI 8510.01, and NAVSEA 9400.2-M.

A.3.5 Assessment. The implemented security controls for the low voltage switchgear design in A.3.4 shall be assessed in accordance with DoDI 8510.01 and NIST SP 800-53A.

A.4 CYBERSECURITY VERIFICATION

A.4.1 Verification. The requirements specified in A.3 shall be verified (see table IX). Any failure to meet these requirements is cause for rejection of the low voltage switchgear.

A.4.2 Additional specified cybersecurity requirements. Any additional cybersecurity requirements as specified in 6.2 shall be verified. Any failure to meet these requirements is cause for rejection of the low voltage switchgear.

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CONCLUDING MATERIAL

Custodians:
Army – AV
Navy – SH
Air Force – 85

Preparing activity:
Navy – SH
(Project 5925-2014-002)

Review activities:
Army – CR, MI
Navy – AS, CG
Air Force – 03, 19, 99
DLA – CC

Civil agencies:
GSA – FAS
MISC – MDA
NASA – NA

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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